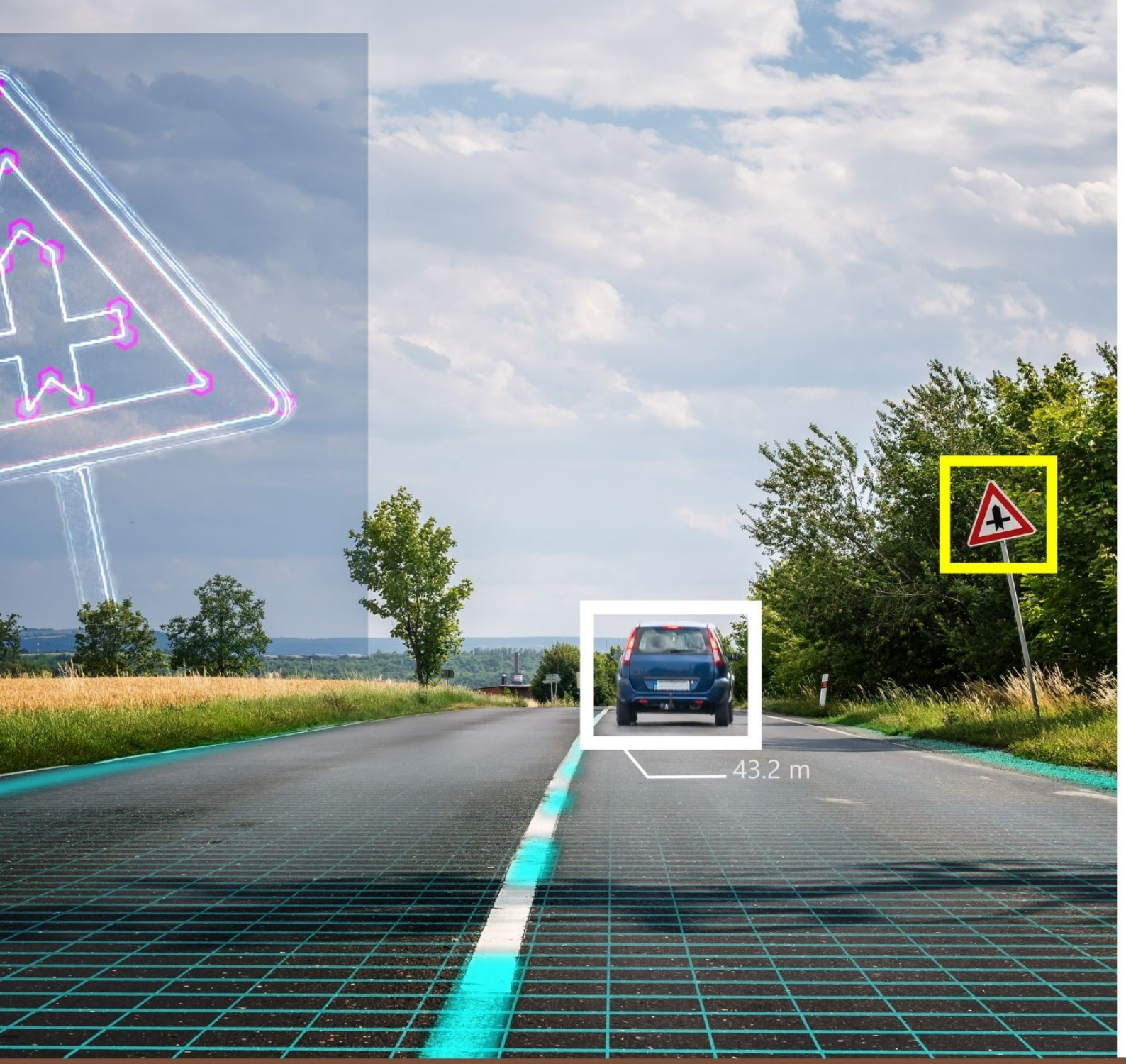


COMPUTER VISION

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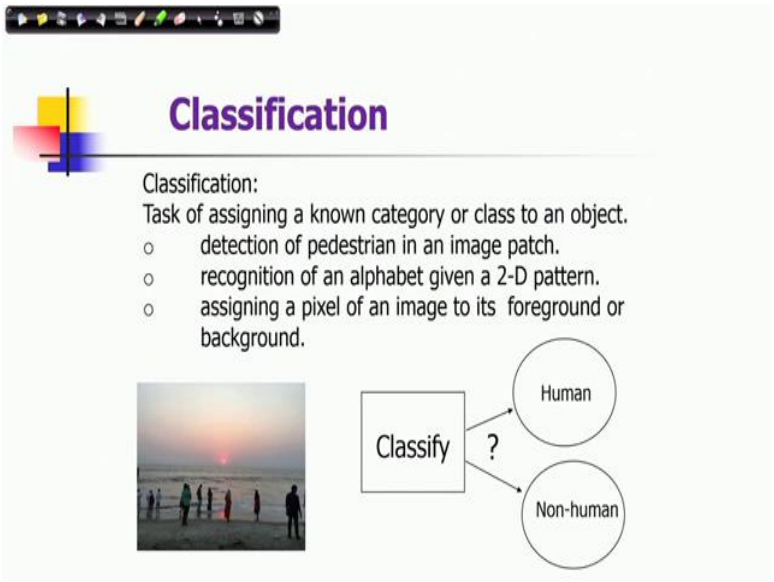
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Lecture - 46
Clustering and Classification Part – I

We start a new topic in this course and that is on Clustering and Classification.

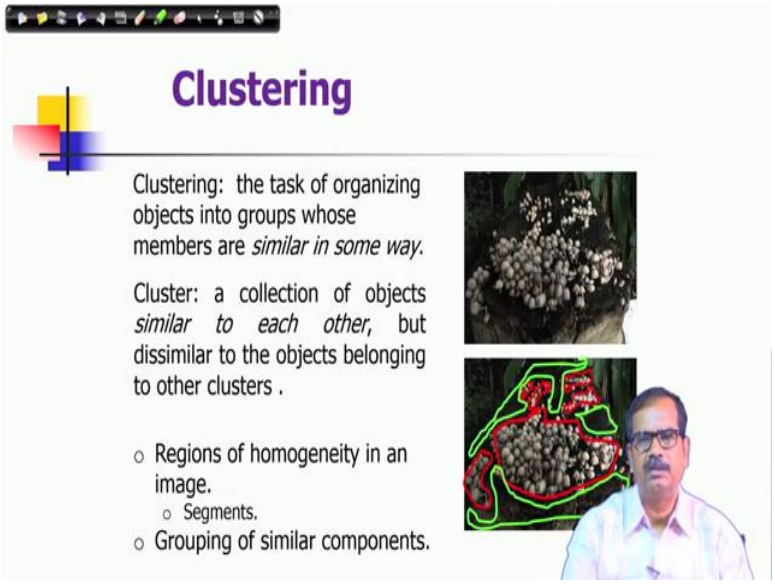
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So, the classification is a task of assigning a known category or class to an object. For example, you take this image and you are asked to classify this image that which are the regions which contains human and or which contains which and whether it is human or non human. So, this is the classification task. So, suppose I have given this particular region of the image and I asked you to classify whether it contains human or not.

And in this way I can in this system I can give various patches from this images from this image and we can ask this ask to solve this problem. There could be other kinds of problems also like detection of pedestrian in an image patch, recognition of an alphabet given a 2-D pattern, assigning a pixel of an image to its foreground or background. So, these are different other classification problems and in this way you can define infinitely many types of problems. So, this is a nature of a classification problem.

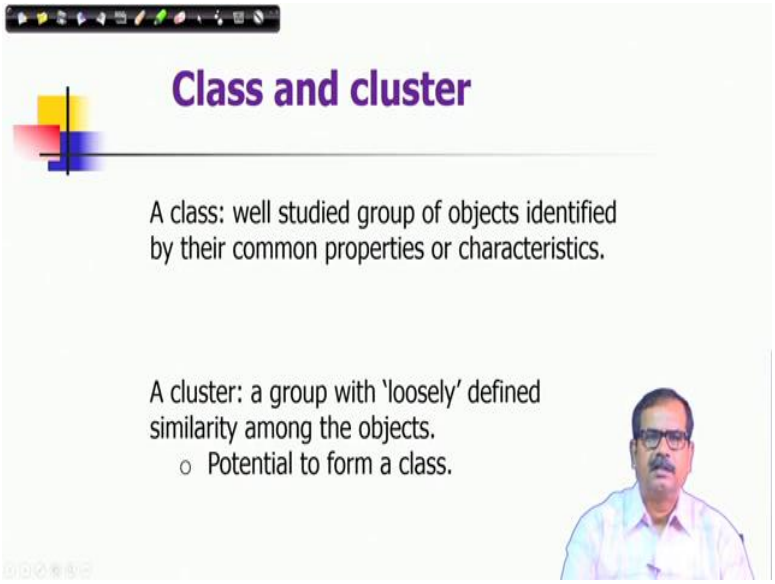
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Whereas a clustering it is a task of organizing objects into groups whose members are similar in some way. So, the cluster is a collection of objects which are similar to each other, but dissimilar to the objects belonging to other clusters. You can consider examples of some of the clustering as finding out regions of homogeneity in an image and which means you are deriving segments this is also a problem similar to clustering.

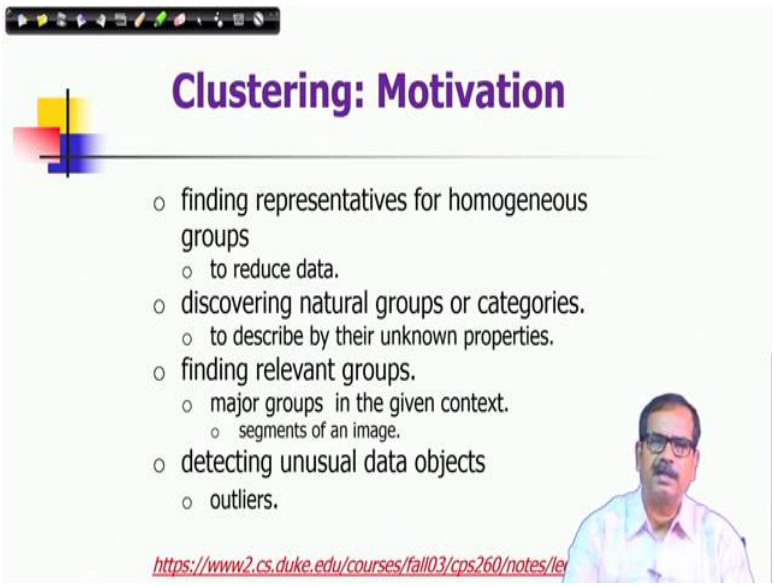
Or grouping of similar components and example we can see here suppose I have given an image of mushroom and in the background there is a humous substance and we would like to cluster the similar pixels or similar regions in this case. So, one result could be in this form here you can see that there are two primarily two regions are two types of clusters are there shown by the green contours and red contours. Again there are different regions of green contours and red contours which can be treated also as segments.

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So, what is the difference between class and cluster? A class is a well studied group of objects identified by their common properties or characteristics where as a cluster it is a group with loosely defined similarity among the objects it is potential to form a class.

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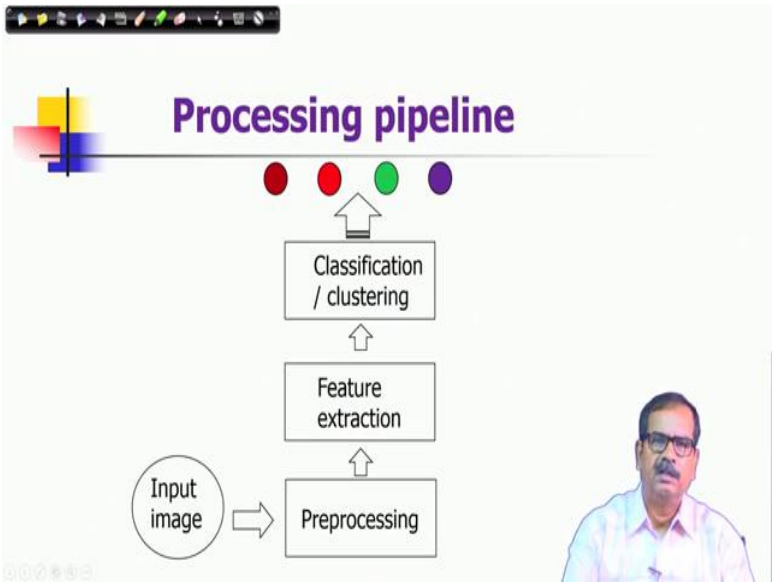


So, what are the motivations of clustering? I have given one example of its application like segmenting images will see different other motivations. Like you can find representatives for homogeneous groups and this would reduce your total data representations you can represent data by smaller set of representative samples we are in those characteristics of

data. Then discovering natural groups or categories that is also another motivation. So, that we can describe them by their unknown properties and finding relevant groups.

So, segmentation is one such example where we would try to draw our attention to relevant groups in the distribution, so in the images. So, it is a measure groups in the given context that we would like to identify like segments of an image. Then detecting unusual data objects, so usual like outliers in data. So, those are also can be detected using this kind of clustering techniques.

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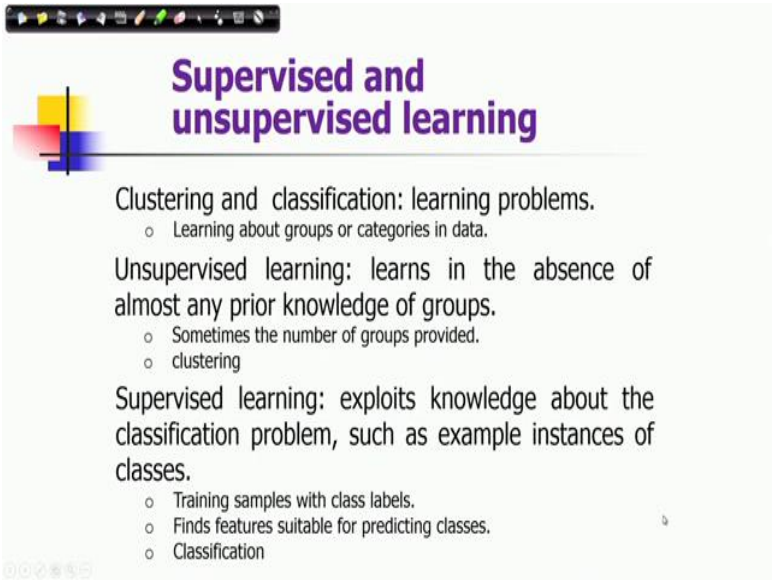


So, in the context of our image and video image processing computer visions also video processing. So, how clustering and classifications in that are placed in this particular processing of information. So, as you can see that it is a very high level process of information. So, at the lower level from the images we would like to derive certain representation of data in terms of feature extraction.

So, it should pass through the process of pre processing feature extraction and then you can get the represent the objects or images image patches in by feature descriptors or feature vectors that you have discussed on already in previous lectures. And we know different techniques of driving deriving the points of importance relevance in an image and then describing their neighbourhoods or also different patches or images you can describe by their corresponding feature vectors.

These feature vectors they are the representative of those corresponding class or groups or objects they are used for the purpose of classification and clustering. So, and then you have to though the task is that no you have to assign them certain known or unknown groups whatever that means they should be similar. For clustering groups are not well defined as we mentioned. So, in the example with different colors we are trying to represent those different kinds of classes or groups.

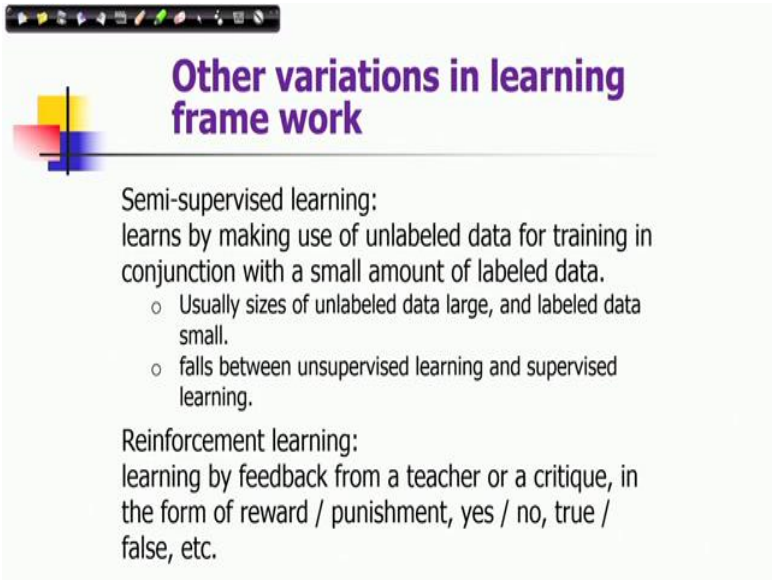
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So, we can see the approaches of you know clustering classification which are essentially learning problems. Now, there are two approaches supervised and unsupervised learning when you consider its a learning problem. So, here we are learned we are learning about groups or categories in data. And when we say it is an unsupervised learning then it learns in the absence of almost any prior knowledge of groups almost I told because sometimes know the number of groups that information is provided.

Now, this problem is like clustering. So, in clustering we used to unsupervised learning and in a supervised learning it exploits knowledge about the classification problem such as example instances of classes. Say in this case training samples with class levels are provided for solving this problem and it finds features suitable for predicting classes. So, this problem supervised learning is also used in the use solving classification problem.

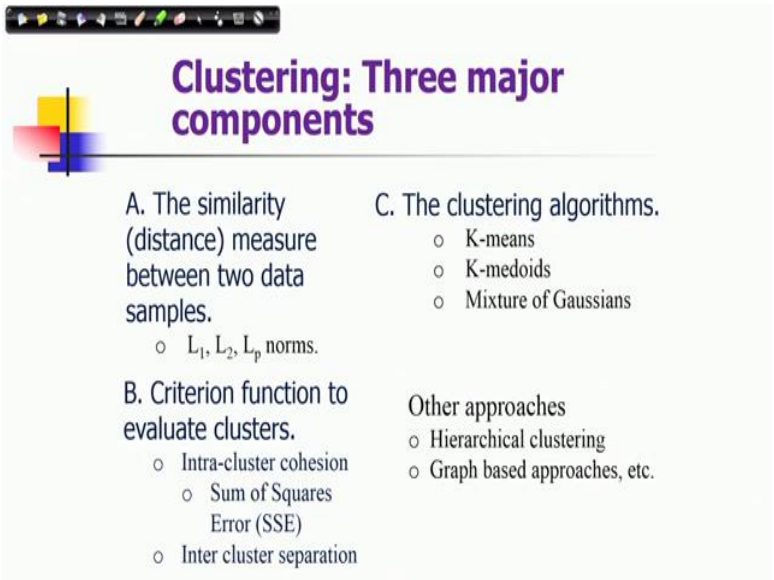
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Now, there are other variations in this learning framework you have semi supervised learning and reinforcement learning like in semi supervised learning it learns by making use of unlabeled data for training in conjunction with a small amount of labeled data. Usually the sizes of unlabeled data it is they are large and labeled data they are kept small.

And in this case this framework this learning mechanism it falls between unsupervised and supervised learning. In the reinforcement learning it is learning by feedback from a teacher or a critique in the form of reward or punishment yes or no true or false etcetera.

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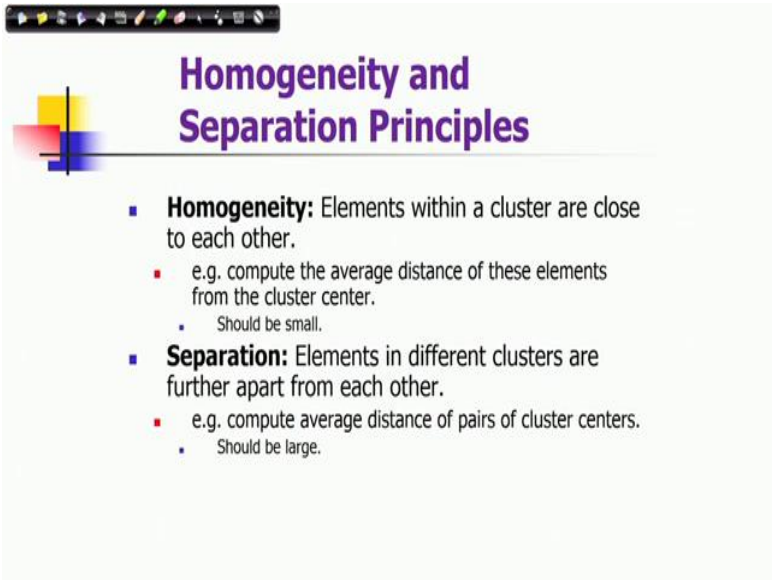
So, we will be discussing now the methods of clustering techniques, we will first discuss methods of clustering techniques and then we will discuss about the classification techniques. So, in clustering there are three measure components; the first it should have a distance measure or which measure similarity between two data or samples. And then there should be some criteria function to evaluate clusters and then of course, the methodology the algorithm by which you should compute you should solve the problem.

So, for defining similarity different distances could be used and some examples of distances we have already seen in various other applications in this particular course also in on various topics. Suppose the L 1 norm, L 2 norm and generalized L p norm could be used and for criteria function or to evaluate clusters there are two particular properties which are looked at it which are looked for in having a good clustering solution.

One thing is that there should be a good intra cluster cohesion which means the members of the cluster they should have good homogeneity property. And one of the measures in this case could be sum of squares of error of deviations from that property or there should be inter cluster separation; that means, groups are also well separated well discriminated.

About the clustering algorithms in this particular lecture or in this course will be learning three different clustering algorithms; one is K means and then K medoids and mixture of Gaussians Gaussian technique. But there could be various other approaches like know there could be hierarchical clustering techniques graph based approaches etcetera will be considering only this three in this course.

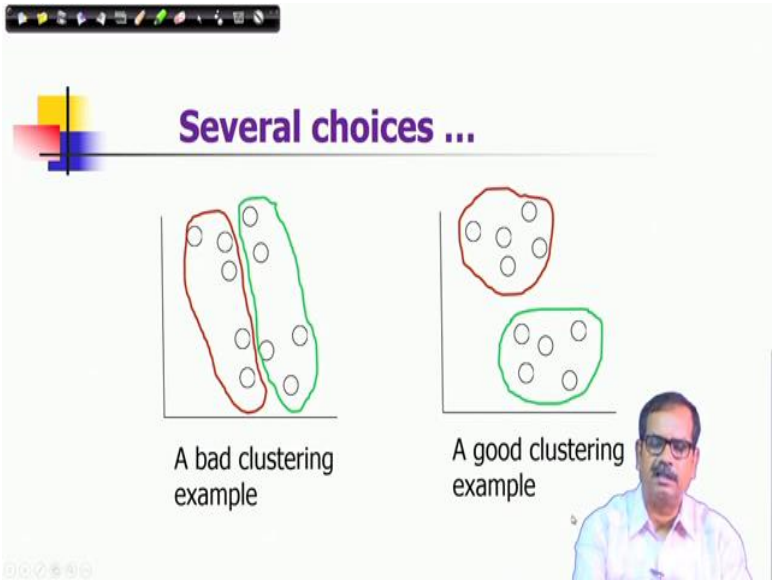
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Homogeneity and separation principles that we discussed when we are considering evaluation of cluster clustering evaluation of clusters those are derived by technique. So, homogeneity as I mentioned it is elements within a cluster which should be close to each other. And for example, you can compute the average distance of these elements from the cluster center. So, if this distance is small then the cluster is good it preserves the homogeneity it has good homogeneity property.

Whereas this separation property in clusters are the elements in different clusters they should be further apart from each other. So, in this case may compute average distance of pairs of cluster centers and those centers should be no should be placed apart they should. So, this distance should be large for having a good separation property.

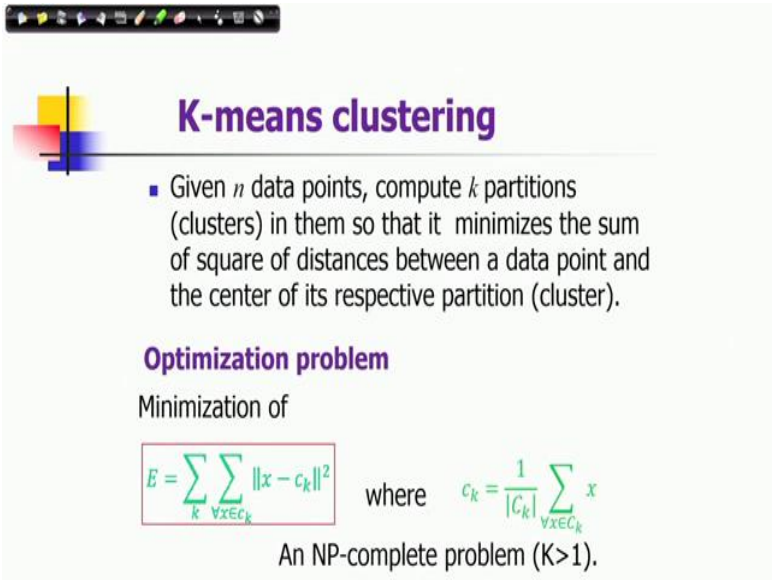
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There could be a several choices of clustering or partitioning, so its a non trivial partitioning problem in that sense. Suppose I give you this data distribution its a 2 dimensional data points which are shown by this circles and locations of the data are shown in has a 2 dimensional coordinate space. So, these are the elements.

So, one could one kind of partition could be like this and the other kind of partitioning could be like this. Now, by considering the two property homogeneity and separation; the first one is a bad clustering example where this is the second one is the good clustering example.

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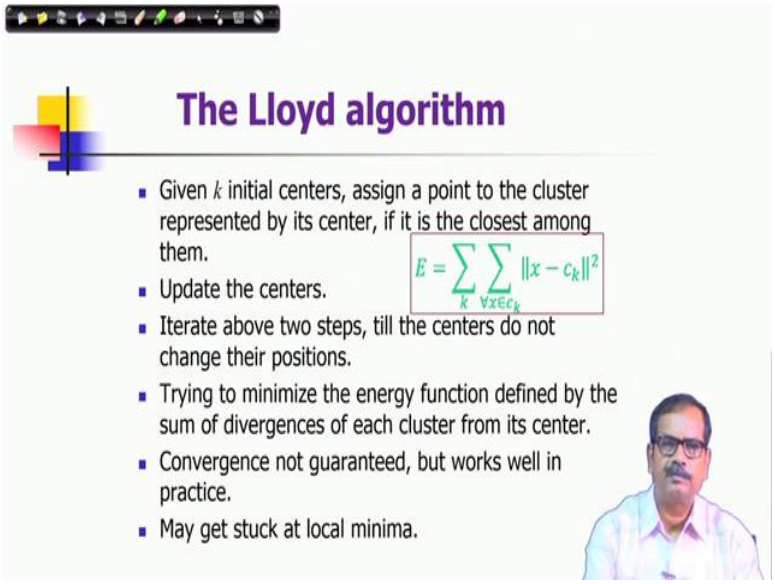
So, the first technique that I will be discussing here it is called K means clustering technique and in this case the problem is that if I give you n data points then you need to compute k partitions though partitions are actually clusters. So, that it they it minimizes the sum of square of distances between a data point and the center of its respective partition or cluster.

So, this is essentially an optimization problem where you can consider this mathematical formulation of the optimization function. So, you see that optimization function which you require to minimize this is actually sum of square of deviations or distances from the mean of a partition which is denoted here by ck and to the points which are included in that partition. And if since there are k partitions you perform this job k times I mean you have summing all those components of k partitions.

$$E = \sum_{k} \sum_{x \varepsilon C_k} ||x - c_k||^2$$

Now, this problem is an n NP complete problem provided K is greater than 1 because if there is if there is K it is just a centre of the cluster compute the centre of the data that itself will give you the solution.

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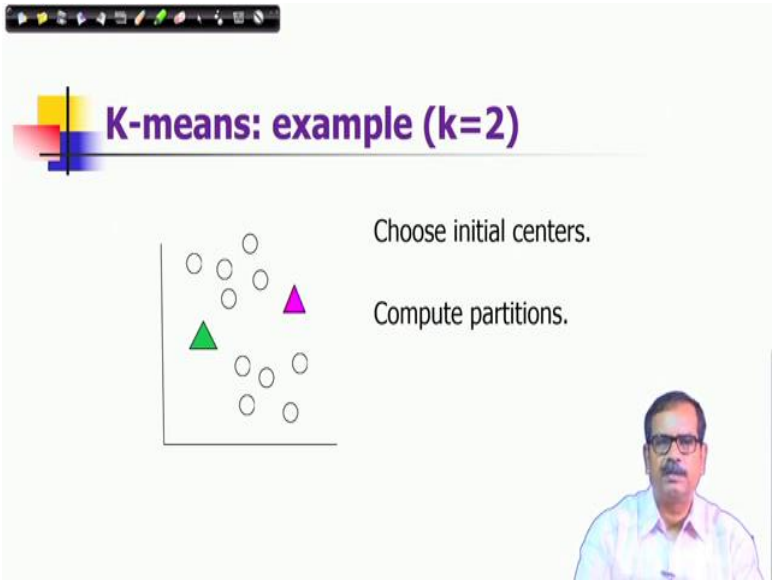


So, the algorithm for K means clustering the its a very famous algorithm and which is known as a Lloyd algorithm by the name of the inventor of this algorithm. So, it is given as k initial centers assigns given k first you consider randomly you can choose k initial centers and then you can based on that you can partition them by assigning the nearest center to a particular points.

So, the cluster of the by assigning that that cluster whose center is nearest to a point. So, it is a closest among them and then we can update the centers once again the partitions so, centers gets updated because once you get the partition and you can compute that center that would be definitely different from what you had earlier. And you can iterate these two steps till the centers do not change their position a very simpler approach, but very effective.

So, what it is trying to do it is trying to minimize the energy function defined by the sum of divergence says each cluster from its center it is the error function what I discussed in the previous slide. So, this is this function what we are trying to minimize. But this method the convergence not guaranteed, but works well in practice and it may get stuck at local minima that is another problem.

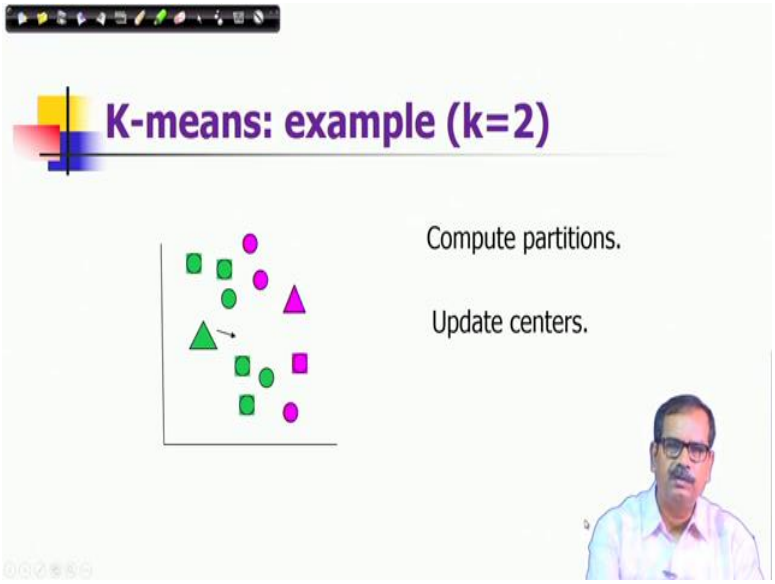
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So, let me show you figuratively how this approach works the essence of this computation. So, we consider once again the distribution of the points in a 2 dimensional space and points are shown by circles their locations are shown because putting center of the circles. So, what we can do that we can choose initial cluster centers.

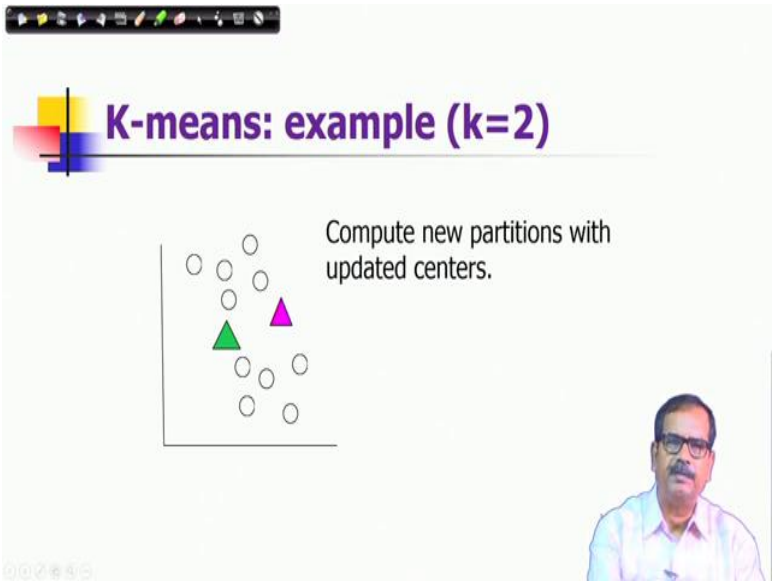
For example, say these are the two initial hypothesize has two cluster centers. Then what you should do? You have to now compute the partitions which means you have to consider the points which are closer to a particular cluster center. So, the corresponding cluster level should be given to those points. So, here the levels are shown by colors for the purpose of visualization.

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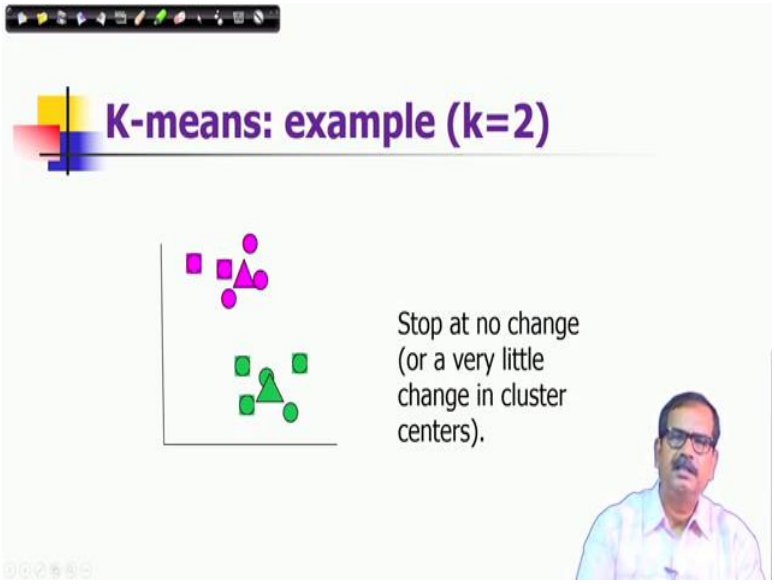
So, I will show the leveling by color. So, you can see these are the set of points which are found to be close to the cluster shown by the color pink and the other one by the color green. So, now we have a two partition once again best on those centers and now you have to you should again update this center. So, you can update this centers which means now these centers they should move inward because of the configuration here. And you get this is the updated positions again you perform the partitioning with these two new positions.

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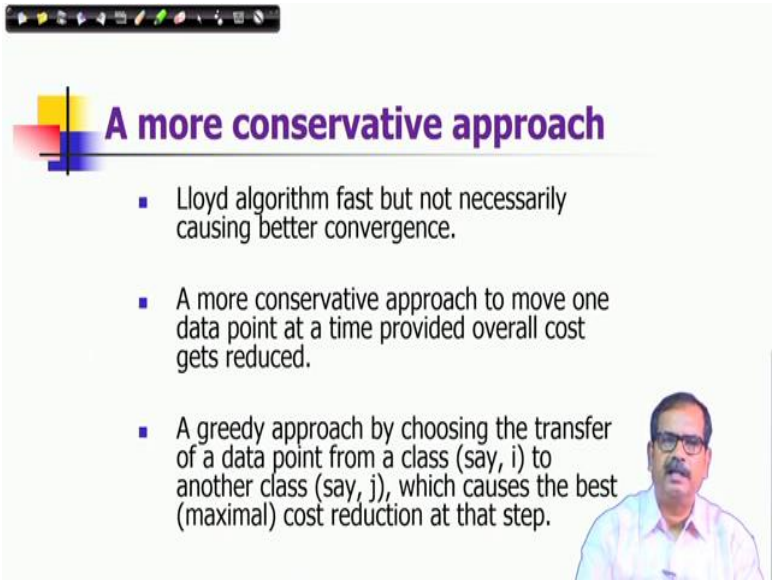
So, compute new partitions with updated centers. So, now, you will find that no these are the set of points which are being clustered which are closer to the corresponding centers and they are shown by their colors. So, you would repeat this operations you update centers and move the cluster centers no update no cluster centers once again you perform the partitions and you will find that you get a new partition; new partitions here and you can you should update the centers and it will be moved further.

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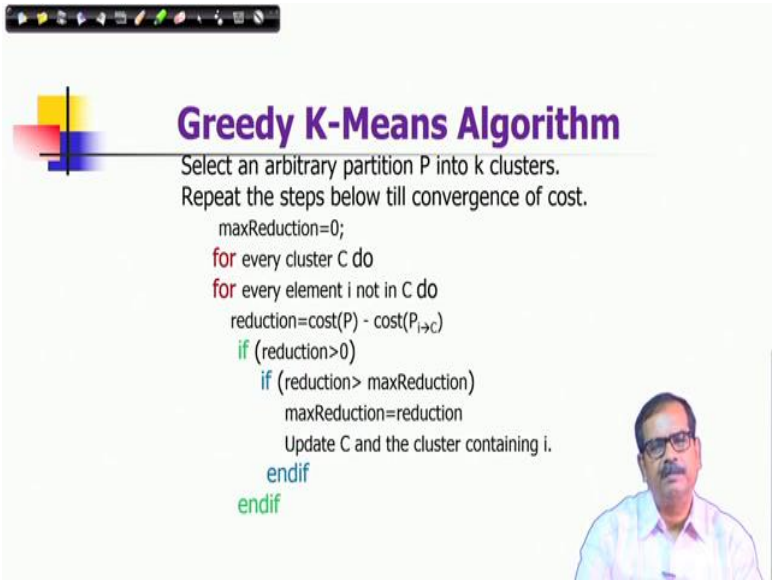
And in this way you will find that after few iterations there is no change or a very little change in cluster centers and then you should stop.

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So, this is how the K means algorithm works. There is about conservative approach of this particular algorithm because Lloyd algorithm is a first is first algorithm, but its not necessarily causing better convergence. So, a more conservative approach is to move one data point at a time provided overall cost gets reduced. So, its a greedy approach and what it does in the principle of this approach is that it chooses the transfer of a data point from a class i to another class j which causes the best cost reduction that gives the maximal cost reduction at that step.

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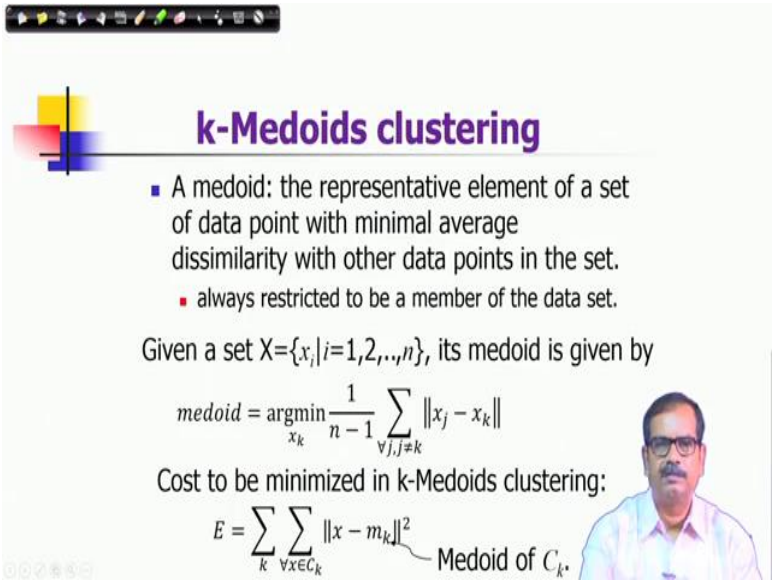
So, we can describe this computations in an algorithmic steps say select an arbitrary partition P into k clusters the like the K means clustering algorithm and you should repeat the steps till convergence of this cost. So, let us consider we keep track that what is the reduction at this step if there is no reduction called some iterations then we should not after some iterations then know we can stop.

So, initially it is initialized with 0 and then for every cluster C and for every element which is not in C you perform these operations. We perform the reduction that is a difference between cost of partitioning of P which was there earlier before the transfer and what is the cost after the transfer; that means, you have recomputed once again the centers and partitions.

Now, this can be done efficiently by simply considering the clusters which are affected by this transfer by recomputing their means. Rest other partitions levels of the rest of the assignment remains same and then you can quickly compute this cost. So, if this is if this there is a reduction and if this reduction is greater than the maximum reduction, then you I mean first thing if there is a reduction then you should move update C and the cluster containing.

So, which means in at every iteration we are considering for all elements what is the reduction of cost and choose that transfer which is a maximum and at that iteration only that transfer is used and go on doing this. So, its a very slow process its not so fast as Lloyds algorithm, but its convergence is better which means as I mentioned that K means algorithm may get stuck at local minima similarly this algorithm also it is not guaranteed that will get a global minima, but it should get a better local minima by this process.

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So, that was about K means clustering; a variation of K means clustering is called k medoids clustering. So, let us understand what is defined by a medoid.

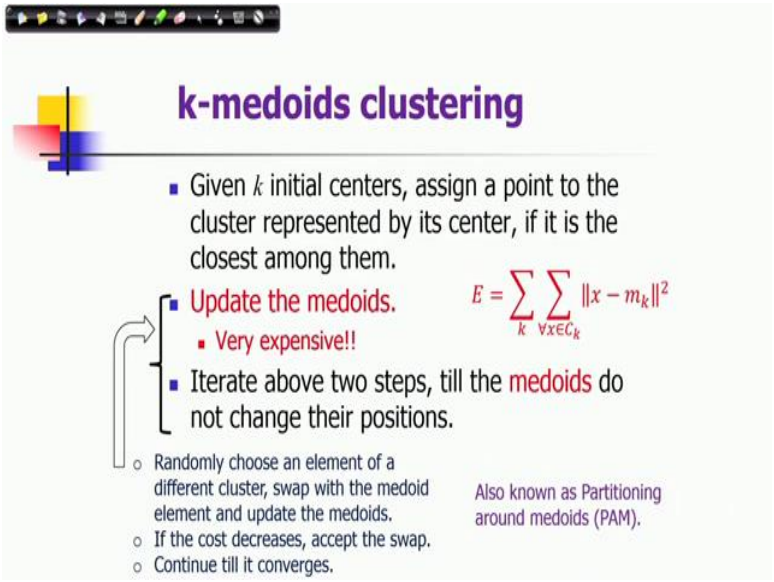
$$medoid = argmin \frac{1}{n-1} \sum_{i,j \varepsilon C_k} ||x_j - x_k||$$

A medoid is the representative element of a set of data point with minimal average dissimilarity with the other data points in the set you con you consider the data points are there all vectors here. So, it is nothing, but the median vector what is defined conventionally in single processing community.

So, this defination mathematically can be represented in this way that given a set X, its medoid is given by this is what you compute the average of the distances from that point to other points and consider that element in that set which has the minimum such distance. So, that is how the medoid is defined.

The advantage here is that it is like median of a set what we get. So, in the median of the set it is one of the element of the set similarly medoid is also one of the member of this data set. So, the cost to minimize in k medoid clustering it is almost structure is same instead of the cluster means now we have replaced means by the cluster medoids in the expression. So, you can see this is almost similar expressions where these are the medoids or clusters k.

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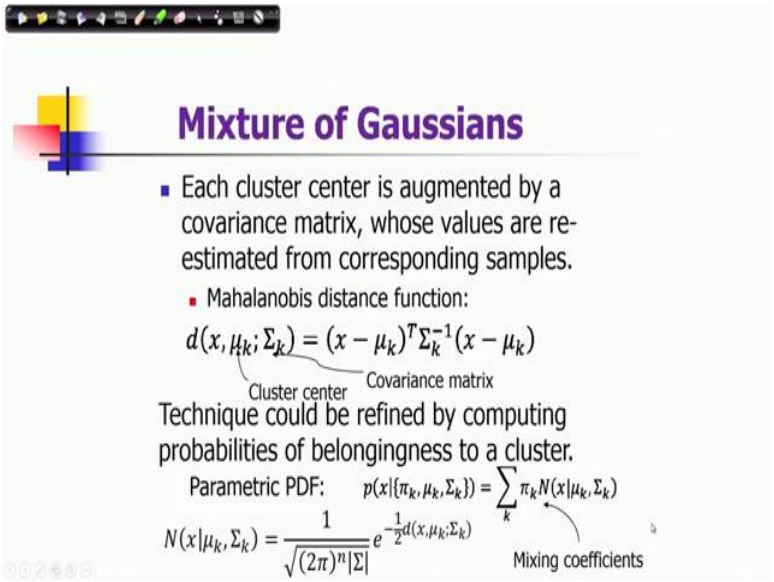


So, k medoids clustering could be almost similar to k means clustering as you can see that this is the same algorithm except the fact that instead of updating cluster centers or means we are updating the medoids. So, iterate above two steps till the medoids do not change their position. So, the corresponding changes in this algorithm they are highlighted by this red color.

But the problem is that this updating of medoid is very computationally expensive it is not so simple like updating centers as we have seen for every cluster we have to compute the corresponding median vector by solving that problem. So, we need to no which we can have some other variations to make this computation faster; one of the approach is that we can consider computing medoid of only two clusters instead of all.

So, randomly choose an element of a different cluster and swap with the medoid element of one of the clusters and update the medoids if the cost decreases, then only you accept the swap and then you can you continue this operations till it converges. So, this algorithm is also known as partitioning around medoids or PAM.

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The last technique of the clustering methods which will be discussing in this particular no subject that is a use of mixture of Gaussians and by analyzing the probability density functions of the data in terms of representing it as a mixture of Gaussians. At the outset we can consider some similarities with the K means clustering technique. Say you consider a cluster center is augmented by co variance matrix, it is not a center is the mean of the data, but let us consider also your using its co variance matrix and know for every at every iterations you are updating corresponding means and co variance matrices.

$$d(x, \mu_k; \sum_k) = (x - \mu_k)^T \sum_k^{-1} (x - \mu_k)$$

And the distance function could be used like a Mahalanobis distance function. So, this is a example this is expression for Mahalanobis distance function as we can see it takes the takes into consideration of the co variance matrix also not only the mean and so this is the cluster center and this is the co variance matrix. So, you can apply like K means algorithm, but of course, this is not the technique of mixture of Gaussians, but these has some similarity. What we can do instead? We can refine it by computing probabilities of belonging into a cluster.

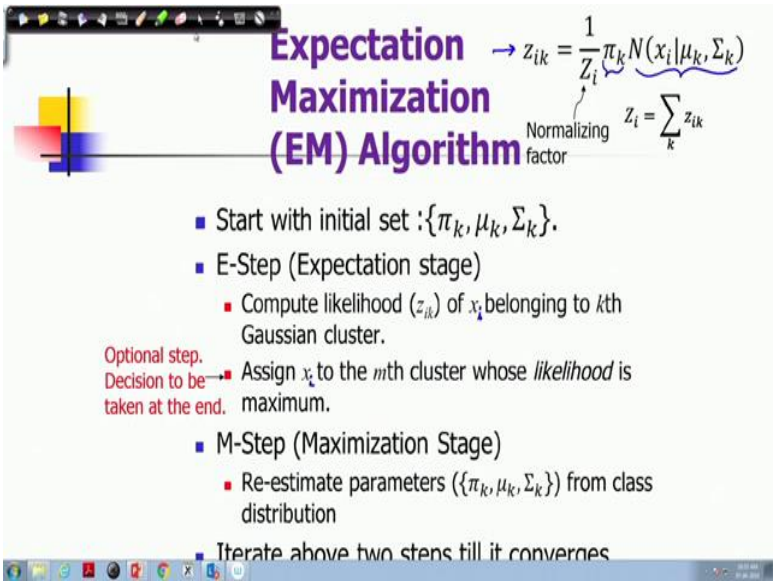
$$N(x|\mu_k, \sum_k) = \frac{1}{\sqrt{(2\pi)^n |\sum|}} e^{-\frac{1}{2} d(x, \mu_k; \sum_k)}$$

So, instead of crisply defining the membership of the of a cluster for every element at every iteration we maintain a probability of belongingness and continue doing this things till no we get a good probability density function which defines the you know which satisfy which describes the corresponding distribution of data consistently describes it.

So, what is this parametric probability density function is described by mixture of Gaussian distribution. So, this is a description here you can see that probability density function of a point data point x is given as a mixture of know normal distributions; so weighted sum of normal distribution.

While weights are given by the parameters pi k and each normal distribution has its own center and co variance matrix; that means, they are denoted by μ_k and $\sum_k$. Just for the completeness I am also providing you the expression called normal distribution in multi dimensional space. And the mixing co efficient are here it they are called this pi k.

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So, there is an algorithm which is known as expectation maximization algorithm and by which you can estimate this probability density functions and get those mixtures get those Gaussian distributions. And each Gaussian distribution component is representing like a cluster here and so elements which have higher probability due to that Gaussian mixture distribution those elements are assigned to those clusters. So, that is a process.

So, what we can do, we can start with an initial set of this parameters any arbitrary parameter you can choose and then the expectation stage what we do we compute the probabilities likelihood of x of the data to a particular know kth Gaussian cluster. So, for each cluster center we compute its likelihood which means know I have to compute simply the probability given that distribution and which has to be multiplied by.

So, this is the likelihood which we need compute and you can see that we are computing the probability of corresponding probability of xi belonging to this distribution this is given by the probability distribution function, it needs to be multiplied by the mixture coefficient pi k and zi is the normalizing coefficients because no you are considering k classes.

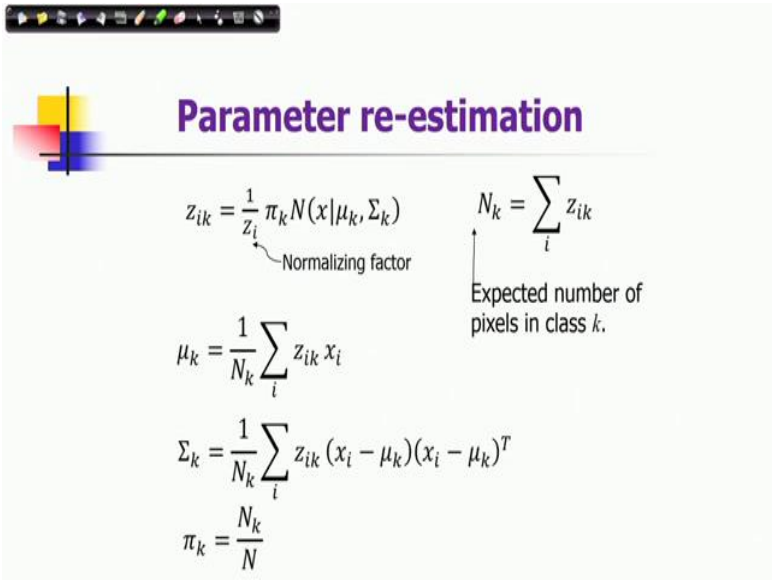
$$z_{ik} = \frac{1}{Z_i}\pi_k N(x_i|\mu_k, \sum_k)$$

So, we are considering the ith pixels and this is how we compute the likelihood. And then we assign the this pixel ith pixel or xi we can assign it to the mth cluster whose likelihood is maximum.

However, now this assignment is an optional step it is it should not be taken at this at every iteration only at the final stage when you computed all the probabilities of likelihoods of every know points then only you assign the cluster which as the maximum likelihood and the maximization. So, this is the expectation stage.

So, you are computing the likelihoods and you from there know you get a redistribution of as redistribution of data. In the maximization stage once again with this likelihood of data you re estimate the parameters of the Gaussian distribution. So, you continue this process. So, this re estimation process let me explain more in an elaborated way.

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So, as we have seen that we can compute the likelihood of the data this is the expression by computing the probability using the probability density function called kth Gaussian distribution in the kth Gaussian component multiplying with the corresponding mixture coefficients and then normalizing with respect to all the classes or components clusters.

$$\mu_k = \frac{1}{N_k} \sum_i Z_{ik} x_i$$

$$\sum_k = \frac{1}{N_k} \sum_i Z_{ik} (x_i - \mu_k)(x_i - \mu_k)^T$$

$$\pi_k = \frac{N_k}{N}$$

So, if I have these likelihood we can consider the strength it is as if it is representing the number of points in a cluster. So, it is not a discrete number as you can see it is sum of probabilities of all pixels whose sum of likelihood of all pixels which may which is not a not an integer, but it is an estimate of the number you mean probability estimate of the number.

So, if the likelihood is 1 and 0, then you get integers integer numbers, but in this case we will get an estimate of that number and this is how it is defined it is. So, it is expected

number of pixels in class k which is which has a fractional component also here in this framework. So, now you compute the mean of the distributions you compute the mean of the cluster k by considering the corresponding say likelihood as the as weights for a particular data points.

So, it is weighted know mean of the data points belonging to that cluster. So, the belongingness is defined by likelihood that is how you get the mean of the cluster in the same way you get the co variance matrix of the cluster. So, this is how the co variance matrix is also computed. So, now, you see that from here you get atleast two sets of parameters of mean and co variances and finally, the mixture coefficients that pi k those are also found as a fraction of total no N here N is the total number of pixels in the image.

So, N k by N now total number of elements in the data points its not only image its a data points. So, N k by N will give you the corresponding know probability of a or prior probability of a class k. So, it is acting like a mixture coefficients. So, you iterate this process and through iteration you know you can converge you can after convergence as I mentioned then you assign the highest assign the corresponding cluster which is represented by a corresponding Gaussian component to a pixel which as a highest likelihood to that component.

So, this is how the Gaussian mixture of Gaussian method could be used for clustering. So, let me stop here and will continue our discussion on classification in the next lectures.

Thank you very much for your attention.

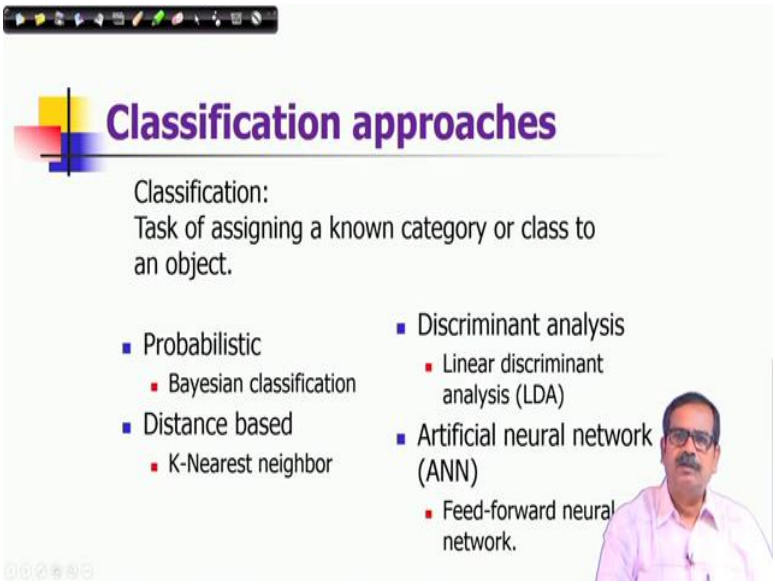
Keywords: Clustering, K-means clustering, K-medoids, expectation maximization.

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Lecture - 47
Clustering and Classification Part – II

We continue our discussion on Clustering and Classification of data. And, in the last lecture we discussed various algorithms for clustering of data points and in this lecture I will start discussing on different classification approaches.

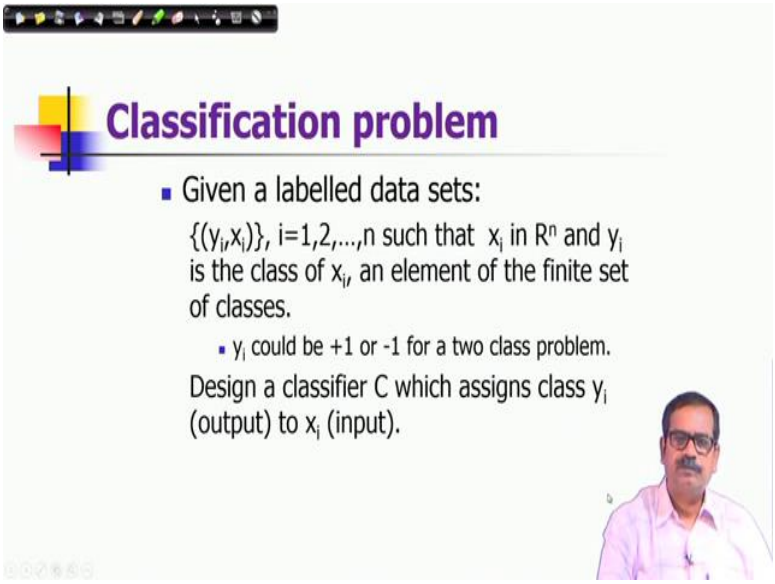
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Now, classification is a task of assigning a known category or class to an object. There are various approaches out of them few are shown here. It could be probabilistic approach, distance based, discriminant analysis best approach, you can use also artificial neutral network for classifying data points.

In this lecture or in this particular topic under this course I will be considering a few examples of each of this approaches. They are for the probabilistic approach will be considering Bayesian classification techniques, particularly will discuss about Naive based Bayesian classifier. For distance based approach will consider K nearest neighbour classifier and discriminant analysis we will discuss about linear discriminant analysis. And, then in artificial neutral network it is a feed forward neutral network that we will be discussing.

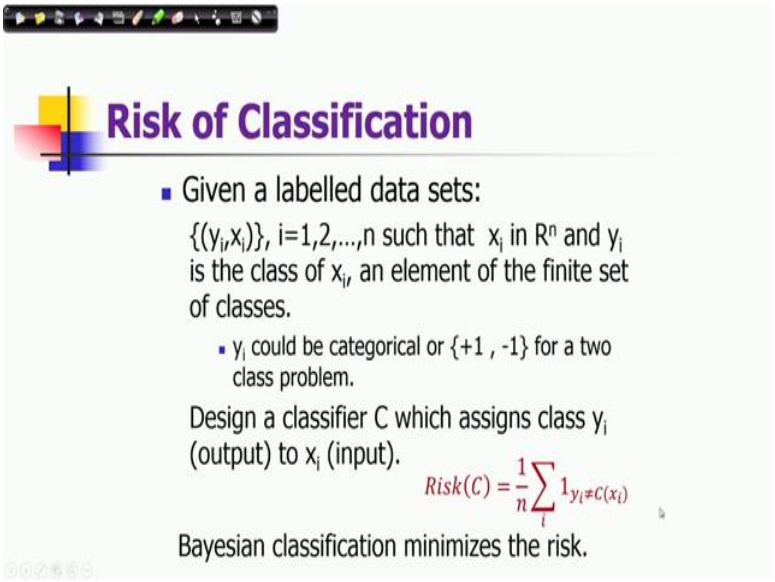
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So, let us start with the problem definition of a classification task. Here you have a labeled data sets given in the form of (yi, xi). Note here xi is the data point, this is a n dimensional vectors as for every object in an abstract way they represented as a feature vector in a n dimensional real space. So, xi is such a vector or such a point in a real dimensional in a n dimensional real space. And, yi is its class, it could be any categorical data. It could be some names of class or there could be numerical representation of those classes.

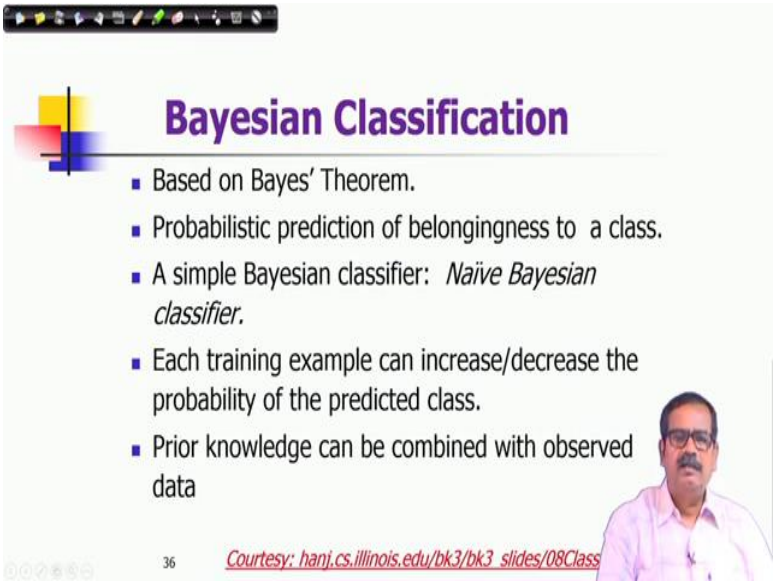
So, the numeric values distinctly identify these classes; however so, we consider of course, a finite set of classes. So, it is an element of a finite data set classes and we can design a classifier C; I mean that is the problem statement that you need to design a classifier C which assigns class yi to xi. So, which should be supported by which should supports its data set ah. So, as I mentioned that for a two class problem say yi could be denoted as + 1 or - 1.

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There is a risk associated with classification, particularly misclassification as you can see that risk is defined as the number of misclassified data. And, it is a fraction of number of misclassified data that is how the risk is defined and incidentally Bayesian classification which we are going to discuss next it minimizes this risk.

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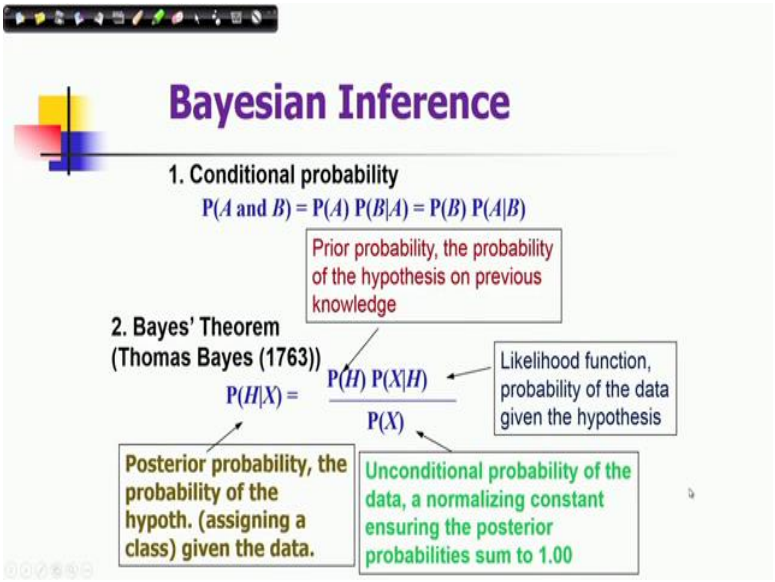


So, let us consider this particular, thing Bayesian classification. So, it is the name comes from a theoretical framework which has been proposed by Bayes and there is a Bayes theorem. We will discuss about this theorem from which this classification rules are set

and classifier set designed. It is a probabilistic prediction of belongingness to a class and a simple example Bayesian classifier that is an example of a simple Bayesian classifier is a Naive Bayesian classifier which will be considering in particular in this lecture.

Each training example in this case it can increase or decrease the probability of the predicted class. So, incrementally you can learn the classifier that is the advantage of this classification technique and prior knowledge can be combined with the observed data.

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So, let us consider the Bayes theorem and the inferacing that could be carried out using by applying this theorem. So, first we know the definition of a conditional probability it is from the joint probability of A and B, we can say that either no B occurs given A or A occurs given B. So, probability of A and B could be expressed as probability of a multiplied by probability of B given A or probability of B into probability of A given B. So, they are all equivalent, they should give you the same probability value.

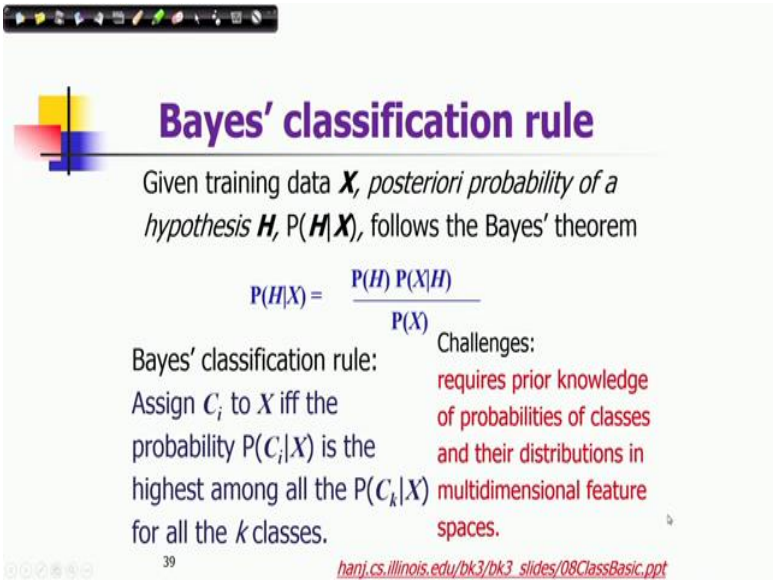
Now, Bayes theorem is derived from this observation that you can see that it has been shown that probability of a hypothesis H given a data X can be expressed as the ratio of this two quantity.

$$P(H|X) = \frac{P(X)P(X|H)}{P(X)}$$

So, probability of H is the prior probability of the hypothesis based on our previous knowledge whereas, probability of X given H we call it as likelihood function. So, in the particular hypothesis if it is true what is the likelihood that X will occur under this hypothesis and probability of X is the unconditional probability of data.

So, it is a normalizing constant in this particular case which is ensuring posterior probability sum should be 1; that means, if X occurs there could be all different hypothesis which may generate this X. So, which are those hypothesis from which to which this occurring of this X belongs to are conditional. And, now if I sum all those possibilities that that should be equal to probability of X. So, posterior probability, probability of the hypothesis assigning the given a data.

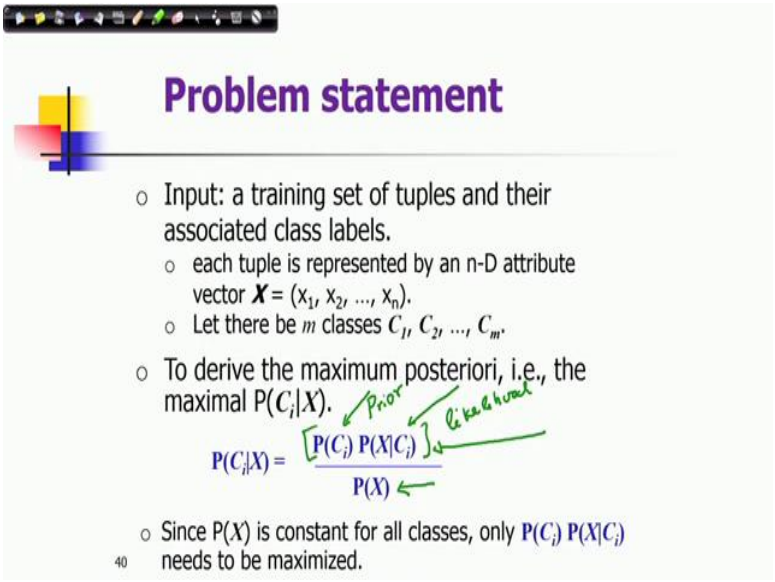
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So, given training data X posterior probability of hypothesis H, it is it can be computed by following this Bayes theorem. We will see that no it is it has significance, it is relevance in terms of designing a classifier; because here ah our objective is to maximize the probability of a class given the data and class is representing an hypothesis.

So, you need to compute this posterior probability here. So, that is what is the Bayesian classification rule or Bayes classification rule. It says that assigns Ci to X if and only if $P(C_i|X)$is the highest among all the other classes all, all other probabilities called all other classes. So, there are challenges of course, in this computation, it requires prior knowledge of probabilities of classes and their distributions in multidimensional feature space.

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So, just to once again summarize the problem statement in a more concrete form, we have an input to the classifier. It is a training set of tuples and their associated class levels and each tuple is represented by an n dimensional attribute vector X. We have shown name of the attributes or the representation has been shown here. And, they are the data points what I am referring in an n dimensional space and let us consider there are m classes. So, as we discussed in the previous slide itself our objective is to derive the maximum posterior probabilities.

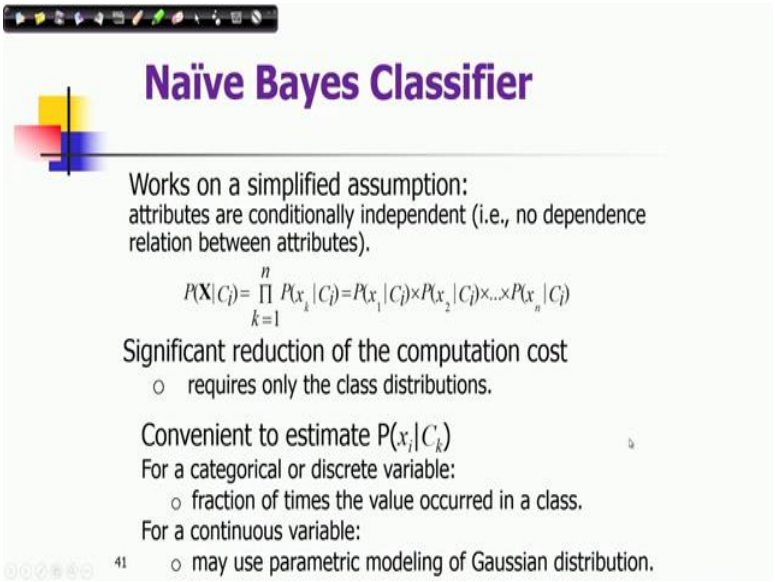
So, for a class the class which keeps the maximum posterior probability given the data so, that class should be no assigned to data. So, we can compute this posterior probability using Bayes theorem of conditional is in the conditional probability rule that you can compute the prior of class Ci. So, this is the prior probability. Now, this computation can be you can get it from the data point itself; some applied information can be obtained from other knowledge from a different source. But, given the data distributions among different classes we can compute that how many times for a data a data occurred or in a class, how many time class instances are occurring in the classes.

So, that fraction can estimate this prior and by observing the data distribution within the class itself, we can compute this probability density function or probability of X given Ci also. So, this called likelihood. So, this term is likelihood and you can see this numerator probability of X actually need not compute. Because, while comparing the values of

probability of C i given X or posterior probabilities; this numerator is constant given the data X this numerator is constant for every classes. So, it is sufficient sorry this denominator is constant for every classes the probability of X.

So, it is sufficient if I compute just the numerator for it or computer these two quantities that would give me the probability no that would give me the basis of comparison. I am not in absolute since computing posterior probability, but in relatively I can compare them by computing those factors. So, this is what that is the summary only product of probability of C i and probability X given C i needs to be maximized that is the problem statement.

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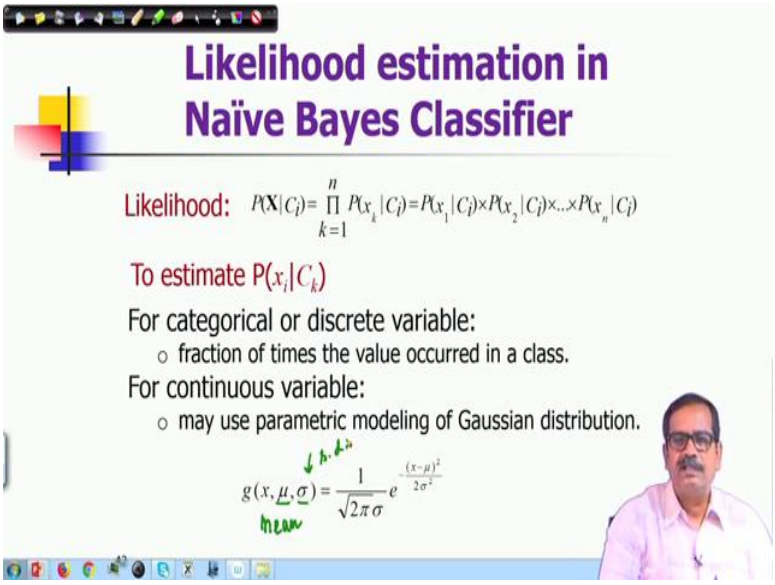
So, now will be discussing about Naive Bayes classifier. It works on a simplified assumption, where attributes are conditionally independent which means that there is no dependence relationship between attributes. And, in that if it is true then you can write the likelihood of X given a class C i as the product of all likelihood probabilities of individual attributes because, attributes are independent. So, that is the advantage in simply you can compute probability. So, all likelihood functions can be defined in one dimension instead of multi dimensional representation of probability density function that is the simplicity of Naive Bayes classifier.

Your handling of data becomes simpler or handling of multi dimensional data becomes simpler for modeling the probability density function. So, there is a significant reduction of the computational cost because, of this and it requires only class distribution that too in

single dimensional feature space. So, it is convenient to estimate every attribute probability of every attribute given a particular class. And for example, for a categorical or discrete variables you can count how many times that values of attributes have occurred given all the occurrences of all other values of those attributes.

And, then you can find out the fraction of its occurrences over all occurrences and that can even estimate of this probability or if it is a continuous variable, then you can model it by any probability density function. Any kind of parametric modelling of density function you can do for example, Gaussian distribution you can and from there by estimating the parameters of Gaussian distribution you can compute this probability. So, you know that in Gaussian distribution there are only two parameters for a single dimensional, one dimensional feature space you have only its mean and standard deviation there.

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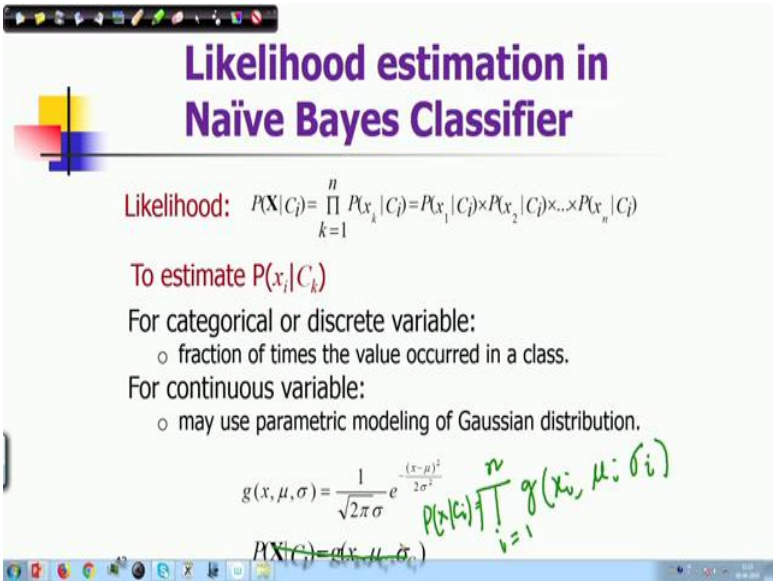
So, the likelihood is expressed in this form

$$P(X|C_i) = \prod_{k=1}^{n} P(x_i|C_i) = P(x_1|C_1) * P(x_2|C_2) *..* P(P(x_n|C_n)$$

So, there is the class condition probability of in attribute xi and as just I mentioned that no these are the methods by which you can do it. So, here this is just elaborating that how parametric modelling of Gaussian distribution could be done in the one dimensional feature space.

So, this is the distribution this is the probability density function of a variable X one dimensional random variable X which follows a Gaussian distribution and you can see that there are two parameters here. One parameter is mu which is called expectation of this distribution or you can in a simple layman's term we can call it mean and this is sigma which is actually the standard deviation of this distribution. So, so given the data you can always estimate these values from mean and standard deviations, these parameters could be estimated.

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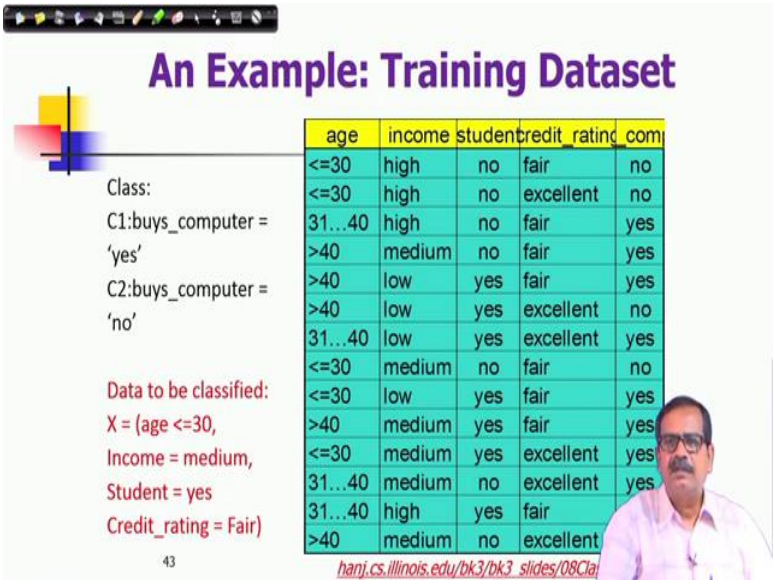


And so, this is what you are considering for it is the product of all this things which has been in this case actually I should write it in this way this is

$$P(X|C_i) = \prod_{i=1}^{n} g(x_i, \mu; \sigma_i)$$

So, this is what is the probability of this variable n dimensional feature given the class Ci this is this is how it should be written. So, there is some problem with this particular know expression.

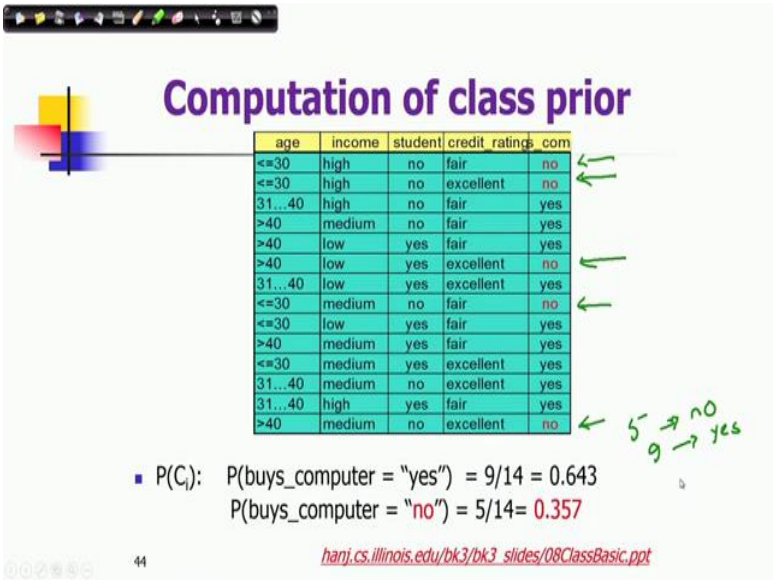
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So, let me consider an example to explain to elaborate this computation, you can see this example here and here there is a data set which has been shown in this particular example. In this data set you have the statistics of the computer buyers I mean whether a person buys a computer or not, you have that kind of statistics. The person's age has been shown here, then the income and whether the person is a student and what is the credit rating and with that background whether that person particularly has bought computer or not. So, there are few distinct values for each variable.

So, these are the attributes age, income, student, credit rating and computer buyers. So, these are the attribute and computer buying is the class, you can see that there are two classes. One buys computer where the value would be yes and other class buys computer the value is and here the problem is that we need a data and a person has this kind of attributes or data associated with the person. What is the probability that the person buys a computer or whether not only probability actually in which class category, we can put this person that it is whether this person should be in the class of buys computer or not. So, this is what is this particular classification problem.

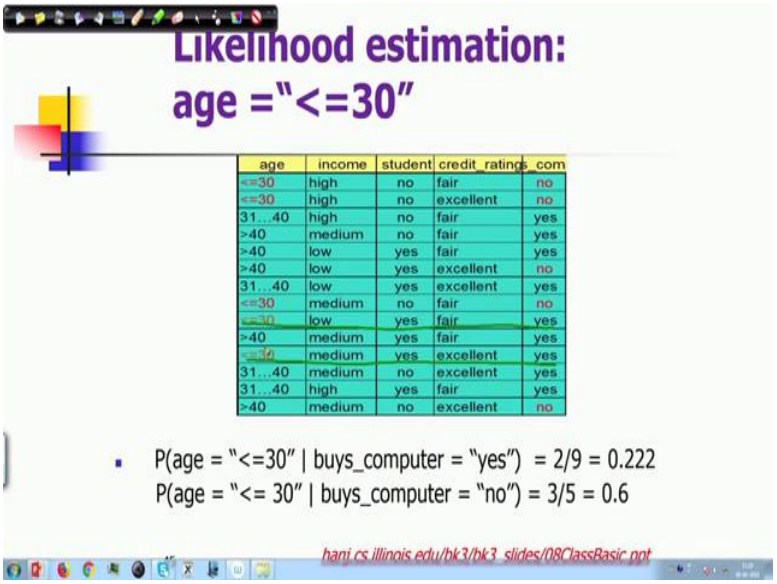
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So, first we need to compute the class prior in this case and you can see that here I have shown with color that how many instances of computer a person buying a computer is there and how many instances a person not buying a computer is there. So, these are the instances which in which cases a person did not buy a computer. So, there are 5 to 4, there are 5 such instances and if I count this number of record these are not actually 14 which means 9 instances there for yes.

So, that is why the probability of prior probability of class of buys computer equally yes would be 9 by 14, that is the fraction that is how this probability is considered and similarly the probability for the other class would be 5 by 14 ok. So, these are the 2 prior probabilities that we can we have estimated from this data, then will compute the likelihood of each attribute value for each class.

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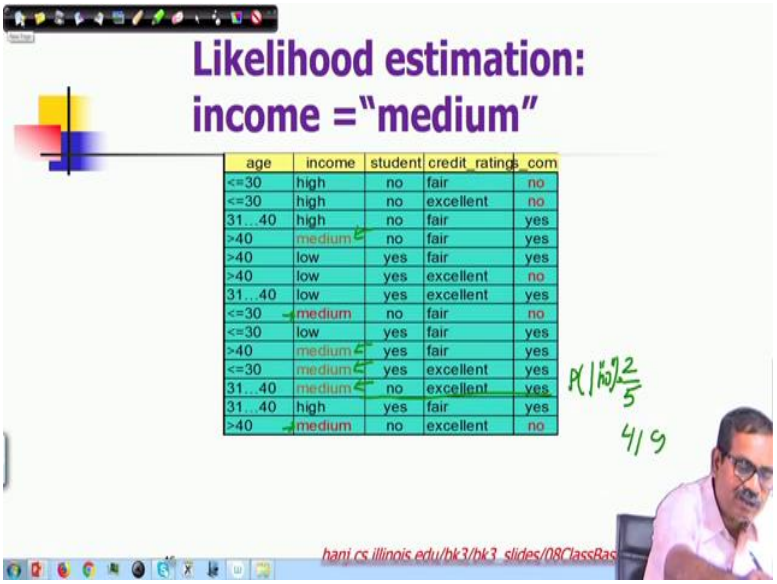


So, now will consider the likelihood of age given the class of buying computer and also not buying computer. So, there what we need to consider here that for each class we have to see that distribution of attribute values; that means, how many times what is the fraction of times age less equals 30 occurs out of all possible values of attribute of age.

So, if I consider say no will see that this is one occurrences of this fraction, this is also another occurrences of this fraction, this is not for no this is for yes. So, there are 4 times this age less than less equals 30 occurs and actual occurrences of no is 5. So, the class likelihood probability of age equals 30 is quite high here, it should be 4 by 5 whereas, the other cases when the class of years. There we can see that there are so, many instances 1 2 3; that means, out of 9 cases only 1 instance of buying a so, this is called yes is there out of 9. So, which means it is likelihood probability is 1 by 9.

So, that is how we can compute these probabilities so, this is yes and this is yes. So, that I 4 is actually 3, it is not 1 is reduced. So, yes has 2 so, 2 by 9 instead of 1 by 9 because 1 9 is to I consider that was no in the class one no. So, it is 2 by 9 that correction should be there for this data 2 by 9 for yes and 3 by 5 for no. So, the probability is 0.222 likelihood probability given the class yes and 0.6 given the class no. So, let us proceed again for the other cases.

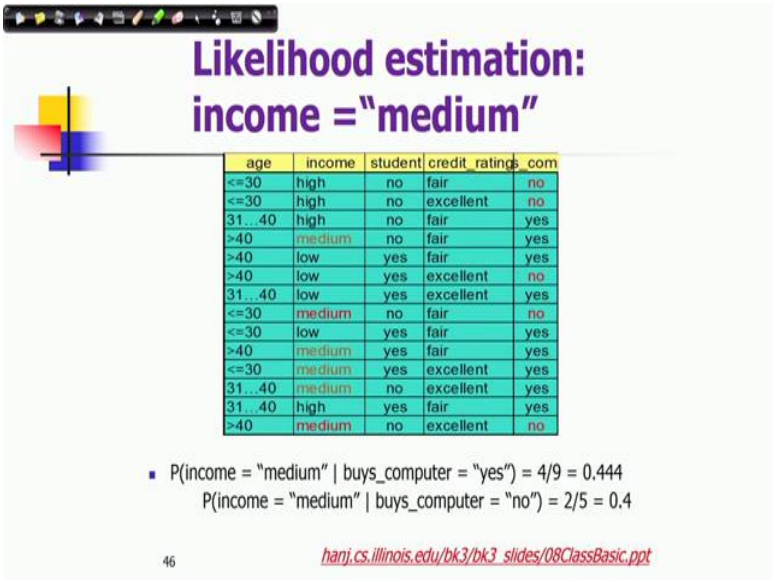
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So, now we are considering the income equals medium, once again will see how many times medium has occurred. So, 1 2 3 here actually I have given colors. So, from the color we can find out for no medium has occurred twice for no and we know that for the class no; that means, the persons those] they are not buying a computer there are 5 instances.

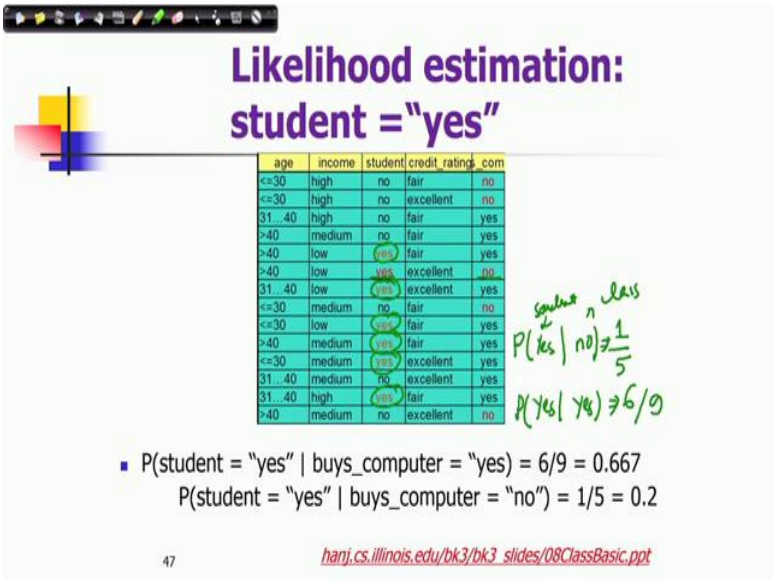
So, the likelihood probability should be 2 by 5 that is the probability of this attribute given the class no, in short if I write and for yes again it has occurred so, this is yes; so, 2 3 4 times. So, out of 9 no instances of yes 4 instances of were of income medium so, it should be 4 by 9. So, that is no value, let us check once again whether I made mistake in the case also let us see.

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Yes 4 by 9 and 2 by 5; so, these are the values, these are the likelihood values.

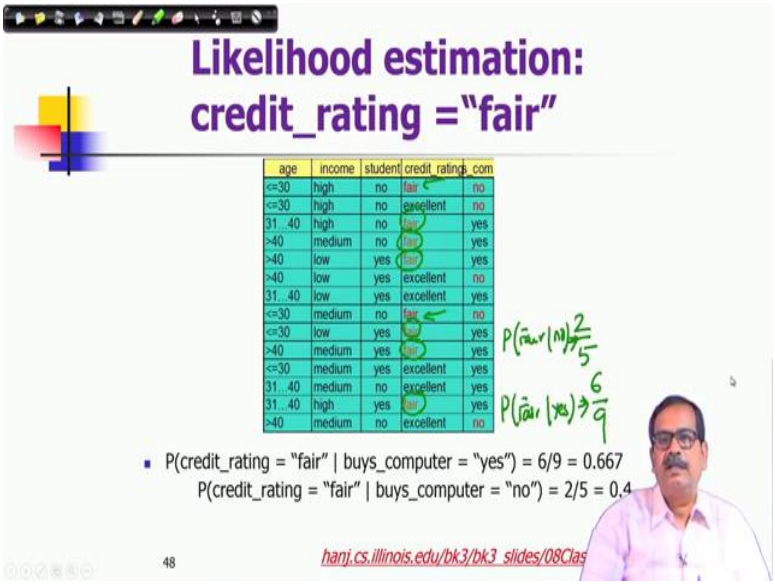
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Will proceed further; so, this time it is student, student equals yes. So, which means the person is a student or whether the person is not a student here. So, here we observe that for one instances in the class no the it is the student and which means that likelihood probability given the class no means not buying computer for a person is student is 1 by 5. So, this is probability yes let me write it in short, this yes means this is for student and this is for class.

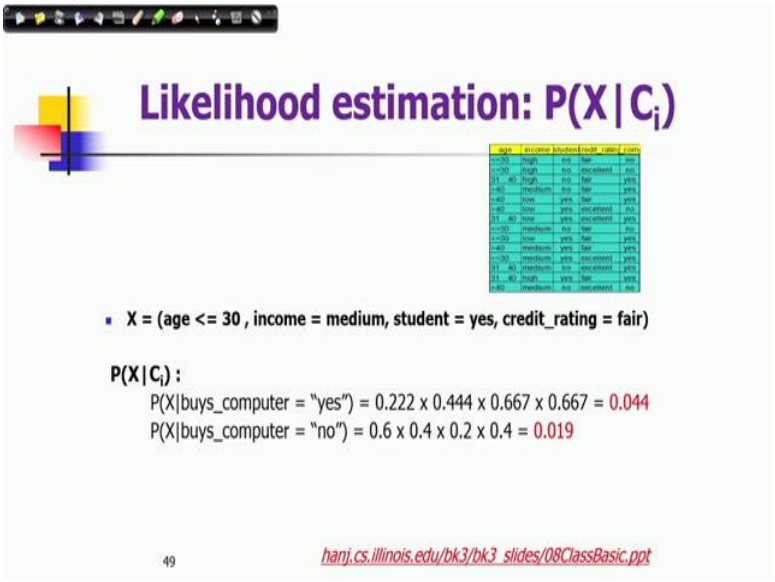
So, this is 1 by 5 and the other cases we have shown that how many times student has occurred; that means, persons who have bought computer how many of them are student or there are 9 person who have brought computers in this data, out of them this is one, this is another, this is another, this is another quite a few. So, this is 6 6 by 9. So, probability a student yes given the class yes so, 6 by 9. So, it is let us see once again yeah. So, this is how the likelihood of student equals yes that attribute value given the classes could be computed.

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So, the next attribute will be considering credit rating where the given the data we have credit rating equals fair. So, in this case we have these are the instances when persons are not buying computer, they have these attribute value of fair credit rating which means that you will have 2 by 5 when it is fair given the class no and for yes you have 3 4 5 6 so 6. So, that is 6 by 9 that is a credit rating here given person's buys a computer. So, these are the values that we can compute estimate likelihood estimates right. So, you have 6 by 9 by 2 by 5.

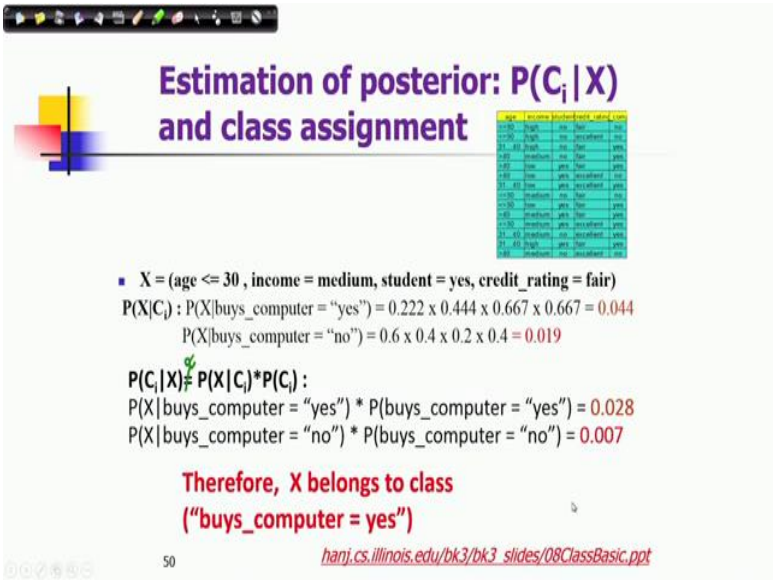
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So, we have computed no all the likelihood values of age less equals 30, income equals medium, student equals yes and credit rating equals fair. So, the likelihood of X given any class any particular class C i for example, the class could be person buying computer means it is the product of likelihood of all those given classes. So, we can compute in this way finally, likelihood of this data given this two classes can be computed.

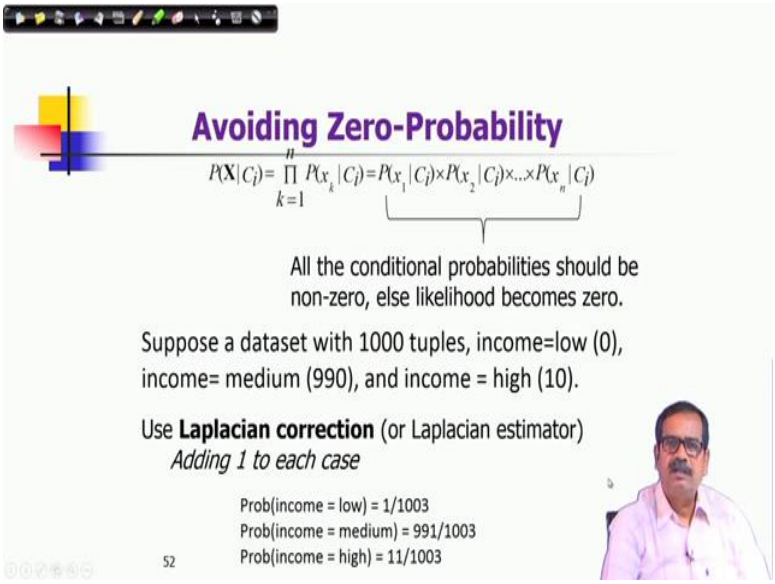
We can find out that the it the value is 0.044, when the class is yes that is buying a computer and value is 0.019 when class is you know buys the computer is no. And then finally, the posterior has to be computed which means each likelihood has to be computed no has to be multiplied with their prior. And then you get the posterior I mean proportional value of course, we are not dividing with probability of X; it is not the absolute posterior value, but you can get the proportional posterior value.

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So, this is how we are getting. So, in this case it is not exactly this, it is proportional in particular this constant when X is given it is a proportional value. So, the posterior of this I should say once again this is yeah, this is the product of this two that is 0.028. So, this value is 0.028 and the other value is 0.007. So, naturally 0.028 is the know greater value and this is out of these two classes this is maximum. So, which means that I can assign this attribute this variable to a class of computer buying class. So, that is the inferencing so, therefore, X belongs to class of buys computer equals yes.

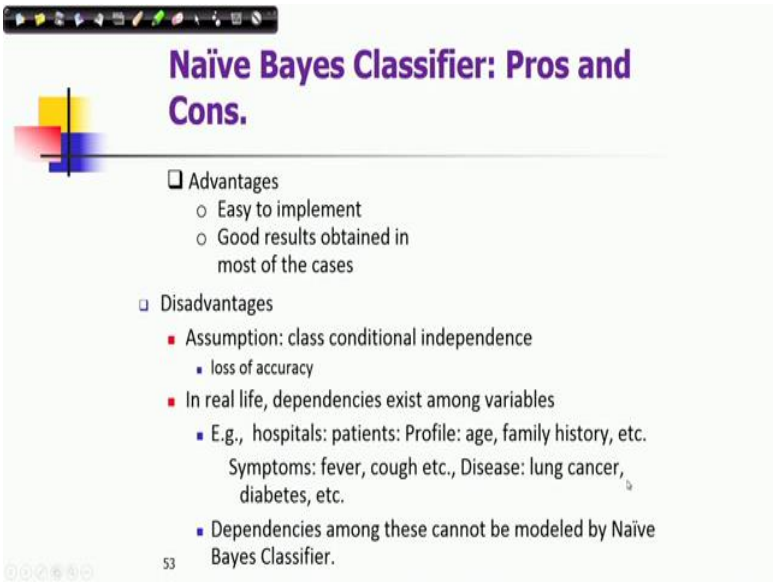
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So, this is the example of learning. So, one of the problem of the computations of this likelihood is that if there is any zero probability; suppose there is no instance at all. It may happen in your data point for a particular class then the likelihood becomes zero. So, or very small of like that is one of the problem. So, to avoid this we need to do certain because, you know it is a observation and observation are noisy. So, we do not except that it would be absolutely zero for every class the likelihood.

So, but it is low, it indicate it should be low ah. So, in that case so, this is one example that now data set with 1000 tuple, income equal to low 0, income equal to 990 and income equal to high 10. So, we use a correction which is called Laplacian correction or Laplacian estimator by adding one to each case. So, instead of 0 we consider the value is 1. So, you will get a very low probability and we avoid this problem of zero probability in computing likelihood.

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So, what are the pros and cons of Naive Bayes classifier? It is the advantage is that it is easy to implement, it good results obtained in most of the cases. There are disadvantages like first it is the assumption is it that class conditional independence which may not be true for a realistic data and if it is not true, then there will lose also accuracy. So, that is what in real life dependencies exist among those attributes or variables.

For examples: hospitals, patients, profile, age, family, history etcetera; symptoms: fever, cough etcetera, disease: lung cancer, diabetes. So, many of the information are related and

this cannot be modeled by Naive Bayes classifier. So, let me take a break here, we will continue this discussion on different approaches of classifications in in the lectures so.

Thank you very much for your listening.

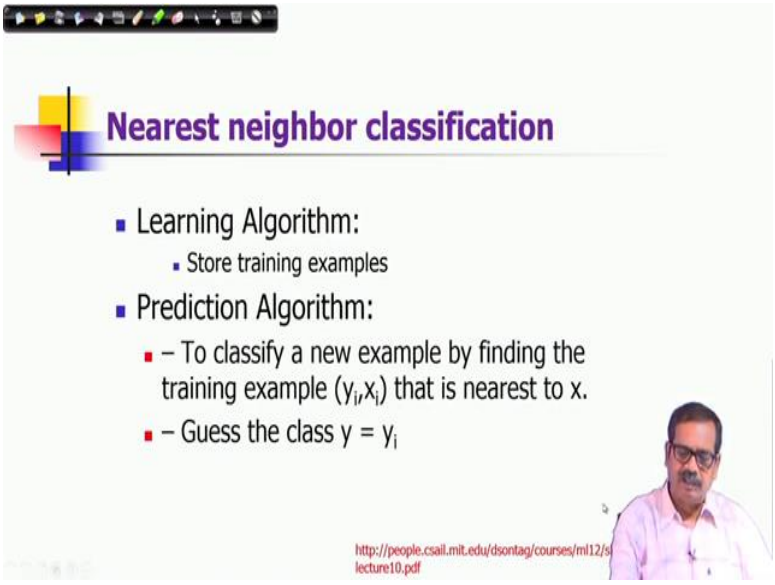
Keywords: Classification, Bayesian inference, likelihood estimation, clustering

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Lecture - 48
Clustering and Classification Part - III

We continue our discussion on Classification of data points. And in the last lecture we discussed a classifier called naive bases Bayesian classifier.

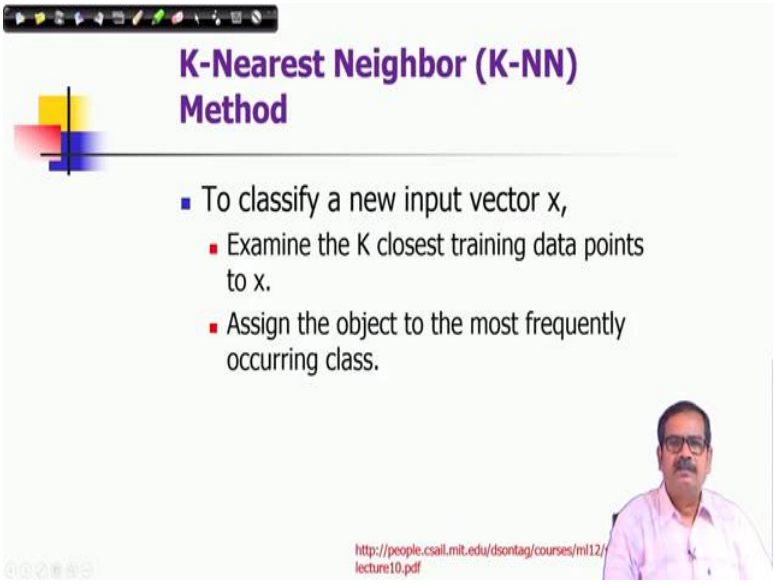
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In this lecture will be discussing about another approach of nearest neighbor classification scheme. And, here the classification approaches quite simple as we can see here. In particular learning algorithm is very simple what you need to do here. You need to simplest towards the training data. In compare to the Bayesian classification you have seen that you need to process the data and perform parametric modeling to get the probability distributions.

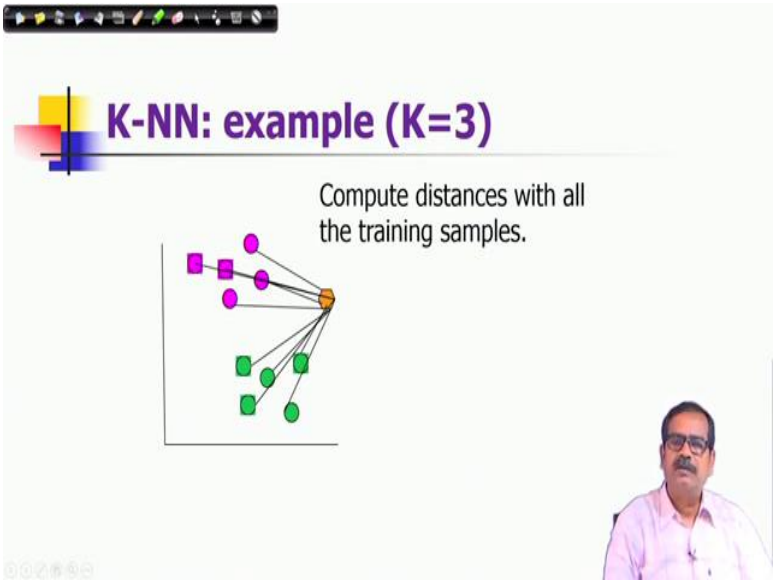
So, but in nearest neighbor classification simply, you store training examples. And then prediction algorithm with goes like this, if you want to classify example by finding the training example that is nearest to x. So, which is one is the nearest, you assign the class to that sample.

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So, one particular method, in nearest neighbor classification is known as K nearest neighbor method. So, instead of observing a single nearest neighbor we observe a set of close neighbors. And, the K most nearest neighbors are called K nearest neighbors. So, in this case to classify a new input vector x; we examine the K closest training data points to x. And then assign the object to the most frequently occurring to us.

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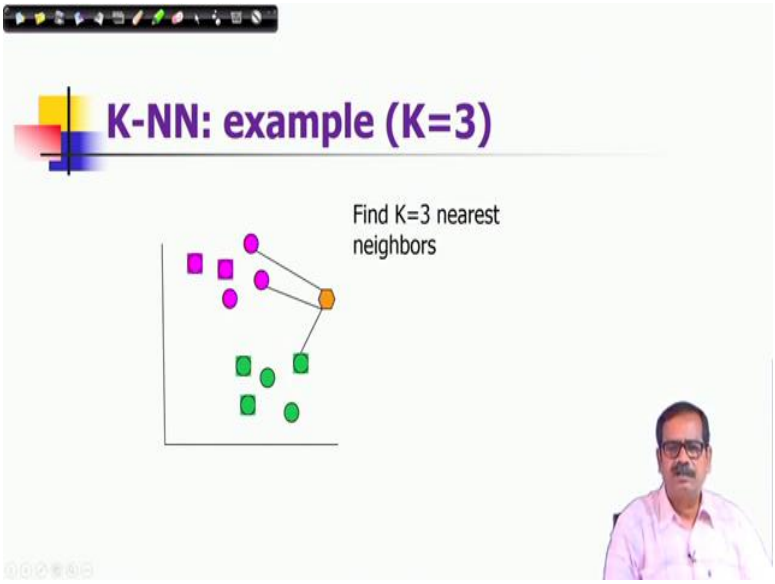


So, here the approach is very simple. Let me illustrate with respect to using this diagram. You can see there are two classes and there are class levels shown by these two colors pink

and green. And suppose you have a query data point it is a 2 dimensional no feature space enquired quarried data point has be shown here.

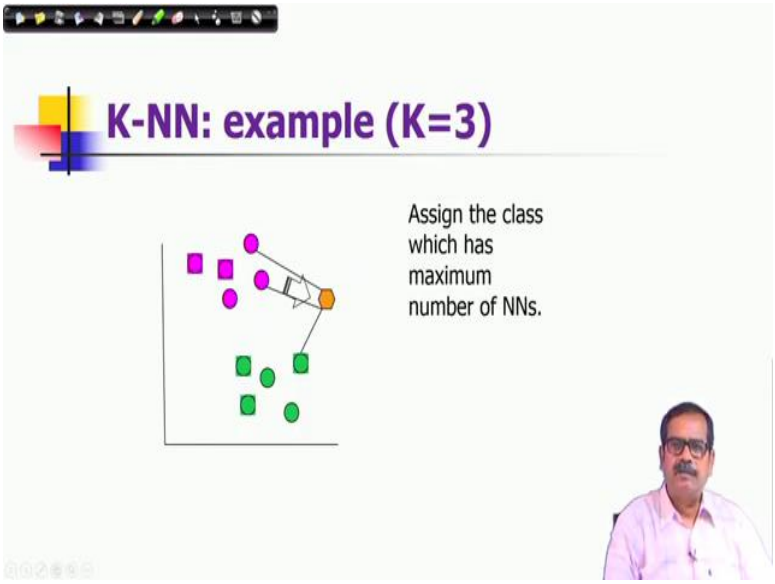
And, then used for computing the nearest neighbors you have to compute distances with all the training samples nearest K close neighbor K most close neighbor. So, you compute distances with all the training sample. And then you can you should solve them and find out three most closest nearest neighbor.

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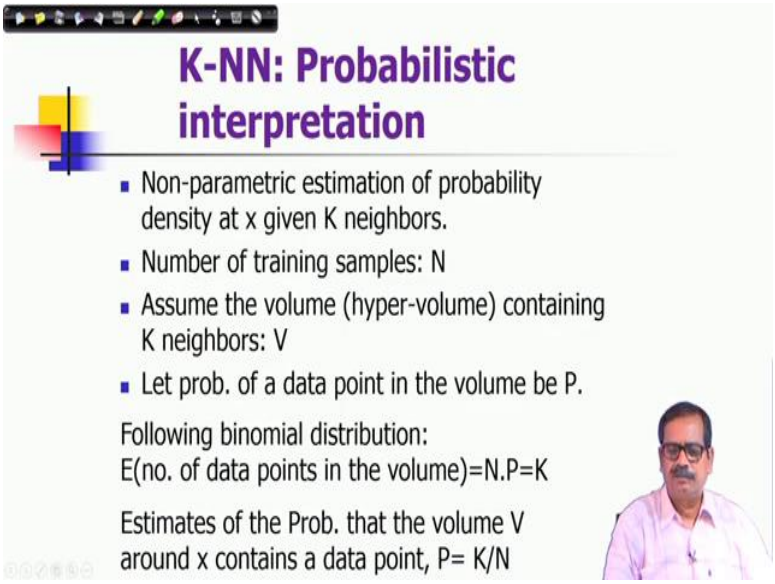
So, you will find that these are the 3 nearest neighbor. So, that is the technical term that is used there.

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And so, and according to the rule what I mention. Assign the class which has maximum number of nearest neighbors of NNs; and we can see that know there are two members from the pink class and one member from the green class. So, we should assign this pink class belong denoting the pink color to that particular query point. So, this is the approach and which is very simple to implement and to understand.

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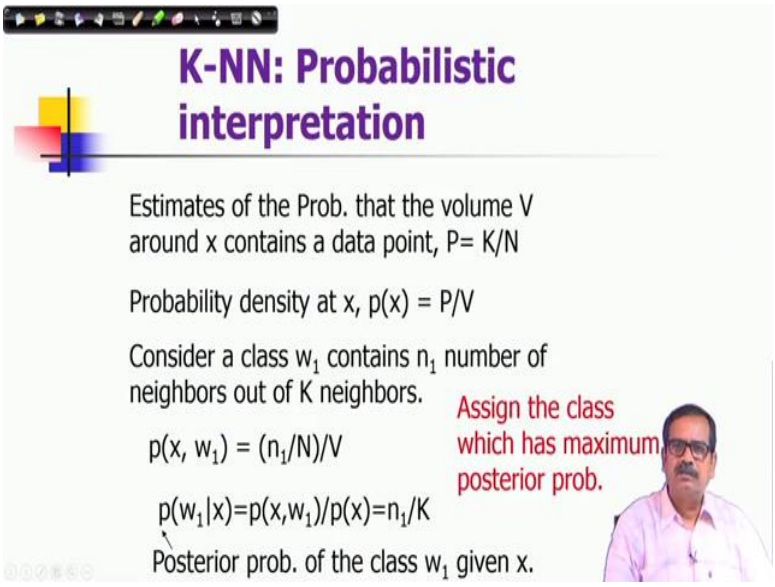


But there are interesting know observation. So, it has a sound theoretical framework though approaches very simple. We can show that actually it is a good relation with the

Bayesian classification rules. So, it is a non-parametric estimation of probability density at x given K neighbors, if I do if I perform these estimations will be doing that. So, suppose there are a number of training samples N. And we assume the volume and in a N dimensional space, we call it hyper value containing K neighbors is V. And let us considered the probability of data point in then in the volume be P. So, following binomial distribution; that means, know how many data could occur.

If it is different kind of its could be modeled, in this fashion that we are sampling the point and with probability P it can occur in the volume V and, if you doing N times this sampling it follows binomial distribution binomial distribution. And, the number of data points in the volume expected number could be N into P and that should be equal to K. So, the estimation of P, is very simple you just get the ratio of K by N or it is the fraction of times it has occurred out of N trails.

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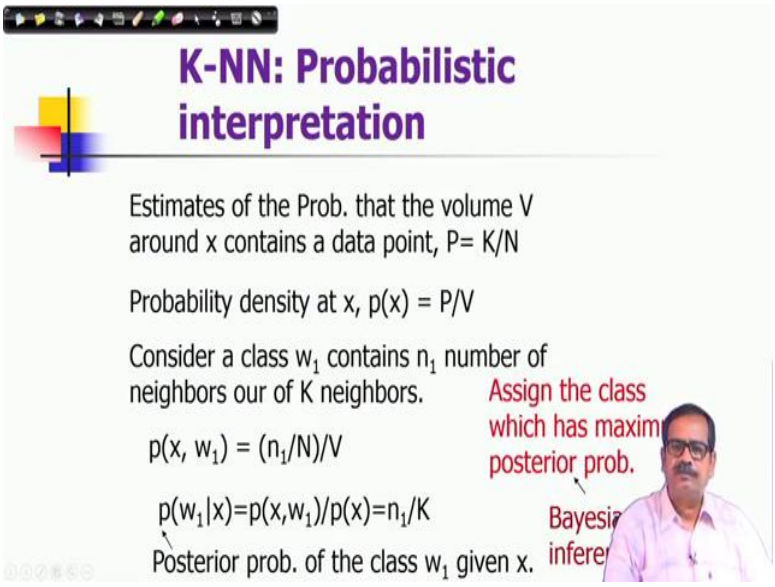
So, this is the is probability that you can estimate here. And then the probability density at x; if I considerates continuous know is space. So, this probability is divided by the volume P that is the know number of data points possible data points here. So, consider a class w 1, which contains n 1 number of neighbors out of K neighbors.

And then the joint probability of x and w 1 will be given by that; that means, first joint probability of interest n 1 has n 1 times that you know it has occurred within that volume. So, it is n 1 by N. And then divided by; that is a density function. So, probability of x given

w 1, which is the posterior probability and which is defined from the Bayesian rule that joint probability divided by probability of x it is n 1 by K.

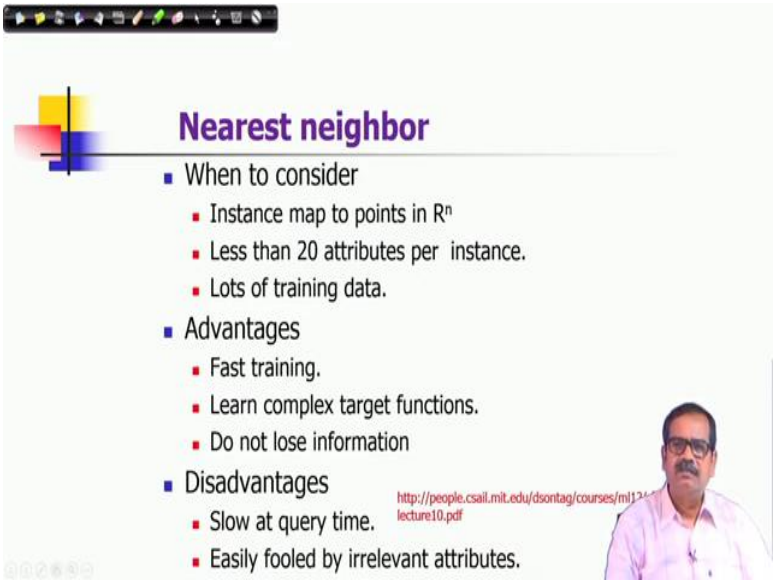
Now, this is the interpretation if you have n nearest neighbors n w 1 out of K neighbors. Then the n 1 by K give the posterior probability of that class given the data point. So, if you assign the class which your for which maximum number of data points of occurred, it is actually in the same as maximizing this posterior probability. So, it is following the Bayesian classification rule as you can see.

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So, this is what and it follows the of Bayesian inferencing rule.

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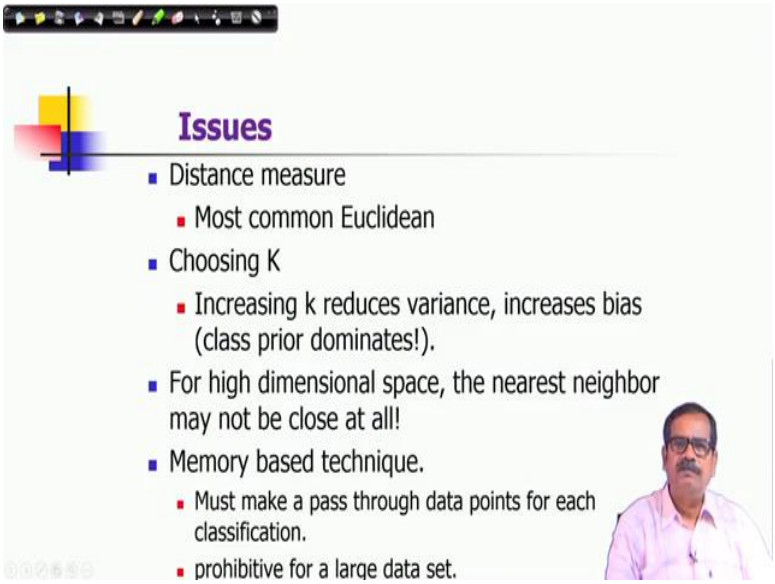
So, when should you consider for nearest neighbor classification? It when the instance maps to points in a in the dimensional space. When you have less than 20 attributes for instance and there are lots of training data. Then those are situations when nearest neighbor classification should be more appropriate. There are advantages as you see that was training what mean it is, it do it means that you have to store the data of course, we will call not storage for that.

So, learn complex target functions the method is very simple, but finally, the function simplicity it is implemented though this method that is not simple. So, if you if you do more detailed analysis, you will be able to understand what are the decision boundaries K nearest neighbors. And those boundaries a quiet complex they are shapes quite complex there not simple.

And it you know it do not lose any information. Disadvantages it is slow at query time it is it is very much competition intensive. You have to compute know all the distances with all the data points. So, which means your query would be very slow. It is linearly proportional with the number of data points.

And if your data point is very large then it is prohibitively very much difficult and easily fooled by irrelevant attributes. So, if there are outliers or rather irrelevant which is not related, but still you are establishing the relationships by nearing relationships.

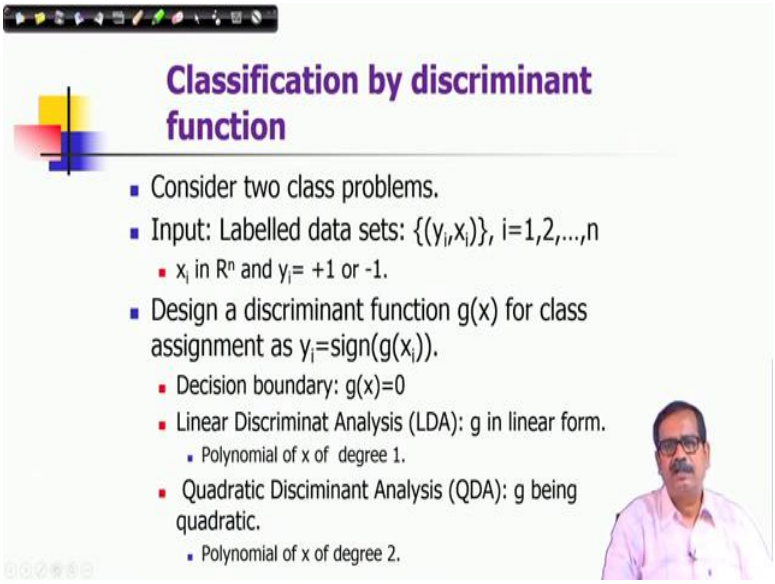
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So, these are the issues, like distance measure you have to choose the most common is an Euclidean distance. And increasing K; reduces variance and increases bias which means not class prior dominates in that case. And for high dimensional space the nearest neighbor may not be closed at all. So, that is another know difficulty using it high dimensional space.

That is why not more than 20 attributes on 20 dimension 20 it is advisable to use this technique. This is just empirical observation memory based techniques. So, it means must make a pass through data points for each classification. So, these are several issues of nearest neighbors. And as I mentioned it is prohibitive for a large dataset.

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So, this is the approach of nearest neighbor approach. The other approach which I will be discussing now, that is called a classification using discriminant functions. Now, in this case we should consider a two class problems to illustrate. There are extensions to multi class problems, but mostly in most cases discriminant functions are very easy to use for two class problems.

And we can extend that discussions for multi class problems. But in this particular course, we will be only considering two class problems. So, the problem statement once again let me revisit with respect to these two class problems. So, we have labeled data sets like there are n such a training samples each one as a level yi each xi.

Once again its say n dimensional feature vector; and the as I mentioned xi is a n dimensional feature vector and yi there are two classes. So, we will be using class representation by this value of yi is + 1 or - 1. Then the problem of designing classifier using a discriminant function is that this function needs to be designed.

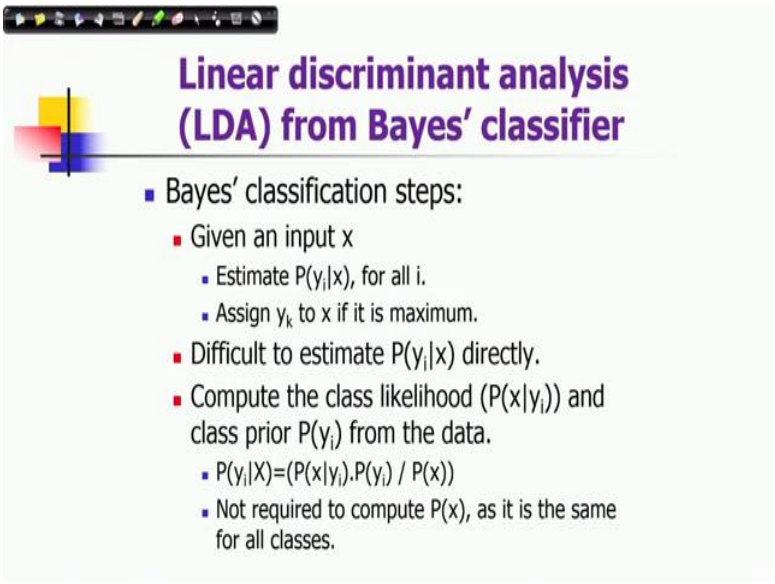
Such that, if I evaluate the function at x I then the sign up that value, which is either plus 1 or minus 1 which means positive or negative should be the class identity that is a definition of this problem. So, should note that the decision boundary is given by g(x)=0.

So, there is a boundary and which geometrically which partitions the space into two regions on one part all the values. If they put the discriminant function there it should be

+ 1 they other part would be - 1. Usually, that is the scenario of discriminant functions. And on the boundaries only and these values are 0.

When we call this analysis as linear discriminant analysis; when this function g is in linear form and it is a coordinated discriminant analysis when g being quadratic. So, the meaning of g is in linear form is that if this is the polynomial of x of degree 1. So, x is a multidimensional vector. So, for each attribute, which is taking part in the expression of functions; the polynomial degree of any attribute should not exceed 1.

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So, it is a polynomial degree 1; and polynomial of x of degree two for quadratic; so, each attribute 1 second degree should not exceed 2. So, let us discuss a particular tie a linear discriminant you know function are deriving of linear discriminant functions from Bayesian classification approaches.

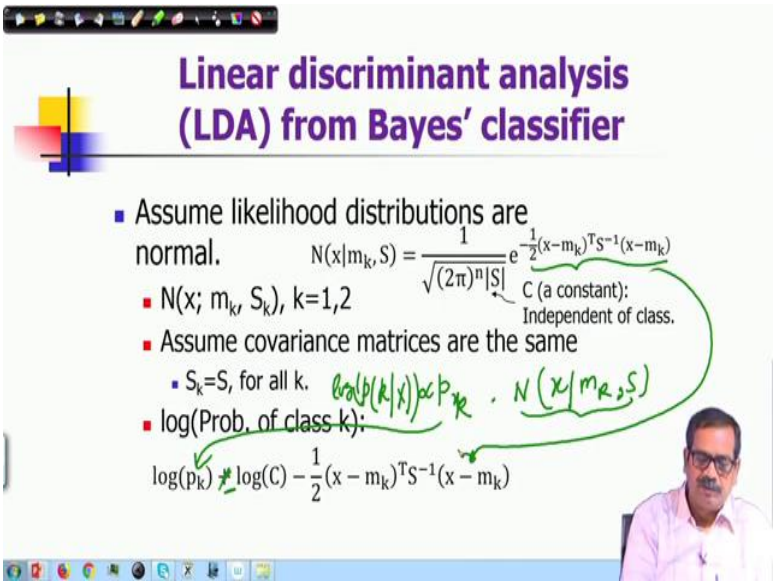
So, we call it as linear discriminant analysis from base classifier. So, the base classification steps as we have seen given an input x; we need to estimate the posterior probability P(yi|x). So, that yi in this case it could be either + 1 or - 1 there are two classes. And then assign that a value y k; that means, k th value of the k th class in to class problem it is just there are two values either y 1, y 2 two x if it is maximum.

Now as we have already discussed that it is difficult to estimate this posterior directly what we should do? We should compute the class likelihood and class prior from the data which

has been shown here. That probably of x given y i that is a class likelihood of given the i th class y i. And the class prior is probably of that i th class itself.

So, this is how posterior is related with them. And once again just to stress that we are not required to compute quality of x. And that is y only computation of those two factors are is sufficient to perform this classification.

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So, now let us assume this likelihood distributions are normal. We have discussed for naive base classifier, where an unlikelihood distance distributions. We have discussed about the discrete and categorical classes. But we categorical attribute to domain actually attribute sets.

$$N(x|m_k,S) = \frac{1}{\sqrt{(2\pi)^n|S|}} e^{-\frac{(x-m_k)^T S^{-1}(x-m_k)}{2}}$$

But if I consider the continuous attribute value in a continuous domain space the attribute value occurs which we assume. We can use the parametric modeling. And let us assume that for every n dimensional endpoints diamond every n variables are in attributes. I mean it is in our combinations is that n dimensional feature vector itself follow a normal distribution.

So, here the independence of attributes search are not considered. So, this is what a normal distribution in a multidimensional space is represented to note here. There are two

parameters though the number of elements in those parameters they are not single as it was for one dimension. One is m k, which is a k th which is a 1 second and dimensional vector, which is the mean of the class it is the vector. And Sk is a covariance metrics of once again N crossing metrics but which is symmetric.

This is how the probability density functions in the multidimensional space looks when it is a Gaussian distribution. Just to simplify these expressions, we considered the denominator part as a normalizing part of this probability density function. And in this case this is a constant. We are assuming that every class has covariance metrics that is an assumption here.

So, assume covariance matrices at the same then only we can do this job. So, this is what Sk equal to S for all k. So, now, we will be expressing this particular probability by using logarithm operations. Because the advantage of logarithm is that the comparisons of values in original domain can be carried out in the log domain. Because, those values are proportional to the log values itself or log values are proportional to the origin and value itself.

And as you can see the normal distributions, one factor is an exponential exponentiation operations. If you take the logarithm operation though so, the power of that explanation operation will be basically computed. And it could make we could make it linear form or it could make it a polynomial form rather it is not always linear.

So, if I perform logarithm so, that is advantage why you should know. Apply logarithm operations over the probability values and compute the log properties or log likelihood values, whatever in there are different ways these measures are named. So, if I perform the logarithm of this expression of the posterior.

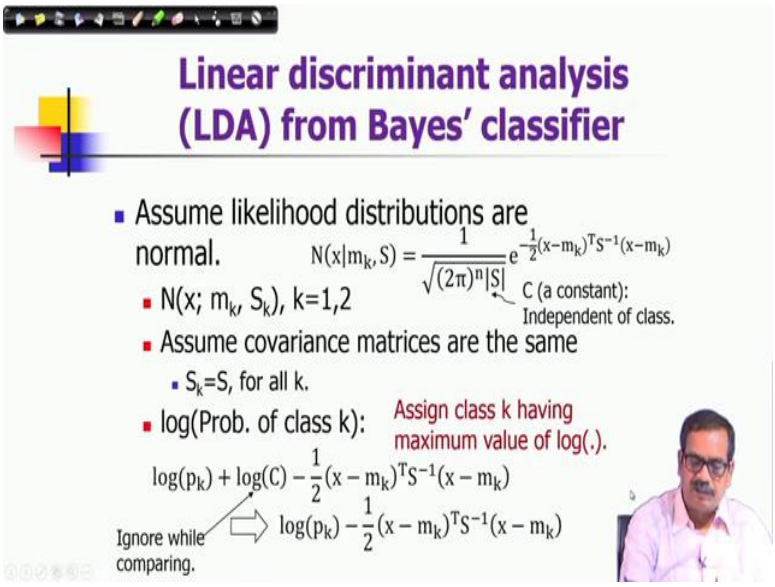
So, once you take the logarithm so, it is a locked p k and this constant. So, the posterior is proportional to p(k) that is the prior probability into. So, posterior p(k|x) is proportional to this value. Now, once I take log of this operation then log (p (k)) comes here. And from here we can see that this is a C; constraint C.

$$\log(p_k) - \log(C) - \frac{1}{2}(x - m_k)^T S^{-1}(x - m_k)$$

So, I should write here - log C; this is in the denominator, but it does not matter in our expression because we can ignore it at it is not dependent on class. It for the comparison so, because we have to find out the maximum of all log of these probabilities and then assigned that class which has the maximum value.

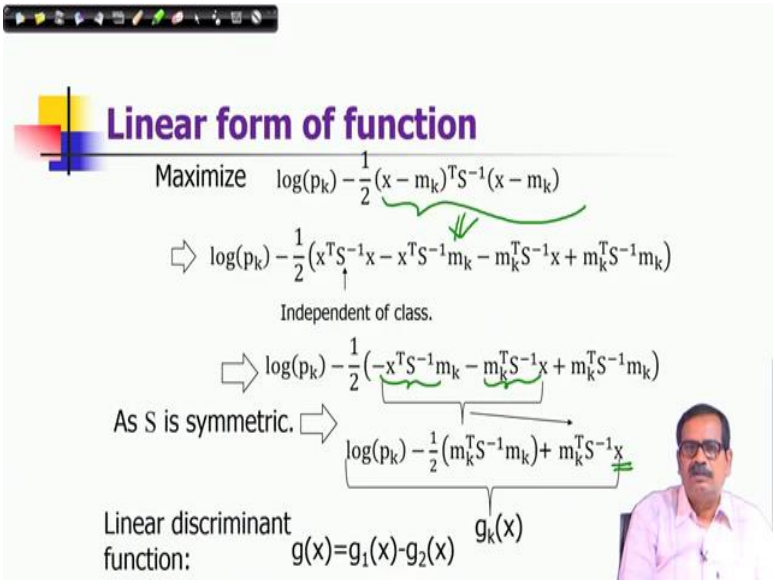
So, in the in comparing those values these constant terms will not matter. And this term is shown here that is a logarithm of exponentiation operations that would be coming there so, this is the plus.

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So, as I mentioned we ignore this while comparing you should note the correction here this plus should be minus according to this representation. So, finally, it is sufficient to compare these values for finding out the class giving the maximum posterior probabilities. And we should assign that class which having the maximum value of this.

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So, we will see that how actually this will give you a linear form of function. We need to maximize this value in between there some steps I will show them.

$$maximize\ \log(p_k) - \frac{1}{2}(x - m_k)^T S^{-1}(x - m_k)$$

And we can see that from that operations I have shown that is that there are two components in that operations which you need to maximize. One is the prior component that log(p (k)); other one comes from the likelihood part ignoring the constant terms.

And, if I expand them just you can perform usual algebraic operations of know multiplications using those matrices itself. So, these are the expansions of this term. So, if you expand it will be coming in this way. You can see that in my multiplications I have considered a very standard algebraic operations.

And, this part is independent of class because the covariance matrix is same for all of the class and given the data x. So, this is independent drop class once again we can ignore this for compression. And on the other hand so, that is why they in this step we are not having this particular know or term; we have only three factors. But since As S is symmetric since As S symmetric, then actually this value and these value they are same. So, simply we can add them.

So, it should be minus to slip let us suppose m k transpose is inverse x that is a value; and then if I multiply with half it would be simply this value. So, that is show the algebraic operation. So, you can see that these two terms we can combine them into a single term because they are the same you simply you can add them. And finally, we get this expression.

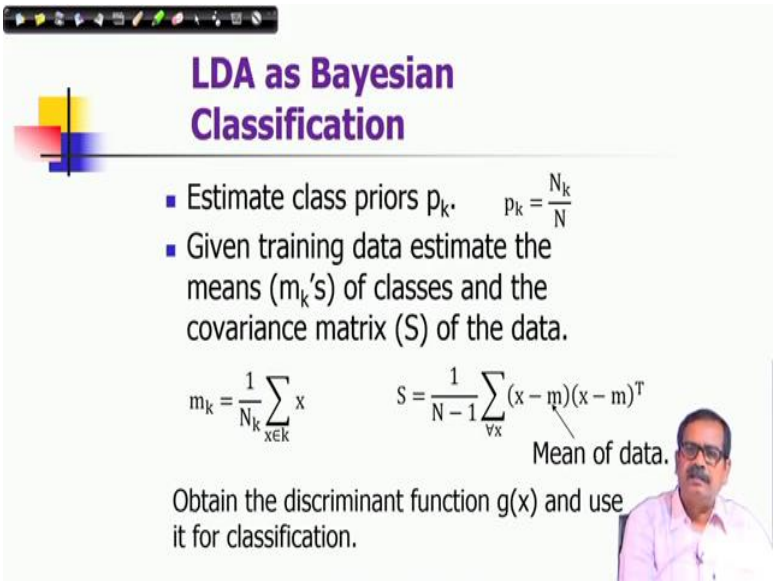
So, in this expression as a it is possible to see that there are class dependent factors these are all class dependent factors. But particularly this for discriminant functions point of view this is a variable x in this discriminant function; and which is a linear form of a discriminant function. So, this is how we can derive a linear discriminant function using a bayesian classification you know approach.

So, this is a discriminant function for these class x if I class k. And linear discriminant function can we consider as difference of these two. Because know the sign of this you have to consider a sign of g x.

$$Linear\ discriminant\ analysis\ g(x) = g_1(x) - g_2(x)$$

So, when g1(x) is maximum or greater than g2(x) the sign would be one positive. And we assign it to class 1 and when g 2 is greater the sign would be negative then assign it to class 2. So, your linear discriminant function is g x and two which is as we can see the form is a linear form.

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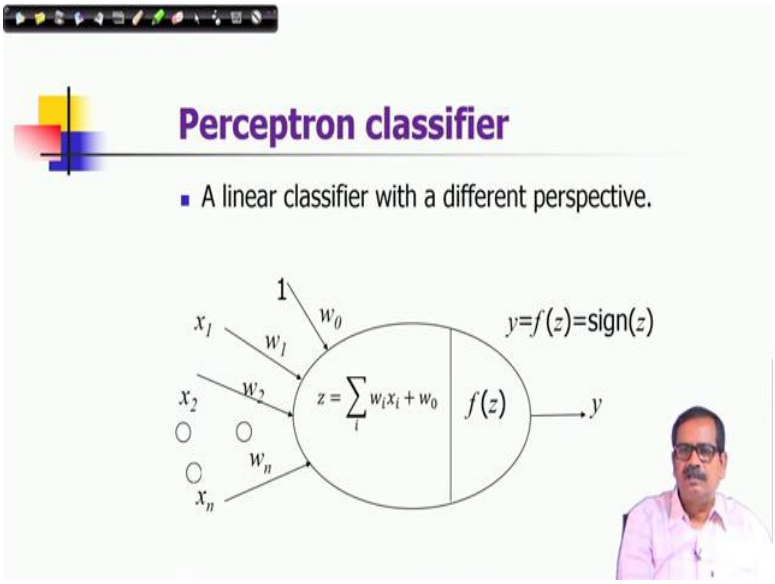


So, just to summarize this approach, that how linear discriminant function could be derived following this bayesian classification what we need to do? You need to estimate different-different class priors that is p(k). So, for estimating class priors you get the ratio of these two quantities N k is considered as the number of instances in class k and N is the total number of instances so, that is how the class prior.

And then given training data estimate the means of classes and the covariance matrix of the data in this way. That this is m k which is mean of know samples in the in the class k in this expression. And covariance can be computed from the whole data. Now, this is one know possible approach for estimating covariance there are different other refinements are there also. So, I am just presenting it for the sake of simplicity of computations.

So, you note m here is defined as a mean of the data and say it is over all mean of the data it is not the class means. So, you obtain in the discriminant function g x as we discussed and use it for classification. So, this is how you can you know perform linear discriminant analysis given the data points up to class problems.

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In the next type of classification, which is also a kind of linear classifier is special kind of know classifier. Or we can see that there are relationships also and but it you can have a different perspective from this. So, let us try to understand this particular classifier which is called here perceptron, you can see there is a node and there is a computational block shown in terms of an ellipse.

So, in this computational block it has its input, where you are getting the getting a feature vector as an input described n dimensional feature vector. x 1 to x n those are the corresponding attributes in that feature vectors. And each attribute to has an weight w; I and weighted combination of this attribute values. Plus there is a constant term or which is can be considered as bias w naught. So, this is a functional form.

So, as says the function is also a linear functional form as you can see. And the class what we would you like to do the classification you have to once again its the same the sign of that. So, it is acting like a discriminant function. So, this z is acting like a discriminant function gx; and then sign op that will give you y.

So, it is nothing but the same problem, what I discussed in the in the previous example of driving example of linear discriminant analysis. So, this is also a and linear classifier. But it has been shown with this perspective it would be clear when I elaborate this aspect know later on. Let me take a break at this point and we will continue this discussion in that next lecture.

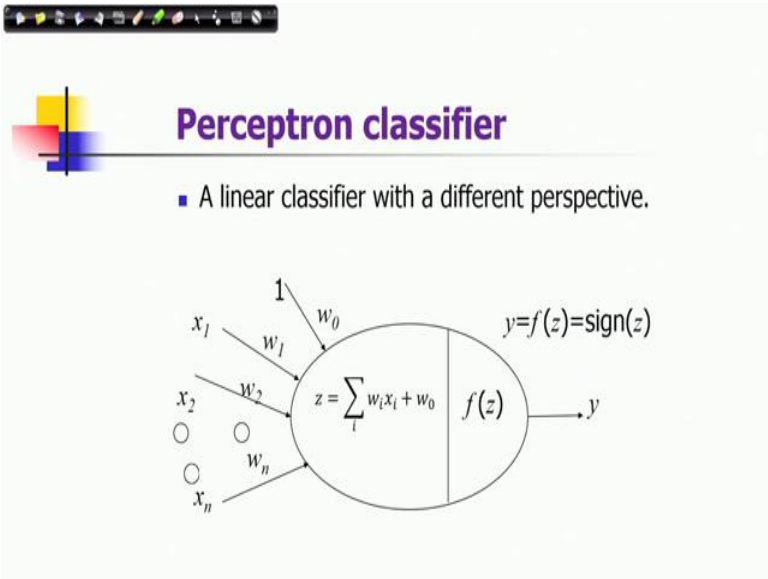
Thank you very much for your attention.

Keywords: Nearest neighbor, linear discriminant analysis, perceptron.

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Lecture - 49
Clustering and Classification Part - IV

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We are discussing about different classification schemes. And today I will discuss about a classifier which is a linear classifier of a different perspective and that is perceptron classifier. So, in a perceptron classifier you can see here that there is a concept of a node and edge representation. So, there is a particular node which represents a node of a classification classifier here. And it has an input, in this case you can see that there are n-dimensional inputs and then in fact the input is also augmented by another dimension one.

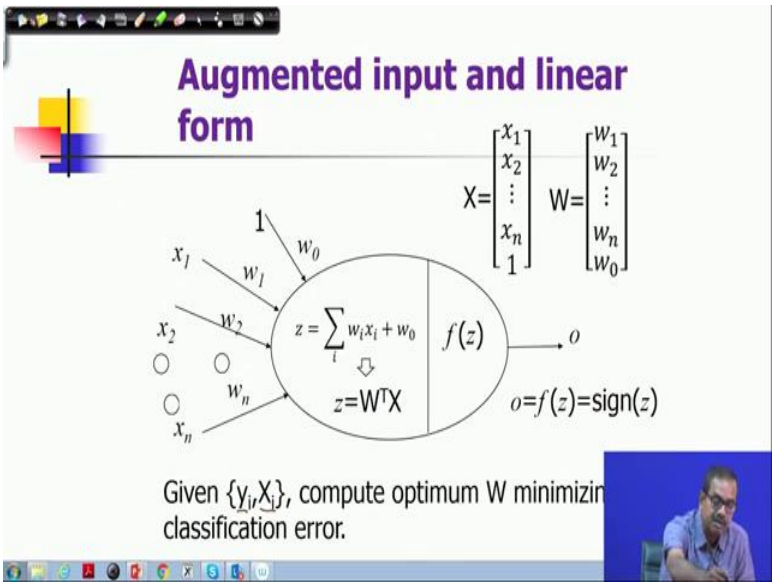
So, this is your input and these are the weights to the particular node. And what we are doing, we are doing weighted sum of this input, and then plus there is a bias. Now, you can also consider as if the input is augmented by another dimension whose value is 1, and then the whole thing can be considered also as a linear classifier.

$$z = \sum_i w_i x_i + w_0$$

We will discuss these things also in subsequent slides that the fact is that now this weighted sum of this weighted input and also add and also it is added with this bias so this value provides also these value is used in a function, and the function value finally, provides you the output of a classifier.

So, depending upon the functional value, the classification output will vary. And in a normal linear classification scheme what we discussed in previous lectures, this function was chosen as a signal functions or sin functions which means if the z is positive, you consider the class level as 1; if z is negative, you can consider class level as - 1, and 0 is little ambiguous. So, for the completeness of the function you can say z is positive for 0 as class level as 1. So, we will see that with this perspective how a perceptron classifier can be understood.

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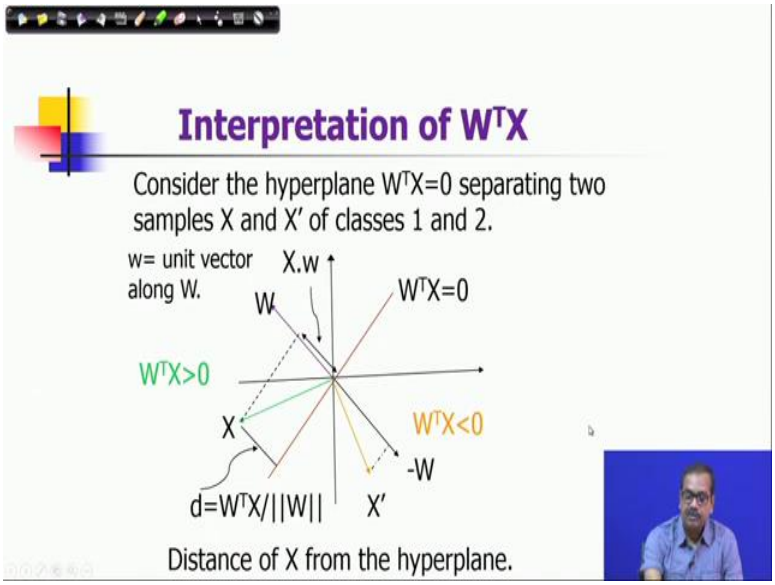
So, as I was mentioning that considering the augmented input, this could be in a linear form. And then we can consider the input as a vector of n + 1 dimensional vector, where the last dimension n + 1th dimension will be kept as always 1. And then and also the weight is also an n + 1 dimensional weights where the bias is included as the n + 1th part as w0. Here as you can see, and then it becomes simply a linear operation.

So, it is it can be expressed as W^TX, which is something like also you can consider from the vector operations as the dot product of the weight and the augmented input X. And this

has been used in a function and finally, output is a functional value of this particular net input which is z. And we are using in this case for example, a sign(z).

So, the problem of classification scheme is that given (y_i, X_i), where y_i is the level of the input and X_i is the corresponding input we have to compute an optimum W which minimizes the classification error. And we have already discussed about this particular type of problem in the previous lecture. So, when you discussed a linear discriminant analysis now with respect to this perceptron classification scheme, let us also revisit that problem.

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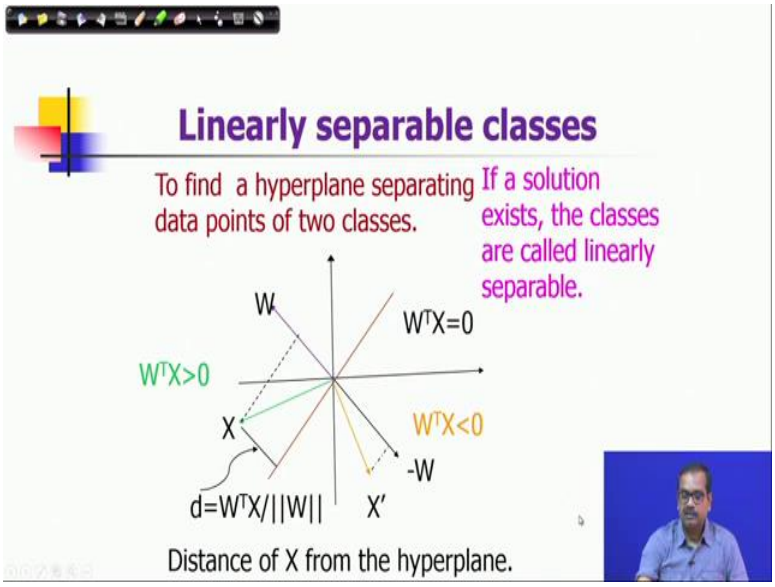
So, in this case, you consider that now this is hyper plane and this $W^TX=0$ is an hyper plane, and it separates two samples X and X' of classes 1 and 2 as you can see. So, in my diagram this is the hyper plane in that is representing as a hyper plane. And you consider the normal to this hyper plane is W. And you note that in this case because of this form this hyper plane has to pass through the origin of the corresponding space, in this case for the simplicity I am showing it in two dimension, but you consider it is a n-dimensional, any arbitrary dimension we have to consider in this case.

Now, consider there is a sample X which lies in one side of the hyper plane, and another sample X' which lies in the other side of the hyper plane. So, hyper plane divides this space into two partitions. So, in this side and if I define W this is a direction of W that vector, so if I take the dot product of these vectors, so this is X.

And so if I take the dot product finally, it is the projection of this vector on this direction that would give you and what it is showing it is basically if you consider that is a unit vector along W, and this is what is X.W, and as you can understand that this should be positive it will form a positive. If it is in the same partition where the vector normal and also the x there lying, then this dot product will be greater than 0. And the interpretation would be that you are measuring the distance of X from this particular hyper plane.

So, the distance from our analytical coordinate geometry, distance function is given in this form, you can see that this is a very standard form how the distance could be completed. Now, if I consider the other sample X' which lies in the other side of the hyper plane, and it is X, it is in our case as I mentioned that X' is of class 2, that means, which is also in a different class then this dot product will be negative because now you are taking dot product with W in that direction, so $W^TX<0$.

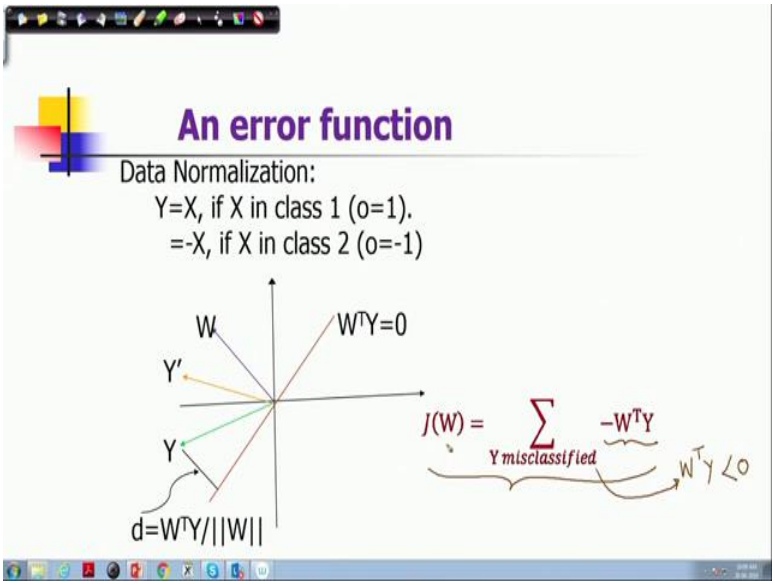
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And so this is an ideal case. So, this hyper plane can discriminate these two samples of two different classes. So, the condition should be therefore a classifier that the samples of class 1 should lie in one side of the hyper plane, whereas the samples of class 2 should lie in the other side of the hyper plane. So, if this condition is satisfied, if your input is given in such a way and if there exists such a hyper plane which can divide the input which can partitions the input in this form, then we say these classes are linearly separable. So, and you can get a solution of hyper plane. So, in that case, our objective would be to find a

hyper plane separating data points of two classes. So, this is what if a solution exists then the classes are called linearly separable.

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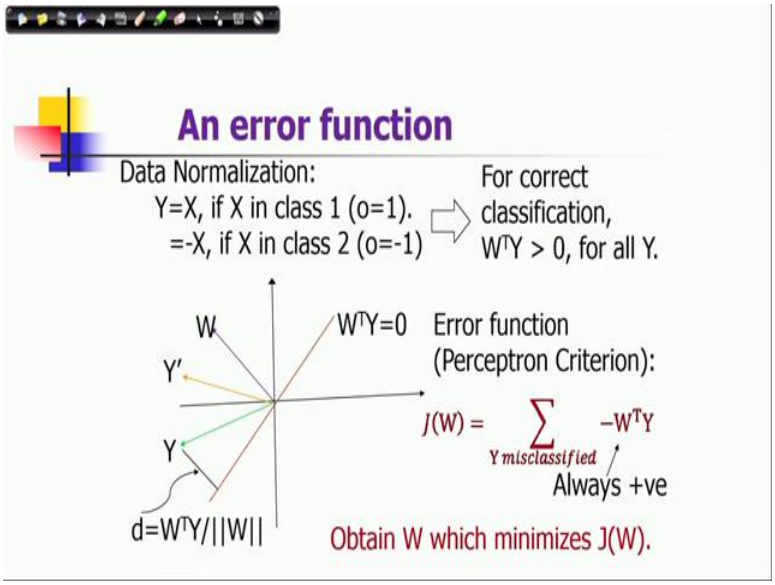
So, we are trying to design an error function. And in this case, you consider the data normalization we will be doing some interesting manipulation of data, so that your problem gets much more simplified. Let us consider the samples or the input samples which belongs to class 1 is X, whereas if it is in class 2 will be inverting this sample will be negating this sample -X. So, in that way intentionally what we are doing even the samples of class 2 also we are bringing in the same hyper plane in the same partition of the hyper plane.

And then with this kind of normalization, so all are input samples which should satisfy the property that all of them should lie in one part of the hyper plane, and we have brought it towards the positive part of the hyper plane, that means, the partition where also the normal also the normal vector also lies in that partition. So, with this kind of operations, they we have satisfied this conditions that all my input sample should be lying in the positive form.

So, with that data normalization we can define an error function where we can see that this is a form of this error function. And if only we consider if Y is misclassified then only we compute this value which means this value has to be always positive because if Y is misclassified with according to your input condition, then for a misclassified sample it would be always negative for a misclassified sample. So, if I take the negation of that, it

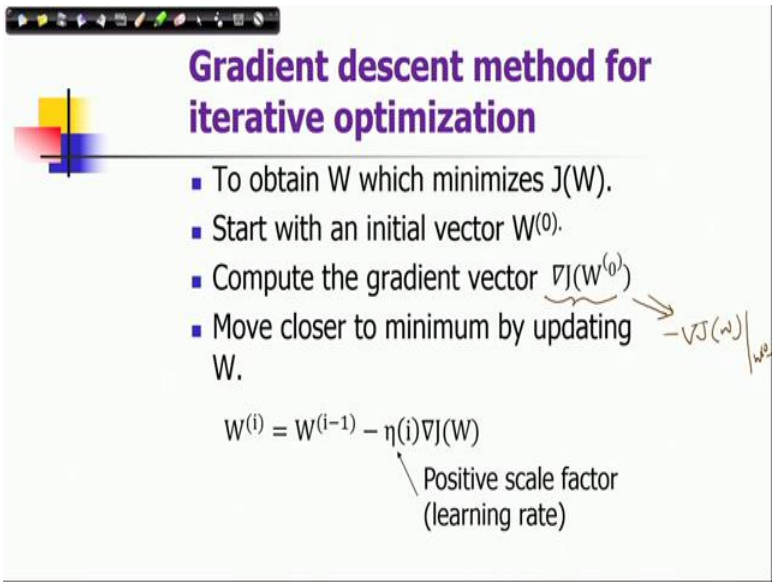
would be positive. So, this error will be always positive. And using this error function, we will be will be designed we will be designing a classifier.

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Let us see how we can do it. So, this is what for correct classification it has to be always positive as we mentioned. And this is an error function which you have defined, and this should be always positive as I mentioned. So, our problem is that to obtain the weight vector which minimizes this particular error function.

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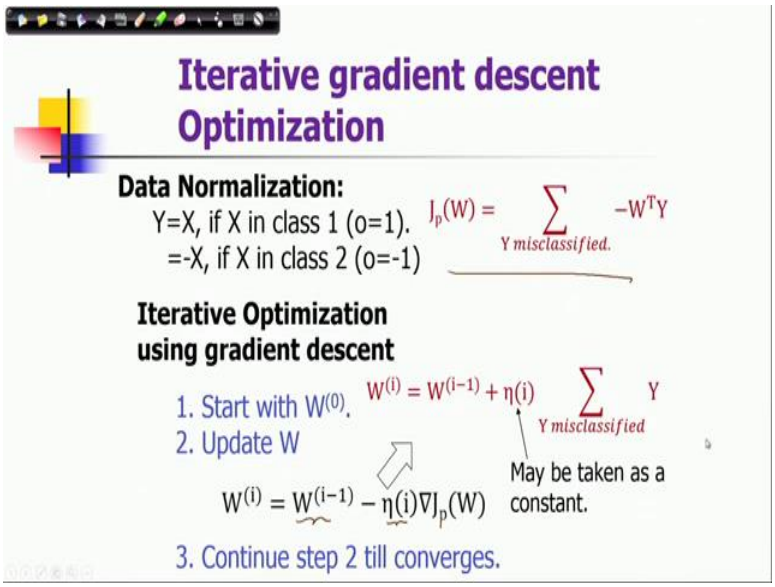


So, we will be using here a gradient descent method for iterative optimization, and the steps should be like this way, we can start with an initial vector which has been shown here, I say W^0 that is the initial form. And then we compute the gradient vector. Suppose, this gradient vector is denoted with this operation we will see how we can compute this gradient vector. So, we are performing gradient over this function. And at the initial, so you note that the functional point is W_0 here. Now, you should move close to the minimum by updating W.

So, you did is following the rules of gradient descent method, you have to move along this steepest gradient directions in the opposite to the steepest gradient directions that means, you should move along $-\nabla J(W^{(0)})$ what has been shown as a symbol say this is a symbol. So, you should move along that $-\nabla J(W)$, which has to be computed at W^0 that is the meaning of this particular symbol.

So, you should move along that, but instead of if what we can do while moving these things will be simply considering there is certain step size by which will be moving it is a proportional factor to that particular value that will be moving. And, this is particular this is called as the positive scale factor or learning it in our weight updation.

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So, we let me further elaborate this particular process of iterative optimization using gradient descent we start with $W^{(0)}$ then update W and W should be updated in this form.

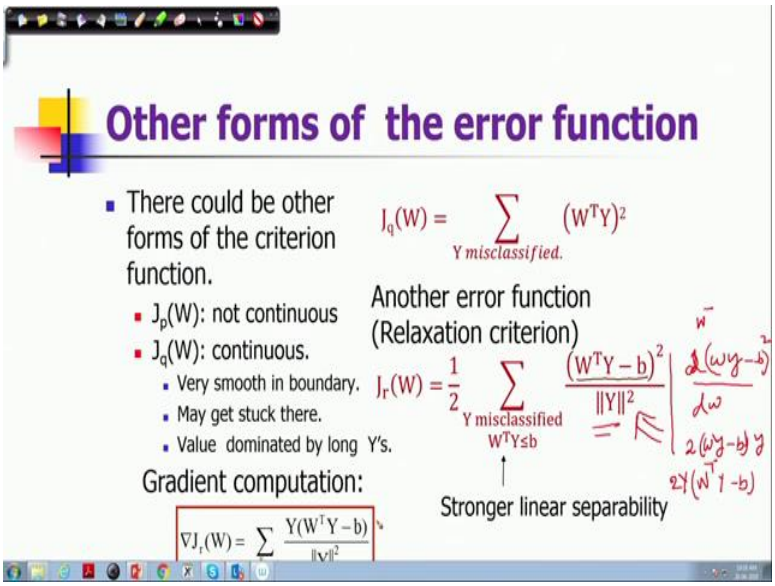
$$W^{(i)} = W^{(i-1)} - \eta(i)\nabla J_p(W)$$

As you can see it is a previous result of the previous iteration of W, and then it has to be moved along the steepest gradient directions multiplied by this scale factor. This is the gradient of this particular function. And we will see how to compute this gradient analytically we can very easily derive these computations, because if I take W^T from here if I take the gradient it would be only sum of Y or sum of - Y.

So, this expression can be modified in this form as you can see that $W^{(i-1)} - \eta(i)\sum_{y\ misclassified} Y$, Y is once again misclassified sample. So, computation is very simple in this regard. What you should do that you start with any W, then you use the classifier to see which samples are getting misclassified, then some of those samples will give you the corresponding update of the weight you should move the weight by adding the sum of those samples with a proportional value.

Now, this value may vary over the steps or you can also use some heuristics to keep it as constant or small. So, you should continue this operation still it converges, so that is what it could be maybe it could be taken as a constant this particular learning rate can be taken as a.

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There could be other forms of criteria function. For example, this is another function where instead of providing it as $-W^TY$, we can consider $(W^TY)^2$. So, the advantage is that the

previous error function what I mentioned that means, it is sum of minus W^TY. Note that all Ys are misclassified only you are taking your defining functions over those values only.

So, if Y is not misclassified that should not come into this functional computations. But this function particularly it is continuous though it has some issues like it is very smooth in the boundary which means there now if you apply gradient descent, you can get arbitrary result, you can get stuck at some location, and also this value is dominated by long Ys. So, some kind of normalization with respect to Y could have been there. So, we can choose another kind of error functions which certain relaxation criteria where this normalization has been considered, and you can see that this is a function where you are using $(W^TY - b)^2$

So, in this case the hyper plane is further moved towards positive side by a distance b. So, it is more stringent criteria for satisfying the linear separability of the classes. And this half is used just to simplify the computations because due to the square term, but the derivative these two will come here. So, 2x(1/2) will get canceled as 1. So, we will also see its analytical form. So, it has a stronger linear separability as I mentioned.

And gradient will be computed in this way. As you can see two half and this squared square term, it gets canceled, and then you can apply simple differential geometries concept of differential geometry here also what you did for the one-dimensional cases one-dimensional functions this can be extended to in many cases particularly when you are when you are using the matrix operations in the linear form, then you will see almost similar rules that applied here. So, this is a function of W as you can see.

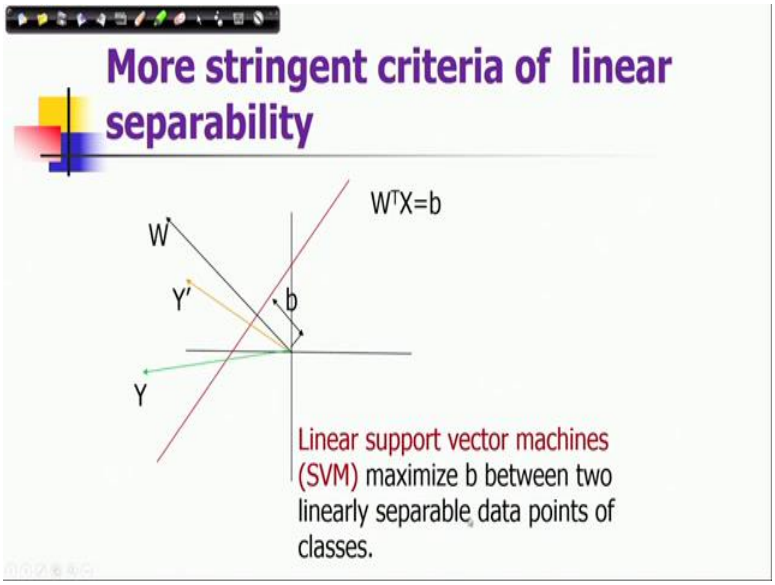
$$\nabla J_r(W) = \sum \frac{Y(W^TY - b)}{||Y||^2}$$

So, if you are deriving over this particular value, then the term Y will come here. So, if I derive this particular value will we can see that like what we did in the in any differential geometry suppose it is a function it is a function of say W.

So, if I perform derivative with respect to W, this value should have been two (WY-b)Y. When you translate it into the matrix form, then you get these operations, then you get this value as you can see why (W^TY-b) that is 2 (YW^TY-b). Of course this is a constant, so this

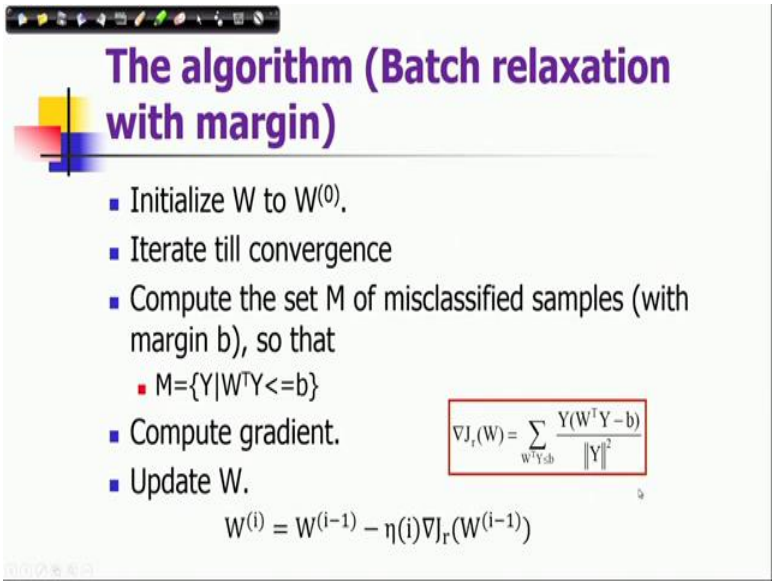
constant term will come here, so that is what you are finding here, so that you can see here that the gradient would be computed in this form.

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So, this particular diagram shows that because of the use of the offset b, you are moving further this separating hyper plane further towards positive partition by distance b and that puts a stringent criteria of a of this linear separability. Just to note here that even support vector machines linear support vector machines that is also that is also a kind of linear classifier and which maximizes this b between two linearly separable data points of classes, so that is the property of a support vector machine.

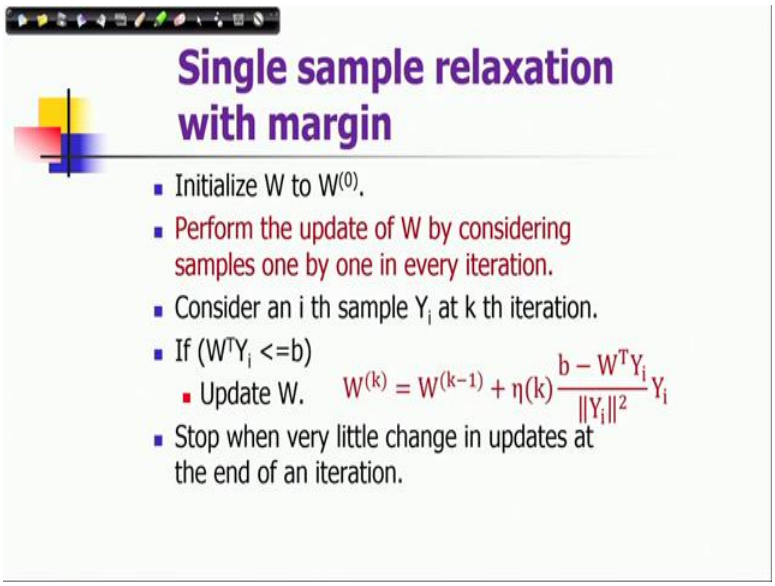
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So, let me summarize this algorithm, and which we denote as batch relaxation with margin. So, in this case we initialize W to W^0, then iterate till convergence. And for that we have to compute a set of misclassified samples where the conditions of classification you know gets violated because of the data normalization all W^TY operation should be positive or rather in this case for stringent classifier, they should be greater than equals b or greater than b.

So, the misclassification is that if it is <=, we consider it as a misclassification. And we are considering those sample Y, and then use those sample you compute the gradient as shown in this particular computational expression, and then you can update W by following this particular rule. So, this is what you should go on doing till you get convergence of your weights.

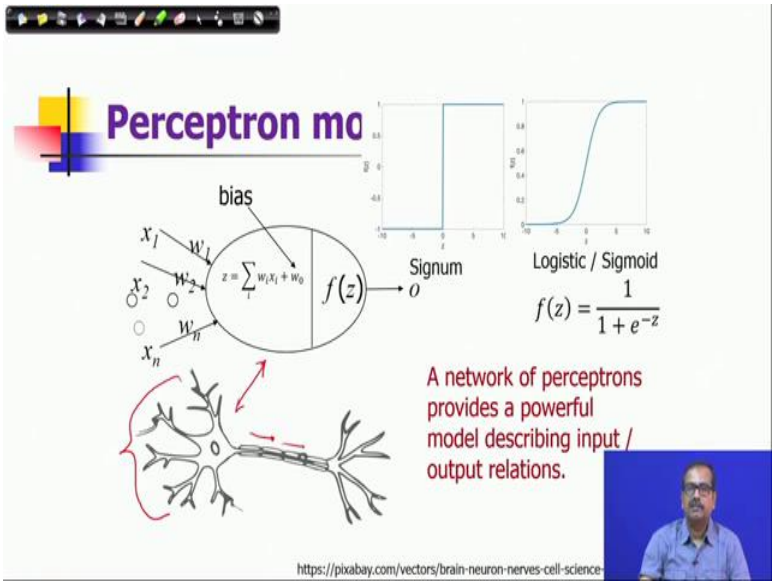
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Now, batch relaxation means now you are performing the classification of every samples in your batch, and then you are computing the gradient function. Now, there is simple simplification of these particular computations, and it has been found you can this is all this also gives the same performance, but it is much simpler and faster instead of waiting instead of considering the whole batch at a time, we can consider immediate updating of W by considering a single sample whenever it is misclassified. So, whenever a sample is misclassified immediately you update the weight W by following the same rule, same classification same gradient computation.

So, this is what you should perform the update of W by considering samples one by one in every iteration, which means if it is misclassified, then you update W. And you go one by one you take all the samples and observe when W is converged. So, this is the expression here now for a single sample computations of updation of W. And you should stop when very little convergence very little change in updates at the end of an iteration.

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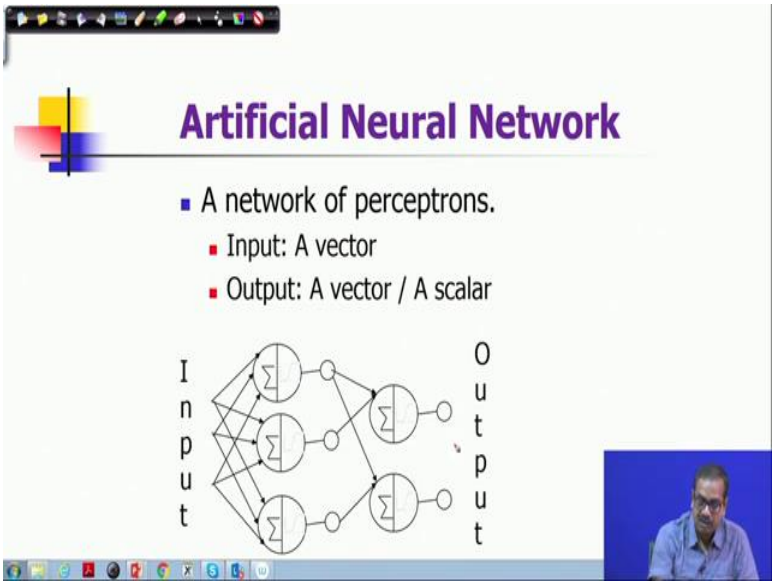
So, actually the perceptron classifier, it models also a neuron what is you know in a in its biological interpretations when you as a biological functions. As you can see in this particular classifier the inputs are given as with the classified the inputs are weighted inputs are taken, and then the output is dependent on as a function of that weighted input. If I consider a biological neuron, you will see that a neuron it consists of several synapses. So, in its input synapses in its input and this could be considered as those exciting synapses, synapses which is coming from which is taken from this input.

And then it processes it this weighted input is processed and then the output is propagated through the corresponding you know path of our nervous system of the neuron. And it goes to the other neurons or the excitation is propagated through this path, and other neurons are connected at the same. So, you can consider the perceptron node is like a single neuron, and it is collecting excitations it getting excitations from different synapses in the input and propagating and generating and output response which propagates through this path and which can be connected further.

So, this particular model is a very strong model. You can consider it as a network of these neurons or perceptrons and you can consider a system that your input and output conditions are or your output is generated through this excitation of these networks there is an input and through this network of neurons finally, the output response is generated and that is this model is known as neural network or artificial neural network model. And various

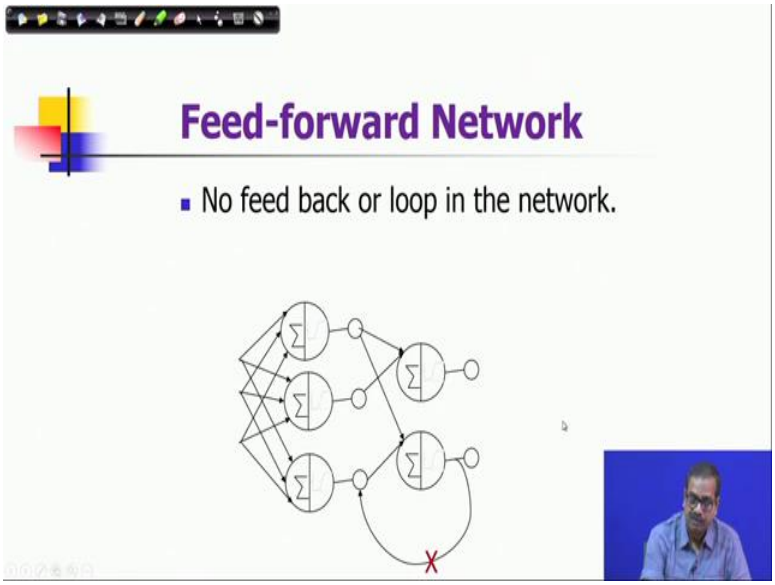
kinds of input-output relationships can be modeled using this generalized model, it is a very powerful model in that respect.

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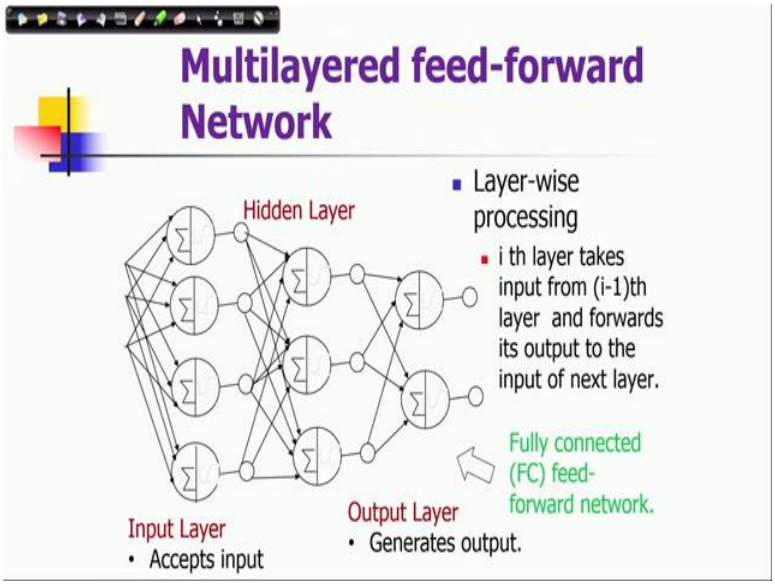
So, this is what we see here in an artificial neural network, it is a network of perceptrons where input is a vector we considered input is a vector and output is also a vector it could be also scalar, that means, you can have several output neurons which is providing you the output or you can consider only single output neuron, then the output becomes a scalar output.

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One of the special case of this kind of network is that they are mean there is no feedback or loop in this network. So, then we call this network is feed-forward network. So, a free loop means from an output you get a feedback to you know one of the one of its input layer sorry input neuron. So, this is kind of loop is not here in this kind of when it is not there in the network then we call this network is feed for feed-forward neural network.

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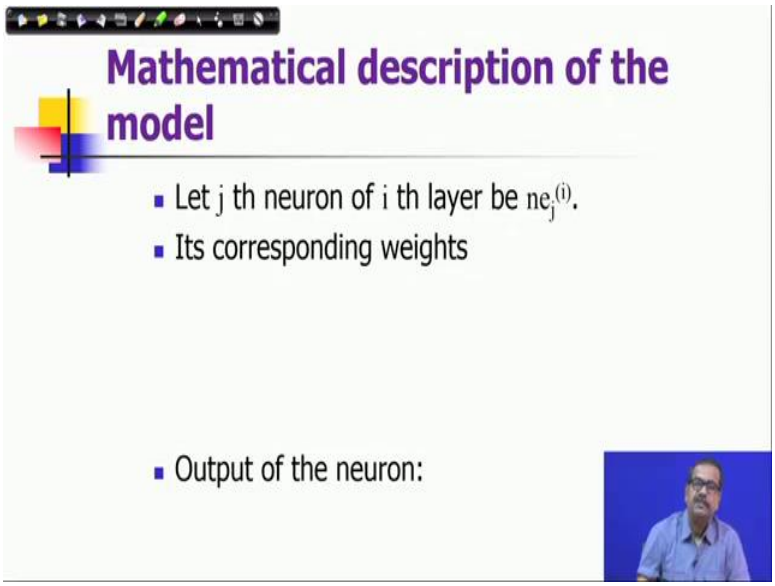
And a very general form of feed-forward neural network is a multilayered feed-forward neural network as has been shown here that there is an input layer which accepts input. So, an input layer which accepts input as we have seen here then there are several intermediate layers those layers are called hidden layer. In this case, we have shown only one particular hidden layer and then there is an output layer where you are collecting the outputs or whether outputs are generated which generates the output.

Now, you can see that in this particular form every layer has certain number of neurons which may vary. And but from a particular layer say from the one input layer to another to the next higher layer now for every neuron the connectivity to all the neurons of the next higher layer is ensured so in this particular model. So, there is a connectivity which means the weighted the there are weights from the responses of a particular neuron, so this weighted response is the input of the input of the neuron of the next layer. So, or every neuron in the next layer gets weighted sum of the outputs of the previous layer and that provides a net input to this layer.

And finally, again as you understand there is also a non-linear function which processes that net input as a weighted sum of the input to that particular neuron and then produces its corresponding output or response which again propagates or which excites the neurons of the other layers. So, this is the kind of model and since all the neurons are fully connected by its previous neurons of its previous layers; so, we call this kind of network also as a fully connected network.

So, as you can see that it is a layer wise processing, and the ith layer takes input from i -1 ith layer and forwards its output to the input of next layer, and this network is called fully connected feed-forward network.

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So, let me take a break here. We will be discussing particularly the mathematical description of this particular model and its analysis so that we can model the we can solve the problems of designing classifiers using neural network or problems of classification is a neural network that we will do in the next lecture.

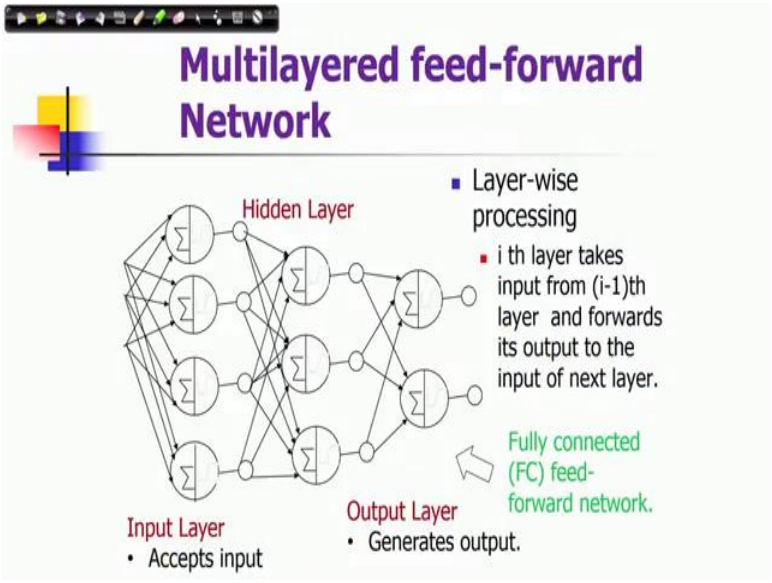
Thank you very much for listening to this particular lesson.

Keywords: Perceptron classifier, linearly separable, gradient descent, batch relaxation, feed-forward

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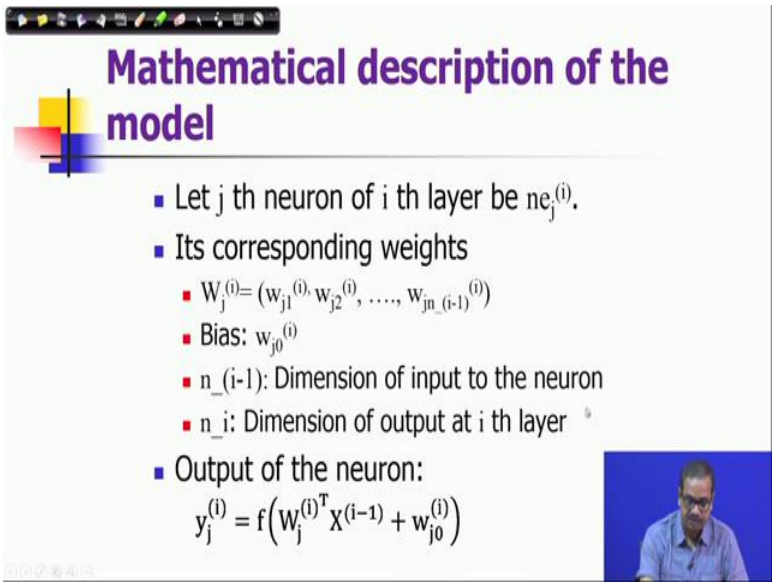
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We are discussing about artificial neural network and particularly it is a multilayer feed forward neural network, that we will be considering and this particular network will be used to model a classifier. So, let us see how we can model a multilayer feed forward neural network. We will be concentrating on a latest model particular neuron of a layer or a particular layer which means let us consider a jth neuron of ith layer.

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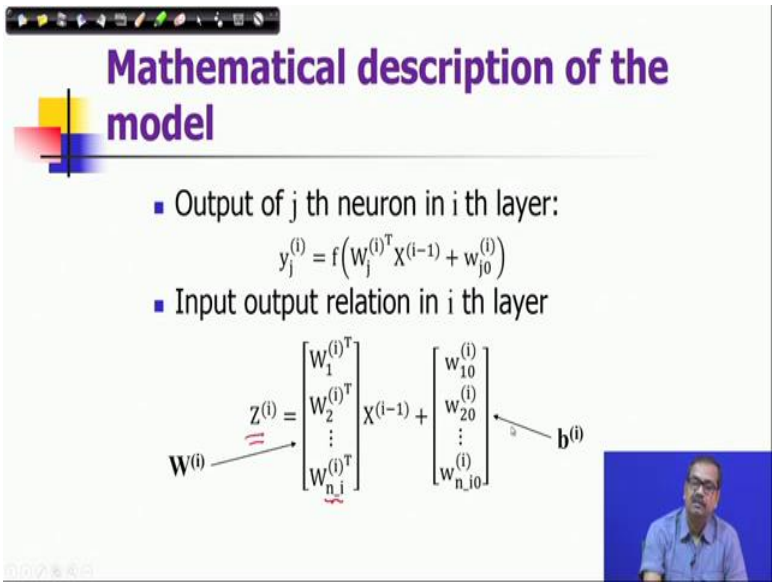


And so, to model this neuron it has its input weights. So, you consider that a weight is a vector and as it is connected to all the outputs of the all the neurons of previous layers. And, if I consider that number of output is given by n_{i-1}. So, where n_{i-1} is a number of neurons at $(i-1)^{th}$ layer so and the bias is considered it as $w_{j0}^{(i)}$. So, this is the dimension of input to this neuron which means also the number of neurons in the output layer for a fully connected network.

And, as I mentioned dimension about output at the ith layer which means that is a number of neurons at the ith layer. So, the output of the neuron can be described in the mathematical form in this way, that it is considered first to take a net input from its input. That means, $X^{(i-1)}$ is actually the vector generated by $(i-1)^{th}$ layer of dimension n_{i-1}. Now, you have the product you have the weighted combination of this input.

And, then it is added with a biasing term which has been considered here and then that becomes a net input to the function which will be a non-linear function which will be generating the output response.

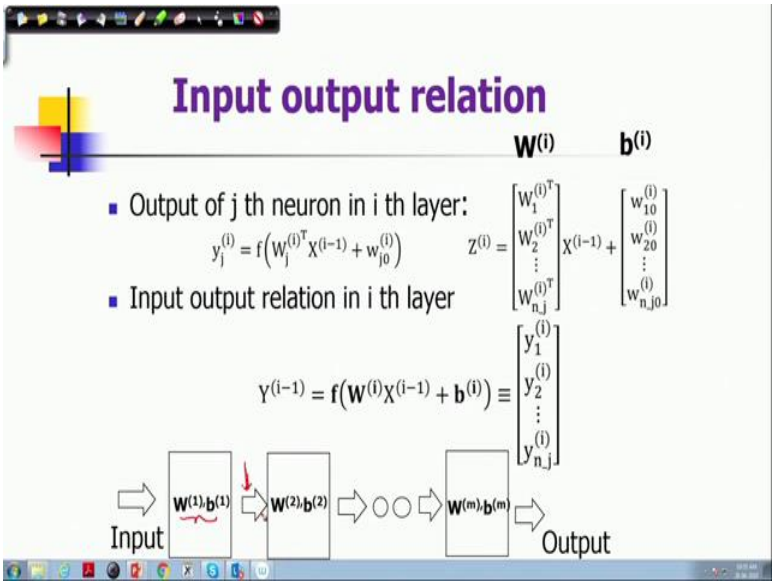
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So, if I consider the input output relation in the ith layer then that can be described in this form, you have in the ith layer you have once again ni number of output neuron. So, for each one there is a weight vector which is connecting all the neurons of the previous layer. So, this is how this relationship can be expressed as you can see that each input with it weight vectors, it is multiplied is it is making a weighted sum with respect to the components of the previous layer and then it is generating the output of the ith layer.

So, note that the dimension here is of ni this output is n i, this is a dimension and these are the biases. So, we can express in brief we can note denoted that this is a finally, this is a weight matrix of at ith layer and this is corresponding the bias vector at the ith layer. So, these are the two parameters which we are related to the ith layer.

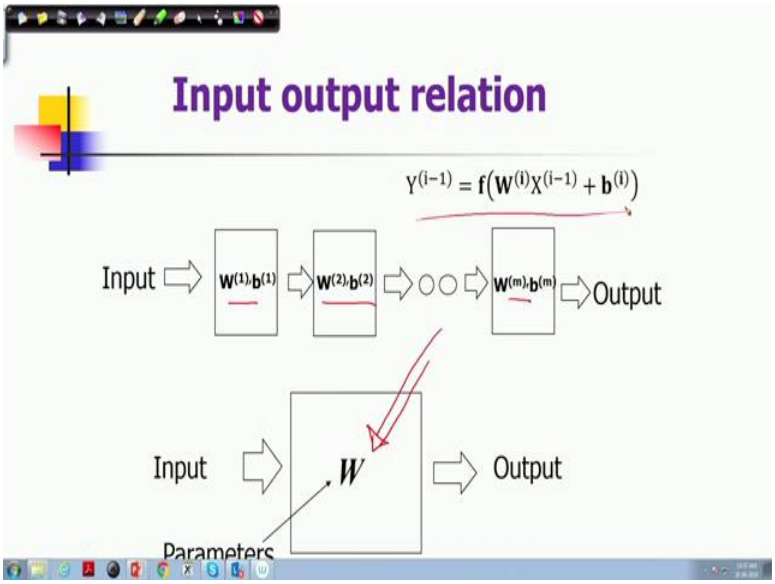
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So, the input output relation can be described to in this form as we can see that in a particular block diagram we have explained here. So, given an input then from this input layer use this parameters $W_1 b_1$ as described, you generate the output for the next layer. And, it in this way you propagates the output to the end of this output layer to the input of the output layer and finally, you get this output. And, at each layer the description of the parameters is in this form, there is a mistake here. It should be ni as per make our description should be align, similarly it should be.

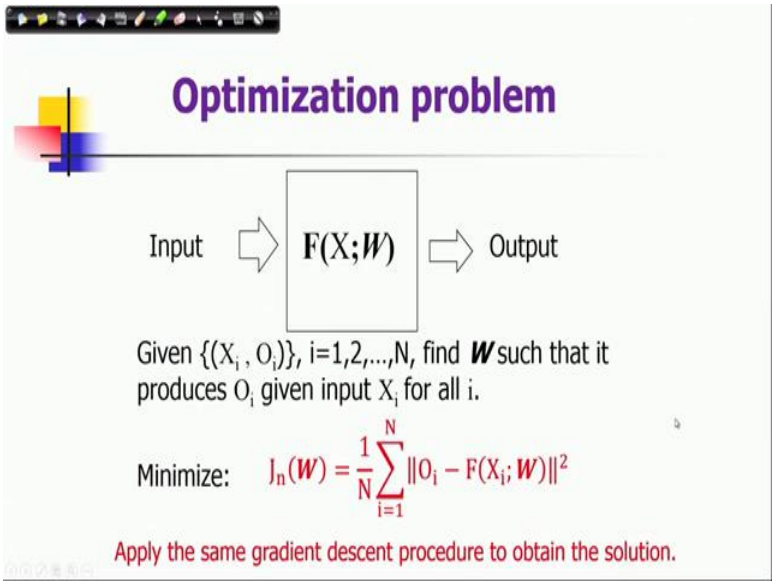
$$Y^{(i-1)} = f(W^{(i)}X^{(i-1)} + b^{(i)})$$

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So, in short the model can be considered the whole set of parameters, see all the set of parameters they define, the collection there are the symbols parameters as it is represent the whole collection of parameters. So, in a model you have the parameters are defined in this model and it has its corresponding input output relation following the functions of the neurons as we discussed before. And, at each functional block this is how it is processing the input.

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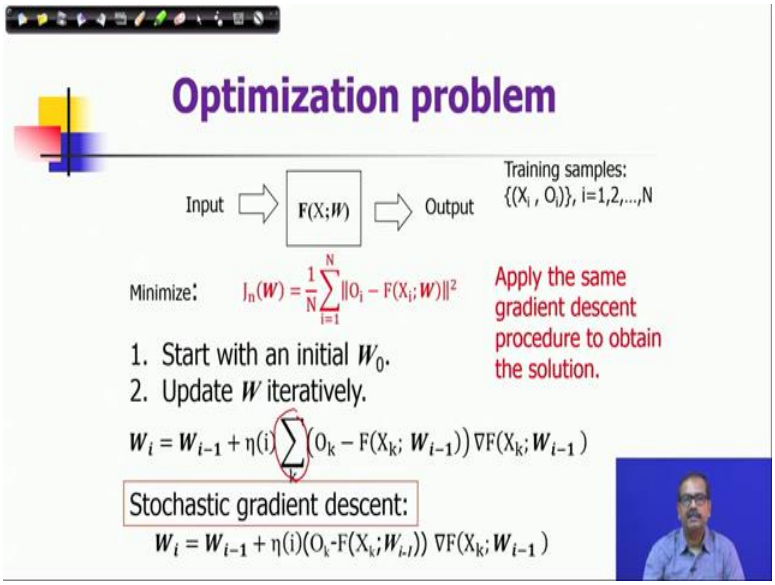


So, when you considered an optimization problem, we are trying to model a neural network in such a way that it satisfies the input output specifications which means that if I give you an input output specifications in that forms of X_i, O_i and these are the samples of n samples. So, you have to find out W such that it produces O_i given input X_i for all i. So, this is the optimization problem and we can use the same gradient descent approve a method to solve this. So, we define an error function which is a defined in this form here.

You can see that it is considered the sum of mean square error, it is basically mean square error given the output and given the predicted output of from the model which is expressed as F (X;W). So, this is a predicted output and this is the actual output. So, the square of this deviation is a single observation and then average over all these observations will give the mean square error. So, this error has to be minimized. So, you have to find out that W which minimizes this error. So, once again we can apply the same gradient descent technique.

$$Minimize: J_n(W) = \frac{1}{N}\sum_{i=1}^{N} ||O_i - F(X_i; W)||^2$$

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So, that is what we can do here. So, the procedure is similar, we can start with an initial weight $W^{(0)}$, then we update W iteratively. So, it is weight means collection of weights as we have described the collection parameters, it considers weights of weights and biases of

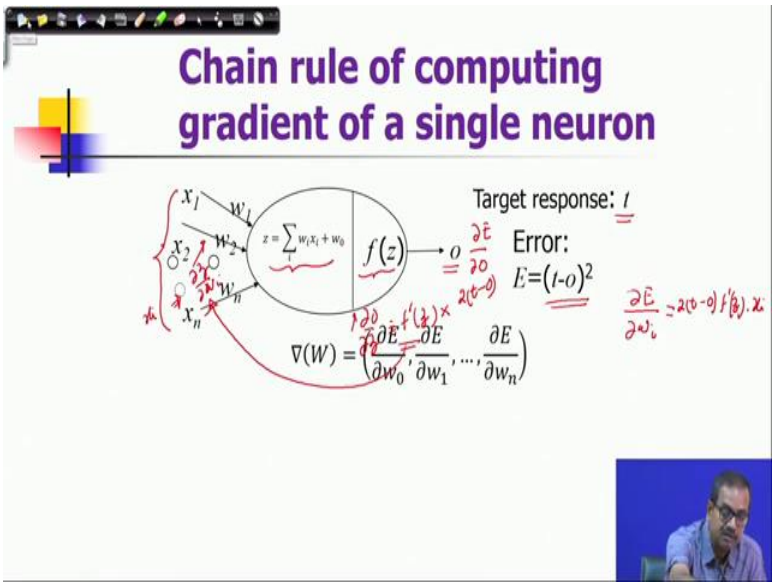
every layer of the artificial neural network. So, the corresponding update scheme can be shown in this form.

$$W_i = W_{i-1} + \eta(i)(O_i - F(X_i; W_{i-1}))\nabla F(X_K; W_{i-1})$$

So, η(i) takes care of all the scaling factors here and as we can see that when you are doing derivation of this particular function with respect to W. And then of course, you have to compute the derivation of F gradient of F with respect to W and you can apply once again stochastic gradient descent instead of instead of considering the O all the samples at it together.

That means, instead of considering the sum you can simply use a single sample like the previously what we have done and we can immediately update this weights and continue doing for every sample. And finally, when weights are getting you know converged; that means, there are a little changes in the values of weights after updation then we can stop.

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So, computation of gradient itself is a task that we need to do it here and here let me first discuss with respect to a single neuron or single perceptron. So, we can apply the chain rule very effectively. So, we know what is a chain rule, let me explain that say; this is the corresponding functional description that. So, this is an input here and you have corresponding weights.

So, this is the net input finally, after considering weighted combination then adding a bias term and then next process next part is that we have to this becomes an input to a non-linear function f. And, then it produces output O and your error function, if your target response is t then error function in this case we considered as $(t-0)^2$. So, this is a square error that is that we are considering.

So now, if you would like to perform you are interested here, you need to compute the derivation derivative of E with respect to individual weights because, we are trying to update the weights. So, how this particular response varies that we need to find out, that gradient and you should move towards or how it is affecting this error.

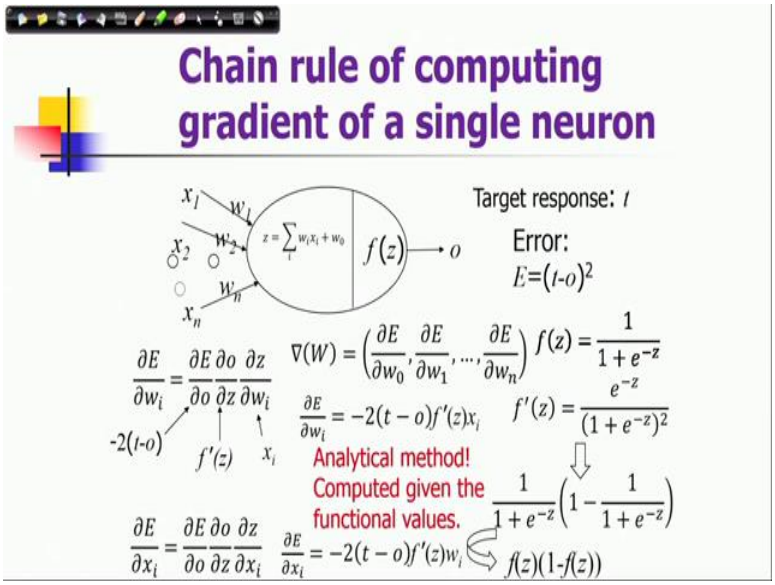
So, you should move towards that direction which will be minimizing the error; so, let us take us gradient directions. So, how do you compute it? Now, by applying chain rule what we can do at this part we can compute

$$\nabla(W) = (\frac{\partial E}{\partial w_0}, \frac{\partial E}{\partial w_1}, \dots, \frac{\partial E}{\partial w_n})$$

$$\frac{\partial E}{\partial w_i} = -2(t - o)f'(z)z$$

So, that is how the chain rule is considered, let me give you the corresponding you know summary here also. So, we need to compute as we mentioned, we need to compute all this gradients.

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And as I was telling you that all these components we have to compute. So, you just multiply all this three things so, you get the corresponding

$$\frac{\partial E}{\partial w_i}=\frac{\partial E}{\partial o}\frac{\partial o}{\partial z}\frac{\partial z}{\partial w_i}$$

Now, let us consider a particular form of f(z), this is a sigmoid function. So, if I take the derivative with respect to z, it will look like this.

$$f'(z)=\frac{1}{1+e^{-z}}(1-\frac{1}{1+e^{-z}})$$

Now, incidentally this can be expressed in this form and which is nothing, but f(z)(1-f(z)).

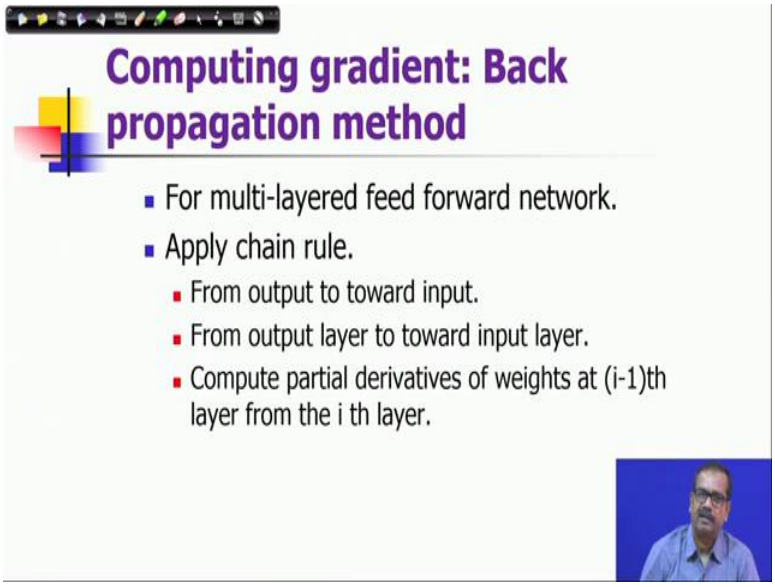
So, this is one interesting simplification, if you consider only sigmoid function. So, sigmoid function has this property and in many cases in neural networks sigmoid function is used and this particular property is effectively used in computation. Yes, if I want to compute $\frac{\partial E}{\partial x_i}$ because, we will see later on also that in some cases we need to compute that and then also we can apply chain rule.

So, you will find that the expressions would be considered here. The interesting part here is at the whole computation can be done using analytical methods. You do not have to do any numerical since you do not have to use the numerical simulation to compute. So,

sometimes for a complicated function what we observe that we give a change to the input and observe the change in the output and then use the ratio of those changes as a derivative.

But, in this case we can directly compute by giving the functional by giving the values at that point. So, only values of x w or weight centre weights and inputs they will be only required to compute this particular function. So, once you compute the functional value, then you can compute also the derivative using those values only. So, you can compute at a point that is what is advantage of this particular method.

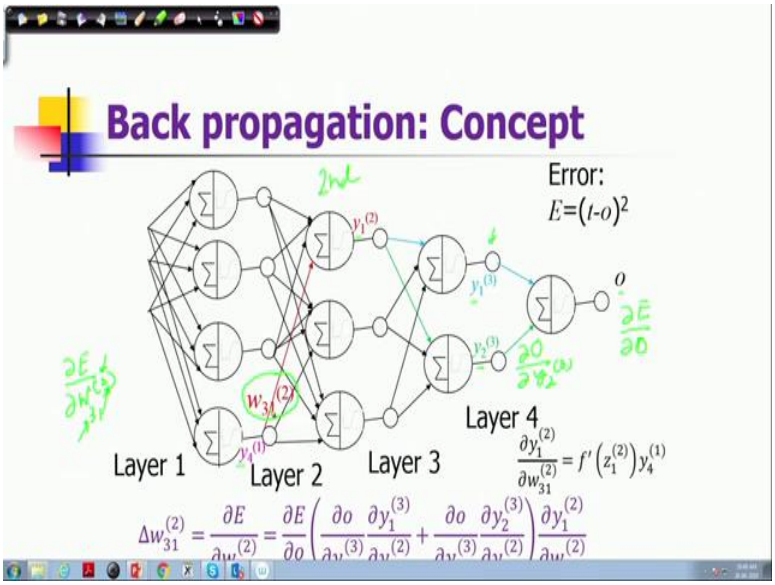
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So, when you would like to compute the gradient of a feed forward network, multilayered feed forward network; we apply the same chain rule. But, in this case since there is a layered computation now, the chain rule has to be also applied following that layered architecture which means that we should compute from output towards the input. So, you should compute all the derivatives from the output end and then you should proceed towards input and compute to successive derivatives.

So, which means from output layer to input player and we can compute the partial derivatives of weights at (i-1)th layer from the corresponding derivatives of the ith layer. We will see how we can do it.

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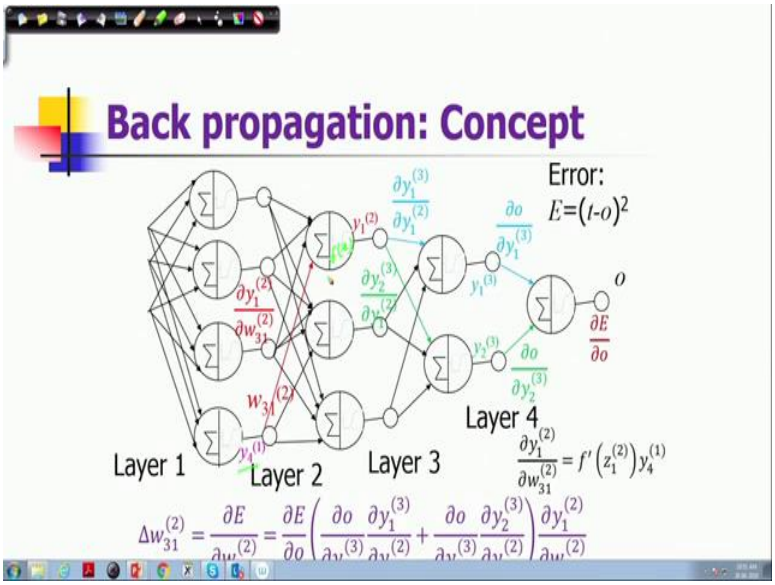


So, let us consider this particular concept here, we would like to compute the derivative with respect to this weight; you note that here in my notation 2 denotes the layer. So, this is a second layer and 3 denotes the jth neuron of the previous layer. So, this is a third neuron of the previous layer and here are the weight 3 1, it is a connectivity of the third neuron of the previous layer to the first neuron of the second layer. So, previous layer is a here so, that is a notation we are using here.

Similarly, these are y's are shown as outputs here. So, actually I have shown all these outputs which are affected by the change of this weights. So, this response is affected by the change of these weights. So, this is generating, this y_4^1 becomes an input to this particular neurons. So, it is acting like an input, but any change of this weight will affect the changes here. So, which means I need to compute here for example, deltaEdelta o then any change here will affect outputs. So, I need to compute delta o delta y 2 3; that means, with respect to this.

Similarly, I need to compute the derivative with respect to this, then I need to compute this derivative of these with respect to this. So, in between there is a functional block here so, we will have to consider that. So, in this way we have to compute. So, only these derivatives are to be computed and then you can find out you know this one.

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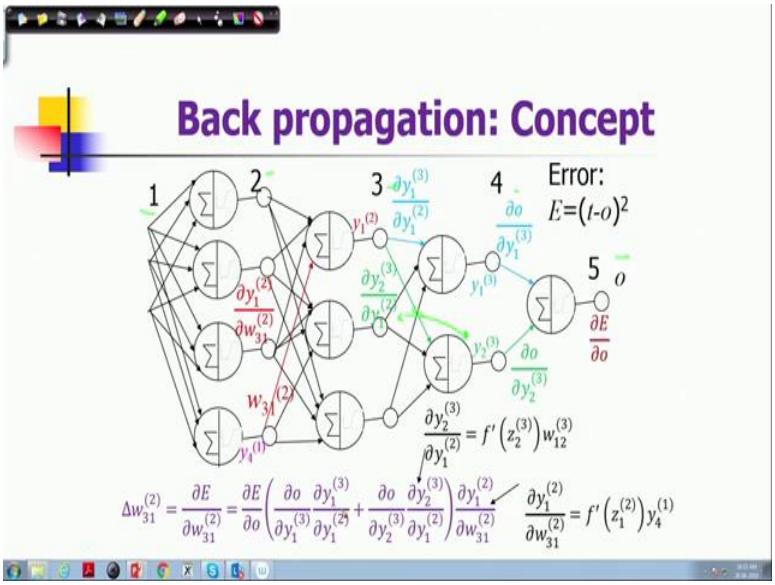


So, this is a relationship with respect to this and you note that the computation of

$$\frac{\partial y_1^{(2)}}{\partial w_{31}^{(2)}} = f'(z_1^{(2)}) y_4^{(1)}$$

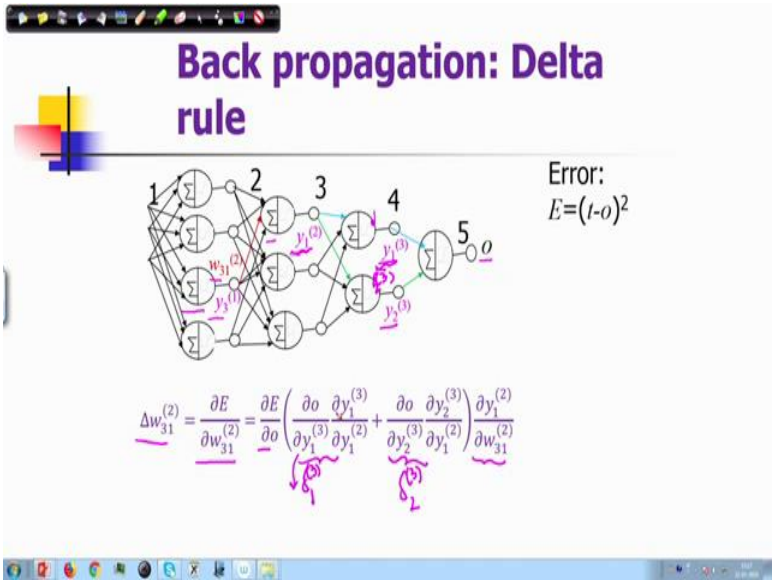
So, this is single neuron, this is the output.

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So, we will continue this computation once again. So, it shows at the top these are the layers 1 2 3 4 5, these are the layers and then it is this computation is carried out in this form. So, $\Delta w_{31}^{(2)}$

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We will be discussing about a particular simple rule by which the gradients can be propagated from output towards the input direction and that rule is called delta rule. So, let me explain this particular rule by which can organize the computations very nicely. So, we measure the error as we discussed that as a square of deviation from the target t to the at response given by the network for a particular input and particular values of weights parameters of that network which is o.

So $(t-o)^2$ is a error and we would like to find out compute the gradient to of this error with respect to a derivative of error E with respect to some parameters. For example, in this diagram we are considering the parameter w31. So, you note here in my notation this is a weight and you can see this 3 is a third neuron in the previous layer and 1 is the first neuron of the current layers. So, the previous layer of neuron this is the first layer and this is a second layer.

So, our convention is to denote the weight connectivity from the third neuron, output of the third neuron of the previous layer to the current layer is in this form. Similarly, the output of the third neuron of the previous layer which is the layer 1 is denoted here y 3 1 and also you can see output of the first neuron of layer 2 is denoted in this form. Similarly,

output of layer three first neuron of layer 3 is denoted here of layer 3, it is also denoted that is the second neuron in this form.

And finally, this is of course, the fourth layer and only one neuron is there and fifth layer is just output response which is redundant and we are not denoting with any other symbol other than o. So, this is a convention that we are following and we are showing all the necessary responses which are affected by the change of weights w31. So, when we compute the gradient with gradient of error E with respect to w31 this weight then these are the variables which will play a role in determining this gradient.

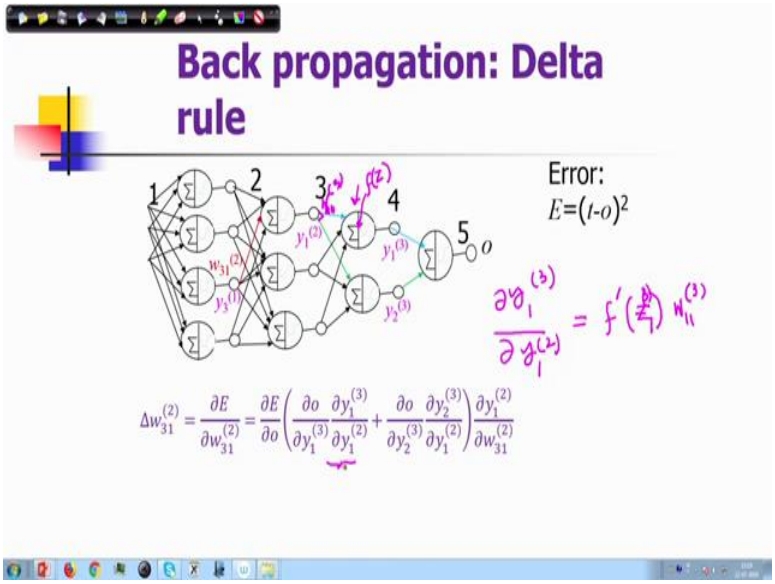
So, let us find out how this gradient value is dependent. So, you can see here we would be like to measure this value and which will give me the corresponding updates of the weight w 3 1 parameter. And so, you applying the chain rule. So, first we compute the gradient of E with respect to output o, then gradient of output o with respect to this gradient y1 and then subsequently gradient y 1 2.

So, this is that value, similarly these two added here. So, because of the linear operations this is other value and finally, when you are computing the gradient with respect to w 3 1, it is t y 1 2 with respect to w 3 1 2. So, these are the components which we need to compute or which these expressions which we need to find out given this particular responses at this instance. So, in this case we have already discussed how to compute the gradient of y 1 3 with respect to y 1 2.

Suppose we know the gradient of output o with respect to y 1 3 and that is we are calling that is a value delta 1 3. That means, this is the accumulated gradient till this point, till the output of the first neuron of third layer we denote in this fashion. Similarly, the gradient at this point will be denoted as delta 1 delta 2. So, it is delta 2 3 so, that is a gradient at this point, that is my notation and that is the definition of delta.

So, delta is the accumulated gradient till a output till the output of certain layer and the corresponding convention of writing delta in this fashion. Similarly, you can write the gradient changes along a edges also particular a edges, we will see how we can do it. So, let us expand this particular quantity; that means, gradient of y 1 3 with respect to y 1 2. So, let us explain this particular quantity which is the gradient of y 1 3 with respect to y 1 2.

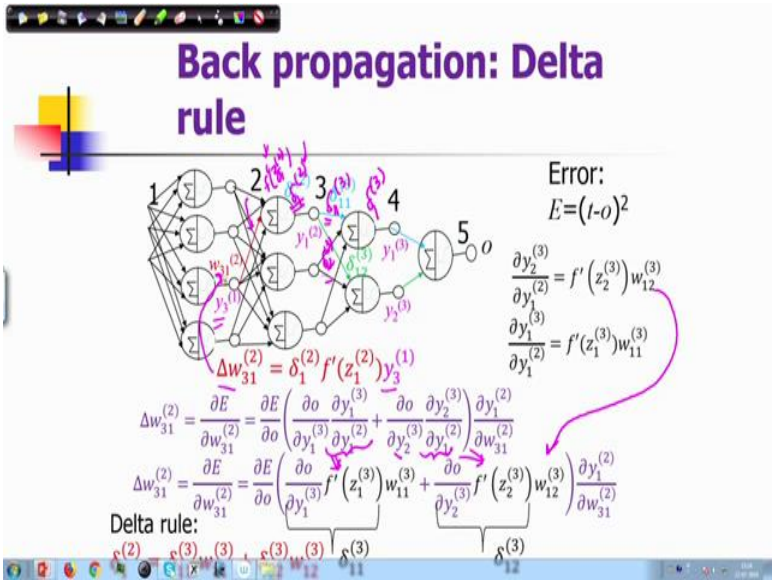
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So, these are how they are related, since in between you have the corresponding non-linearity which is f z and that would play into role. So, we can write it as the we can write this fact as say f dash z and then we know that there is a weight which is also connected here. So, it is since it is a linear combination of this. So, if this weight as per our convention it is w this is a first neuron. So, this is 1 and this is also 1 because this is the first neuron of third layer and this is 3.

So, according to our convention this is weight so, this is linearly related. So, this is a scale y 1 2 is scaled by w12 and then it is contributing to the net input which is a again determining the y 1 3. So, we can write delta y 1 3 delta y 1 2 as f dash z that is also at that point. So, we will see whether we have taken any conventions. So, we can write it also see f dash z 1 and it is layered 3 f dash z 1 3 into w 1 1 then 3. So, this is the expansion, similarly you can do this expansion and later on we will see how the chain rule is defined with respect to this.

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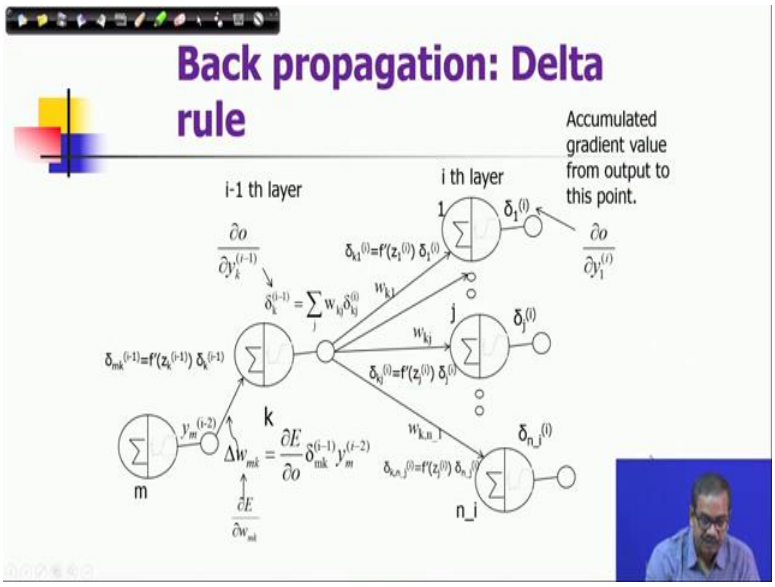
So, let me proceed. So, this is what as I mentioned we will compute this part and similarly we will compute the other one and this will expand. So, as I mentioned this is expanded into this form and this is expanded into this form, that is coming from here. This one is coming from here and this one is coming from here. Now, we define the delta 1 1 as I was mentioning delta 1 1, like I have defined earlier the delta this is delta 1 3 and at this edge this is delta 1 1 3 and that gets multiplied with w 1 1 3.

Similarly, at this stage this is delta 1 2 3 and that gets multiplied with w 1 2 3. So, we will find out how this is happening. So, this is what; this is what is delta 1 1 3 as I have mentioned, this is getting multiplied what with w 1 1 3 and this is w delta 1 2 3 and this is getting multiplied with w 1 2 3. And, that would give me the corresponding here that would give me delta 1 2 so, this is the delta rule. So, if I write it so, we can define the delta rule in this fashion as I was mentioning that delta 1 2.

So, this is what this is delta 1 2 is equal to then. So, this is added this is an this is equal to the weighted addition of delta 1 1 3 and delta 1 2 3 where the weights are the weights of the corresponding edges. So, that will be more clear when I show it here; so, this is a delta rule. So, once you get delta 1 2, similarly in the same way you can also get the delta value here. So, it means it has to be multiplied with f dash. So, from here it should be multiplied with f dash it should be z 1 2 and then you are computing actually the response of this point. So, it is delta y 1 2 delta 3 1 2.

So, it is this multiplied by f dash z 1 2 into the y 3 1 because, when you are computing the gradient at this point in the multiplication factor with respect to this parameter will be this response. So, that is what we will be writing here. So, you can see that it is delta 1 2 multiplied with f dash z 1 2. So, this is delta 1 2 this is f dash z 1 2 in that path and then you are multiplying it y 3 1, that would give you this quantity and which is used for updating this weight.

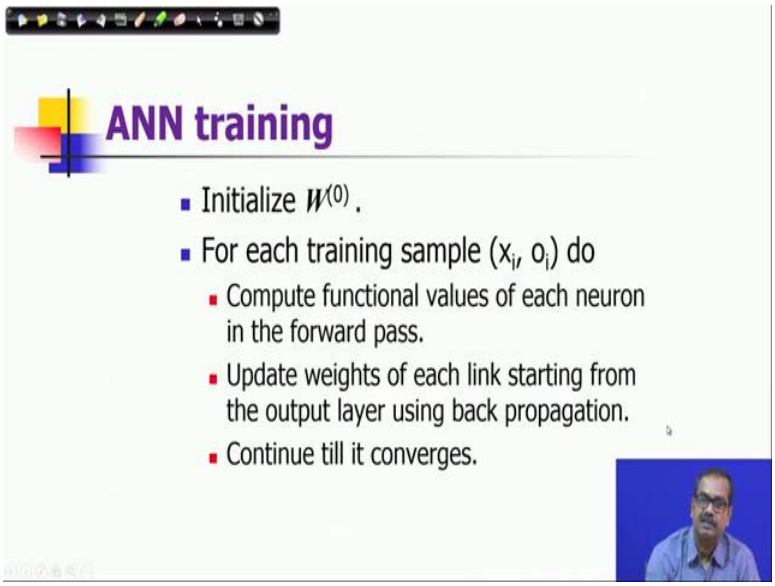
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I think this is explaining this diagram explaining in a bit in a clearer way. So, what it is doing let me see, let me explain; see you compute as I mentioned delta 1 i at this point. So, which is the gradient accumulated gradient value from output to this point, see this is delta j i. So, in that you are computing all these accumulated gradient value and then it is first it is propagated. So, delta k 1 i is multiplied with respect to f dash z so, this is delta k 1 i similarly delta k j i this.

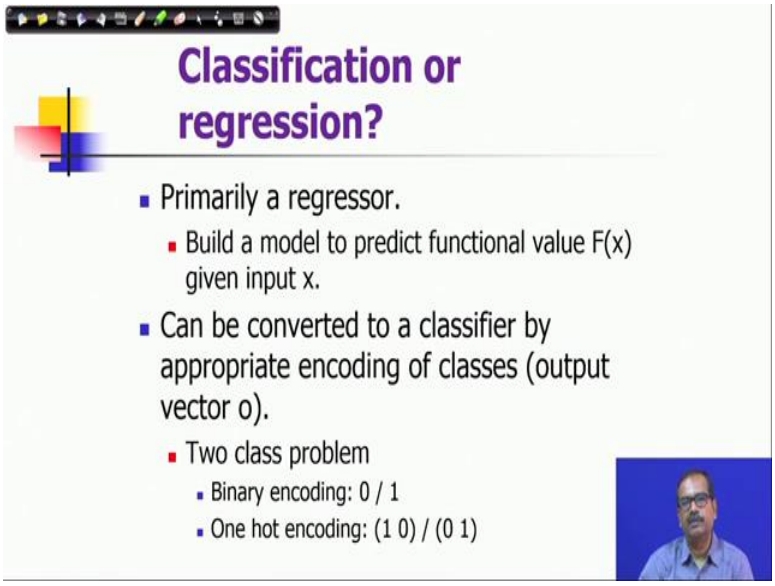
So, you take the weighted sum of all this. So, you take the weighted sum that gives you there the corresponding accumulated gradient value from the output to this point; so, at this point. So, and next you will propagate in this fashion; so, it will again propagate in this fashion. And so, the grad great way the update of weight should be this particular know the delta E delta o into this because it is from the weight output response. So, this is our update, in this way weights are you know updates are computed at every branch. So, this is what is at its semi computing in this gradient.

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So, the in this way you are computing the gradient of with respect to the corresponding weight, with respect from the gradient of the error and that is a how we get the gradient vector. And, then the algorithm follows in the same fashion that for each training sample you compute functional values of each neuron in the forward pass. And, then update weights of each link starting from the output layer using back propagation and then it should continue till it converges.

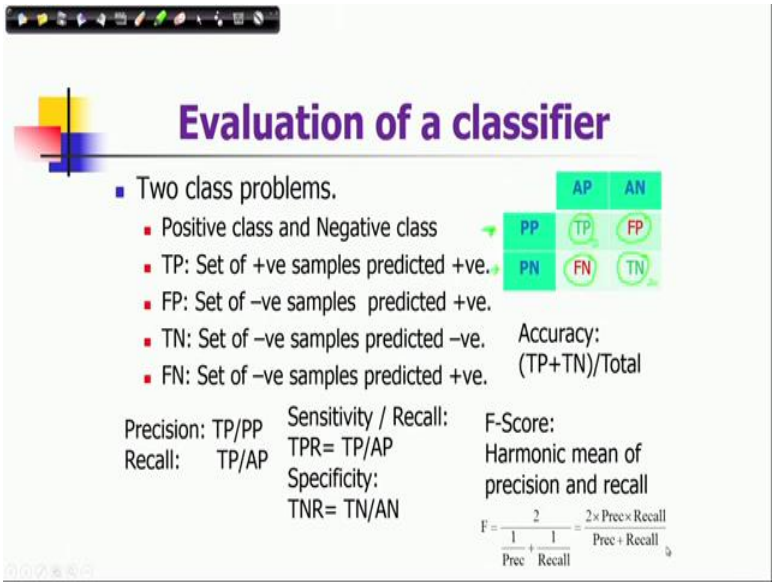
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One of the thing we would like to you know mentioned here that when you are doing artificial using this model whether it is a classification or regression model. Now, you can see primarily it is a regressor, it builds a model to predict no functional value F x given input x. But, you can convert this model, you can use this model also as a classifier by appropriate encoding of classes; that means, your output vector would be those encoding of classes.

For example, if you have a two class problem, you can consider a binary encoding. So, you can consider either 0 or 1 or you can also consider one hot encoding; that means, there are two neurons output neurons one of them will be 1, other will be 0. For the other class other will be 0 another is 1. In one hot encoding it is idea is that if there say n classes that could be represented by m such you know binary variables n bits and only one of them would be 1 for a particular class, rest will be 0.

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It is necessary that when you design a classifier you should evaluate a classifier, particularly for a supervision supervised classification problem. So, we will discuss about you know some methods of evaluation, some measures of evaluation. Consider you have two class problems and there are positive and negative classes. Now, there are several possibilities after classification like here in this diagram in this table I am showing the classification what has been predicted by the model.

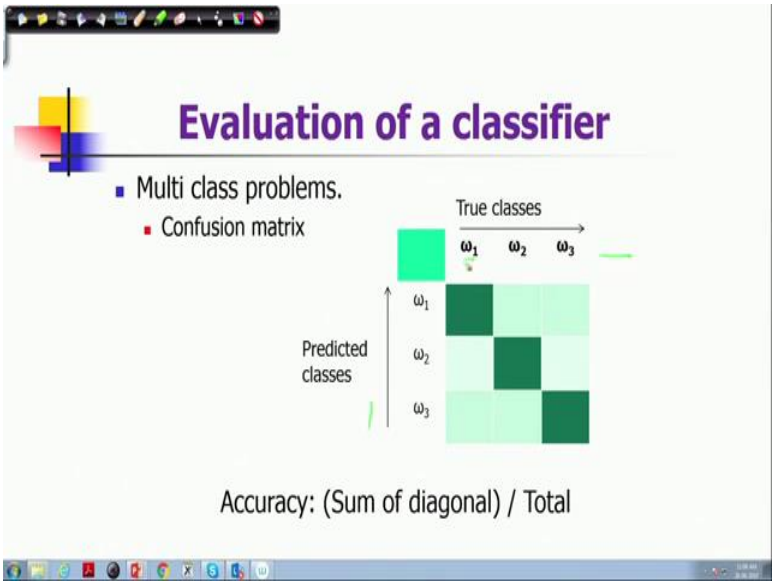
So, you are seeing where it is these are the predictive predicted positive and predicted negative. So, it can predict other positive or negative. So, it could happen that the sample is actually positive and it is also predicting positive that would give you two positive or it is negative, but it is predicting positive. So, it is the class of false positive. When it is positive actually positive, but it is predicting negative then it is false negative and in the other way when it is negative actually, but it is predicting also negative.

So, this is an desirable situation, this is true negative. So, this true positive and true negatives are desirable outcomes those numbers should be high whereas, these two numbers should be less. So, in a measure we use this numbers there are different measures, like if I consider accuracy of a classification we consider what is the total number of predictions which are true either positive or negative. So, it is true positive plus true negative by total number of you know samples which have been tested, there is there are other measures like precision and recall.

So, in precision you see that what is a fraction of predicted positives are true and recall is what is a fraction of actual positives are true. There are other measures like sensitivity which is the same as recall, these are used particularly for the medical world. So, or we can considered it also true positive rate. So, it is true positive by actual positive and specificity is true negative rate which is true negative by actual negative.

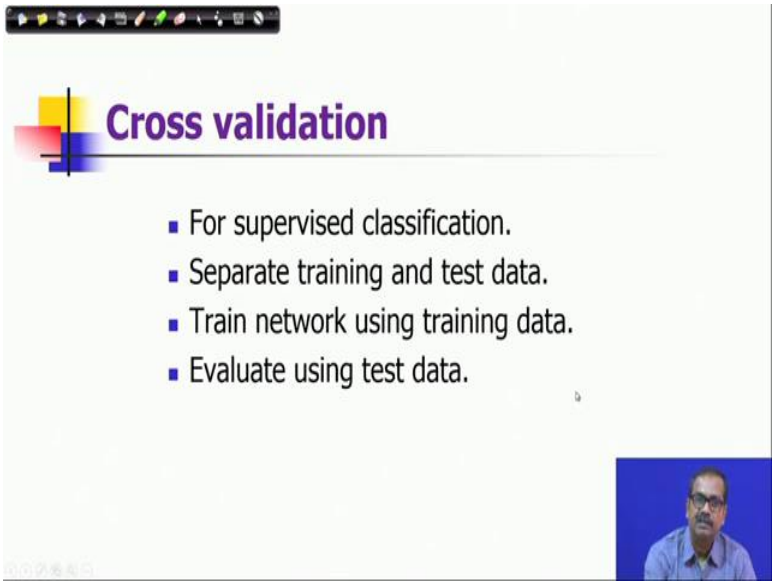
And, you can combine these scores two scores into one F score which is called harmonic mean of precision and recall so, which is has been defined here. So, higher the F score better is the classification, when you consider it in combination.

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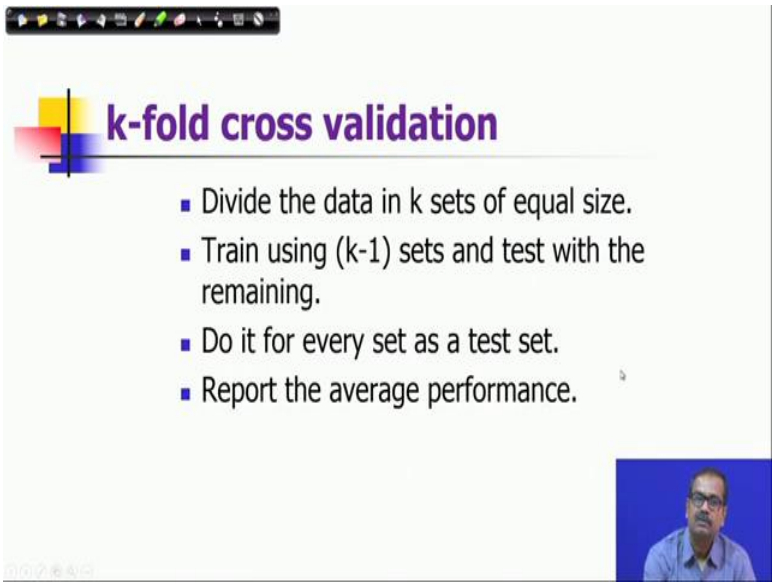
For a multi class problem we can use a confusion matrix where, once again you have you can I have shown here that they these are the true classes and this is the predicted classes. So, in the diagonal term if this numbers are high, those are the desirable outcomes. All other terms there are some kind of errors are there because, it belongs to actual class omega 1, but your prediction is omega 2 or omega 3. So, the accuracy measure which means which will be sum of diagonal by total.

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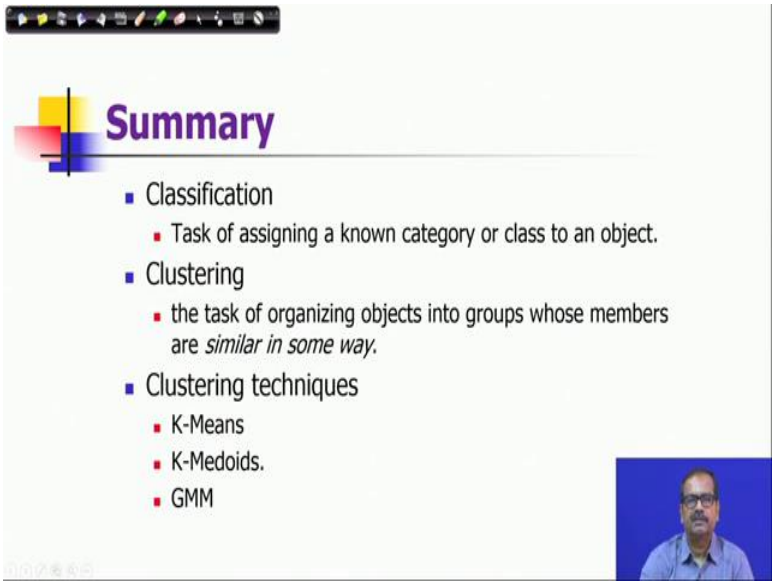
There are methods of you know testing the; testing the performance of a classifier. One method is called cross validation and once again it is applicable for supervised classification. What we do that we separate training and test data. Then we train network using training data and then evaluate using test data.

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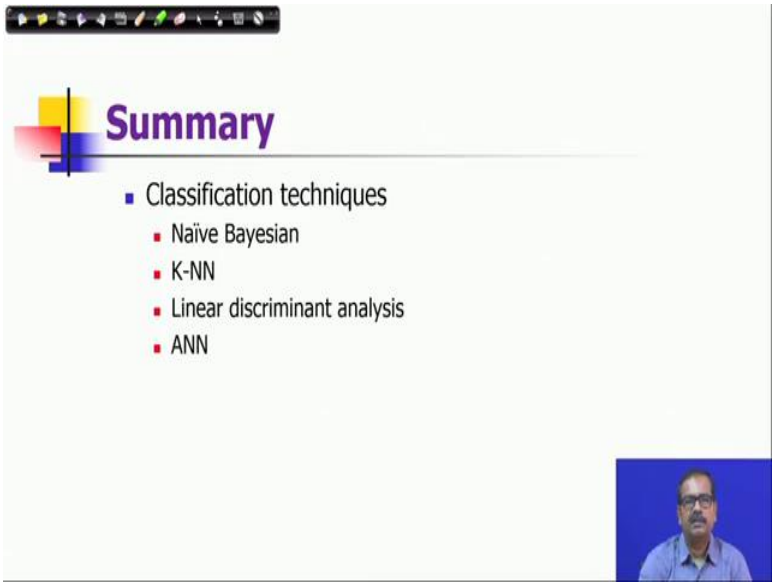
So, for k fold cross validation, we divide the data in k sets of equal size and we train using k minus 1 sets and test it with the remaining. And we do it for every set as a test set and take the average. So, report the average performance.

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So, here we comes to the you know come to the end of my you know talk for particular topic. So, just let me summarize whatever you have discussed under the topic of classification and clustering. So, as we as you know that classification is the task of assigning a known category or class to an object whereas, clustering is a task of organizing objects into groups whose members are similar in some way. So, clustering techniques there are several clustering techniques we discussed like K means, K medoids or Gaussian mixture model.

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And, for classification techniques we considered Naive Bayesian classification scheme, then K nearest neighbor classification scheme, linear discriminant analysis and finally, artificial neural network models. With this let me stop here and we will continue our discussion for our lectures of the next topic.

Thank you very much for listening to this talk.

Keywords: feed forward network, chain rule, back propagation, cross validation.